

Model Analysis

Table of contents

1	Data analysis /Dataframe import and filters	2
1.1	DataFrame import and filters	2
1.2	Metadata	5
2	Data correlation and clustering	6
2.1	Clustermat and cluster list	6
2.2	Features selection	7
3	Model(s) analysis	9
3.1	Metadata	9
3.2	Explanation legend tables	10
3.3	Repartition of train and test	11
3.3.1	External Split: X_train vs X_test	11
3.3.2	Internal CV Folds	11
3.4	Model : Linear	13
3.4.1	Gridsearch	13
3.4.2	Scatter	15
3.4.3	Scatter residuals	16
3.4.4	Learning curve	16
3.5	Model : LGBM	17
3.5.1	Gridsearch	17
3.5.2	Scatter	25
3.5.3	Scatter residuals	26
3.5.4	Learning curve	26
3.6	Model : CatBoost	27
3.6.1	Gridsearch	27
3.6.2	Scatter	34
3.6.3	Scatter residuals	35
3.6.4	Learning curve	35
3.7	Sumup-table	36

4	Plot	36
4.1	Standalone plots	36
4.1.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	36
4.1.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	39
4.1.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	42
4.1.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	45
4.1.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	48
4.1.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	51
4.1.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	54
4.1.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	57
4.2	Compare plots	60
4.2.1	Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3	60
4.2.2	Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11	62
4.2.3	Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5	63
4.2.4	Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4	64
4.2.5	Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11	66
4.2.6	Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4	67
4.2.7	Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5	68
4.2.8	Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3	70

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```

nan_ratio = df[cols_to_check].isna().mean()
valid_features = nan_ratio[nan_ratio <= 0.15].index
valid_features_list = list(valid_features)

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                        feats_to_balance + feats_grouping +
                        feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":

```

```

df["campaign"] = df["datetime"].apply(assign_campaign)
df = df[df["campaign"] != "other"].copy()
df["innova_point"] = df.apply(assign_innova_point, axis=1)
df = df[df["innova_point"] != "no"].copy()
with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
    #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["columnn"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}

```

```

        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: []

1.2 Metadata

Nombre de lignes df	7730
Nombre de lignes df clean	7730
Nombre de features	53
% de NaN	0.0
Taille dataset entraînement	5411
Taille dataset test	2319
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_mean, r_NH3_st_2h_std, r_NH3_st_2h_rms, r_NH3_st_2h_median, r_NH3_st_2h_min, r_NH3_st_2h_max, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_NH3_st_12h_mean, r_NH3_st_12h_std, r_NH3_st_12h_rms, r_NH3_st_12h_median, r_NH3_st_12h_min, r_NH3_st_12h_max, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_temperature_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_rms, r_temperature_st_2h_median, r_temperature_st_2h_min, r_temperature_st_2h_max, r_temperature_st_2h_diff_last, r_temperature_st_2h_slope, r_temperature_st_12h_mean, r_temperature_st_12h_std, r_temperature_st_12h_rms, r_temperature_st_12h_median, r_temperature_st_12h_min, r_temperature_st_12h_max, r_temperature_st_12h_diff_last, r_temperature_st_12h_slope, r_umidity_st_2h_mean, r_umidity_st_2h_std, r_umidity_st_2h_rms, r_umidity_st_2h_median, r_umidity_st_2h_min, r_umidity_st_2h_max, r_umidity_st_2h_diff_last, r_umidity_st_2h_slope, r_umidity_st_12h_mean,

r_umidity_st_12h_std, r_umidity_st_12h_rms, r_umidity_st_12h_median, r_umidity_st_12h_min, r_umidity_st_12h_max, r_umidity_st_12h_diff_last, r_umidity_st_12h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

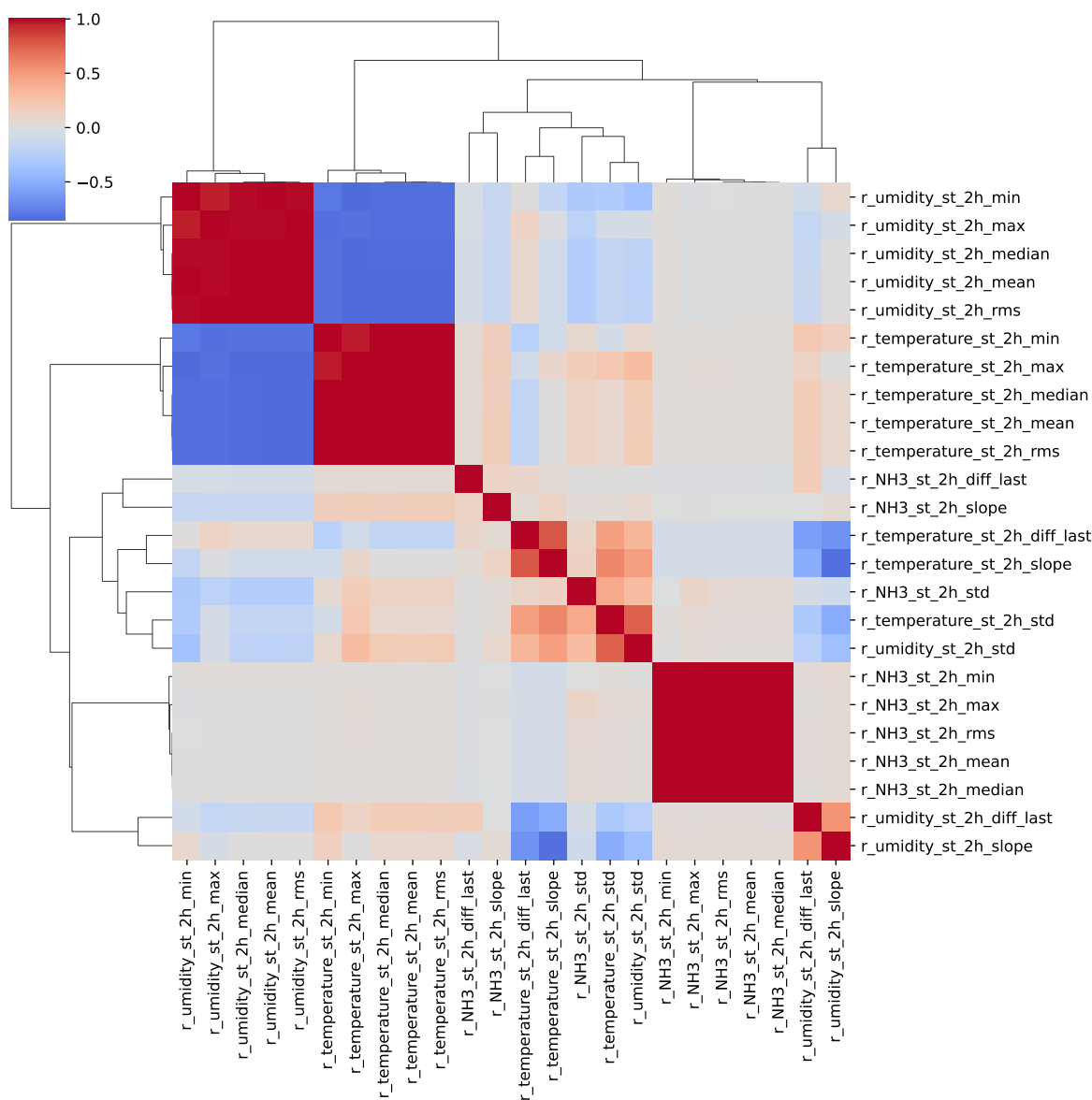
The following section has been done with a threshold choosen of : 0.8

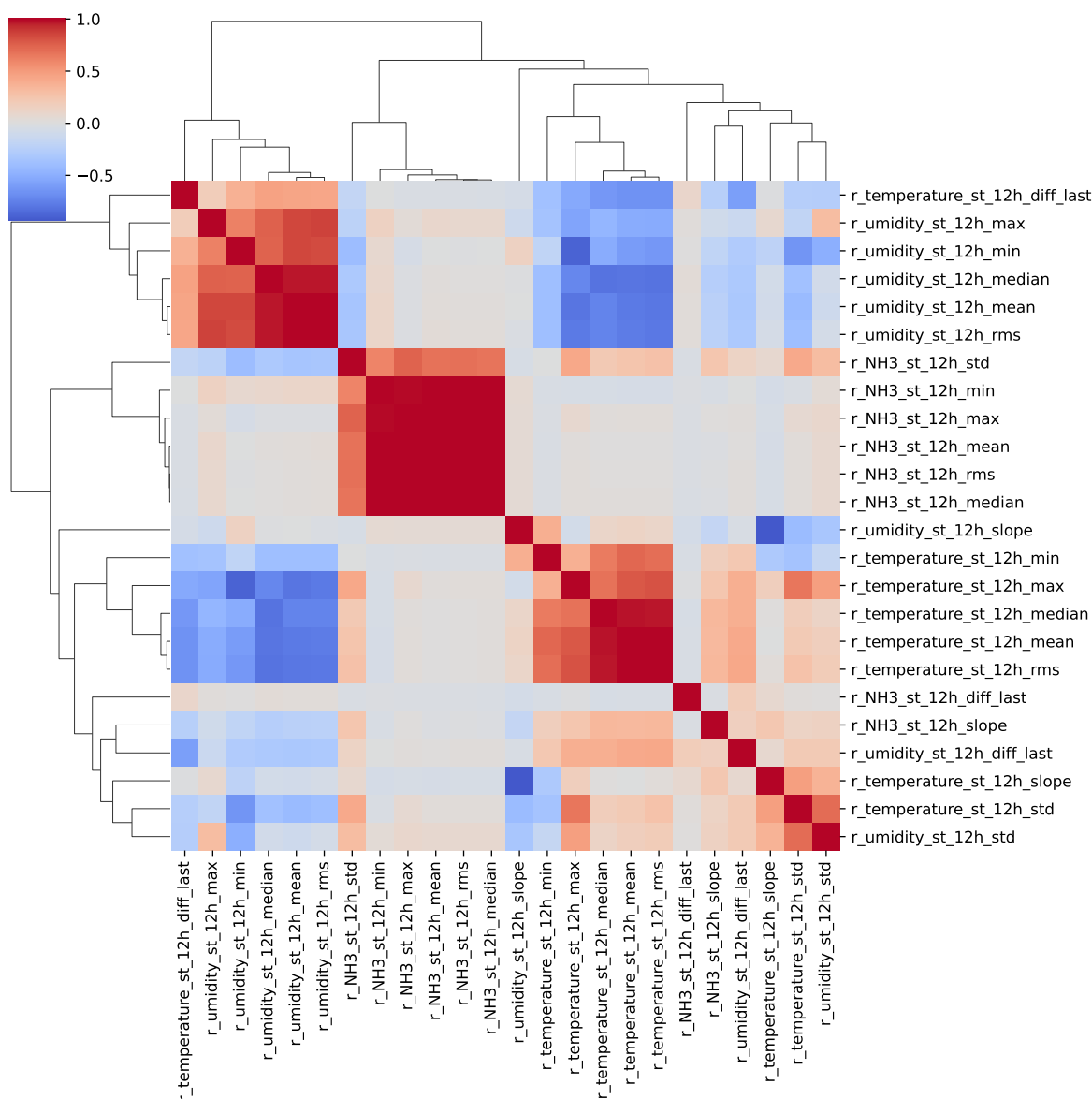
cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, __, T, H, I ; **cluster 6:** r_NH3_st_2h_median, r_NH3_st_2h_rms, r_NH3_st_2h_min, r_NH3_st_2h_max, r_NH3_st_2h_mean ; **cluster 7:** r_NH3_st_2h_std ; **cluster 8:** r_NH3_st_2h_diff_last ; **cluster 9:** r_NH3_st_2h_slope ; **cluster 10:** r_umidity_st_2h_mean, r_umidity_st_2h_min, r_temperature_st_2h_mean, r_temperature_st_2h_max, r_temperature_st_2h_rms, r_umidity_st_2h_max, r_temperature_st_2h_min, r_umidity_st_2h_rms, r_umidity_st_2h_median, r_temperature_st_2h_median ; **cluster 11:** r_temperature_st_2h_std ; **cluster 12:** r_temperature_st_2h_diff_last ; **cluster 13:** r_umidity_st_2h_slope, r_temperature_st_2h_slope ; **cluster 14:** r_umidity_st_2h_std ; **cluster 15:** r_umidity_st_2h_diff_last ; **cluster 16:** r_NH3_st_12h_rms, r_NH3_st_12h_mean, r_NH3_st_12h_min, r_NH3_st_12h_max, r_NH3_st_12h_median ; **cluster 17:** r_NH3_st_12h_std ; **cluster 18:** r_NH3_st_12h_diff_last ; **cluster 19:** r_NH3_st_12h_slope ; **cluster 20:** r_temperature_st_12h_median, r_temperature_st_12h_max, r_temperature_st_12h_mean, r_umidity_st_12h_median, r_umidity_st_12h_mean, r_umidity_st_12h_rms, r_umidity_st_12h_min, r_umidity_st_12h_max, r_temperature_st_12h_rms ; **cluster 21:** r_temperature_st_12h_std ; **cluster 22:** r_temperature_st_12h_min ; **cluster 23:** r_temperature_st_12h_diff_last ; **cluster 24:** r_umidity_st_12h_slope, r_temperature_st_12h_slope ; **cluster 25:** r_umidity_st_12h_std ; **cluster 26:** r_umidity_st_12h_diff_last ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features __X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_median, r_NH3_st_2h_std, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_umidity_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_diff_last, r_umidity_st_2h_slope, r_umidity_st_2h_std, r_umidity_st_2h_diff_last, r_NH3_st_12h_rms, r_NH3_st_12h_std, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_temperature_st_12h_median, r_temperature_st_12h_std, r_temperature_st_12h_min, r_temperature_st_12h_diff_last, r_umidity_st_12h_slope, r_umidity_st_12h_std, r_umidity_st_12h_diff_last





3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_2h_median, r_NH3_st_2h_std, r_NH3_st_2h_diff_last, r_NH3_st_2h_slope, r_umidity_st_2h_mean, r_temperature_st_2h_std, r_temperature_st_2h_diff_last, r_umid-

ity_st_2h_slope, r_umidity_st_2h_std, r_umidity_st_2h_diff_last, r_NH3_st_12h_rms, r_NH3_st_12h_std, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_temperature_st_12h_median, r_temperature_st_12h_std, r_temperature_st_12h_min, r_temperature_st_12h_diff_last, r_umidity_st_12h_slope, r_umidity_st_12h_std, r_umidity_st_12h_diff_last

Len(Features_X): 26

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

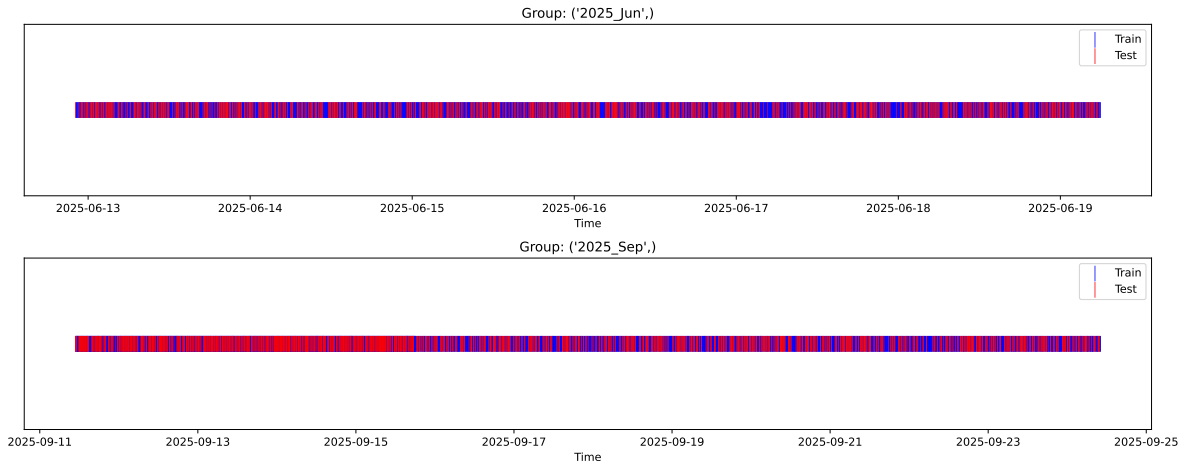
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_{train} vs X_{test}

External Split: Train: 5411 samples Test: 2319 samples



3.3.2 Internal CV Folds

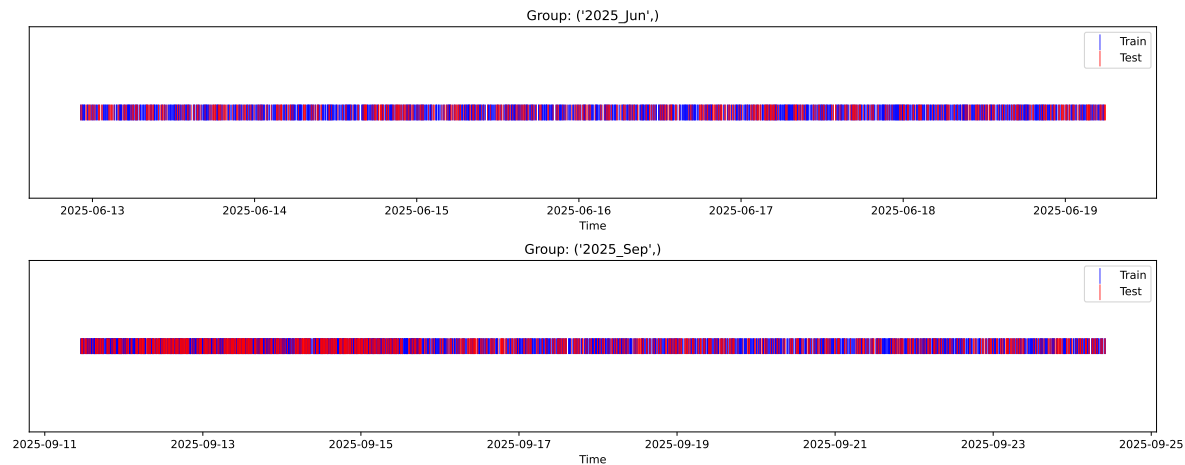
Fold 1/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



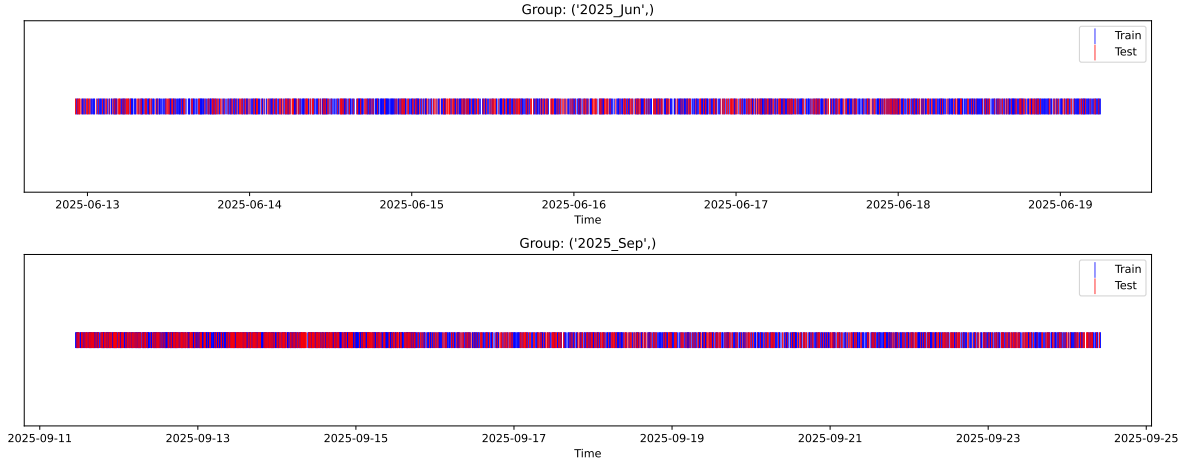
Fold 3/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



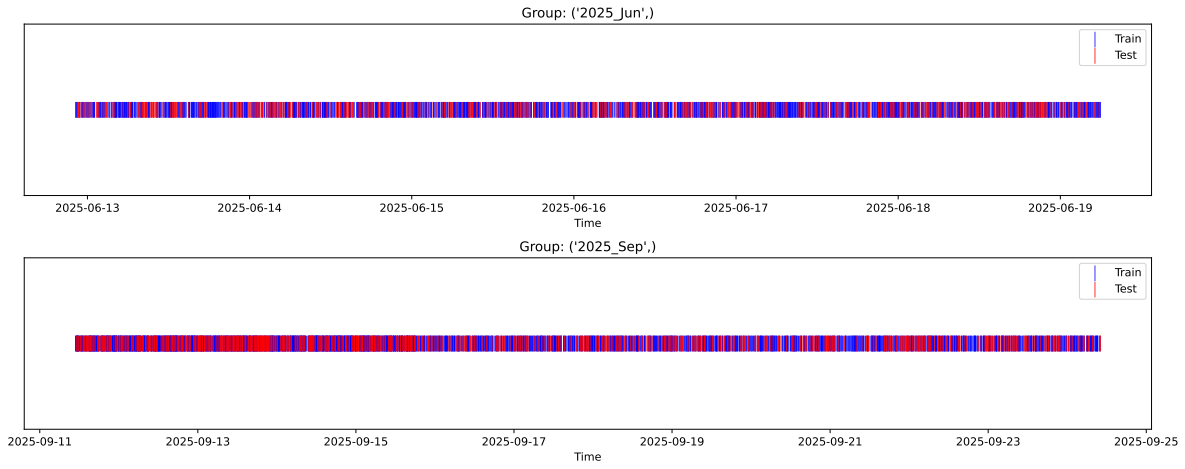
Fold 4/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.342, std=0.800, min=0.959, max=13.845
 y_{test} : mean=2.364, std=0.795, min=1.068, max=7.277

Fold 0 — y_{test} : mean=2.34, std=0.79, y_{train} : mean=2.34, std=0.80

Fold 1 — y_{test} : mean=2.35, std=0.83, y_{train} : mean=2.34, std=0.79

Fold 2 — y_{test} : mean=2.35, std=0.79, y_{train} : mean=2.34, std=0.81

Fold 3 — y_test: mean=2.33, std=0.77, y_train: mean=2.35, std=0.81

Fold 4 — y_test: mean=2.34, std=0.85, y_train: mean=2.34, std=0.78

train scores CV: [0.56457284 0.59145008 0.5647885 0.56362099 0.58044198]

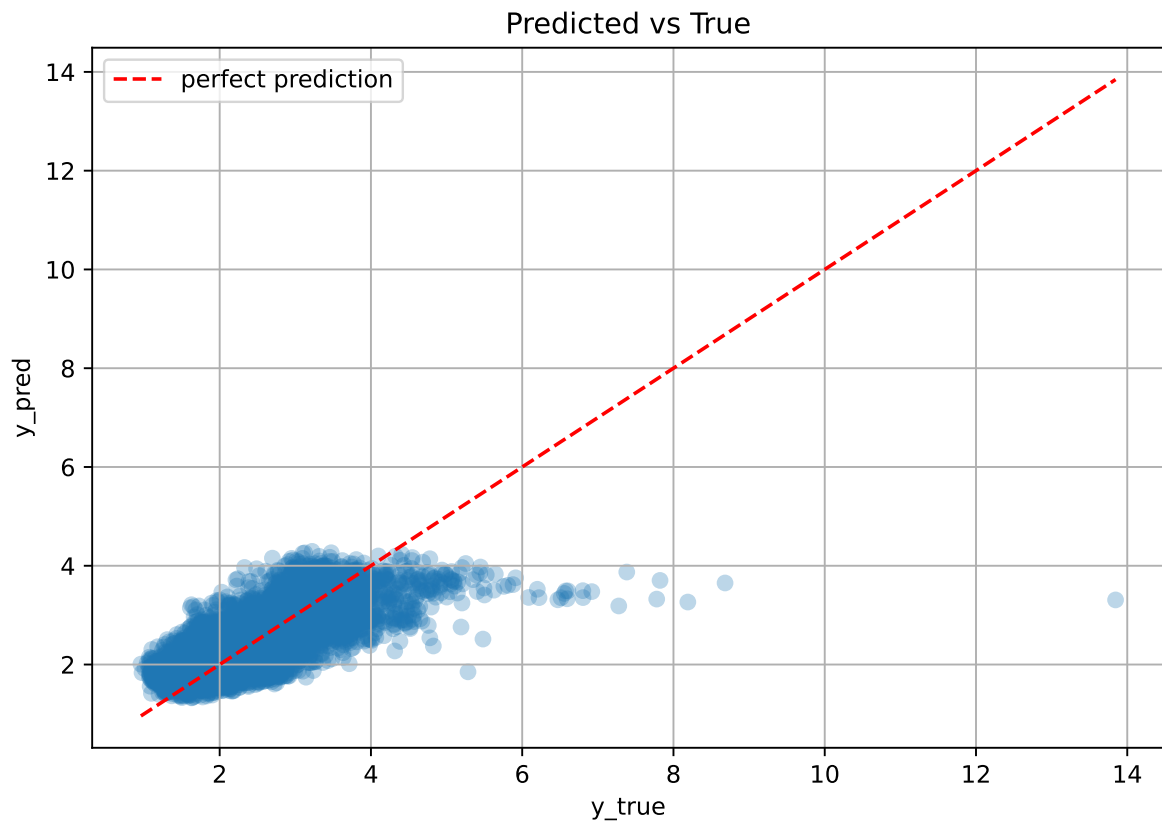
test scores CV: [0.56616807 0.50893293 0.56673854 0.56948298 0.53382236]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.573	0.549	0.0239	0.5891	0.5905	0.0415	1.0755	1.0436

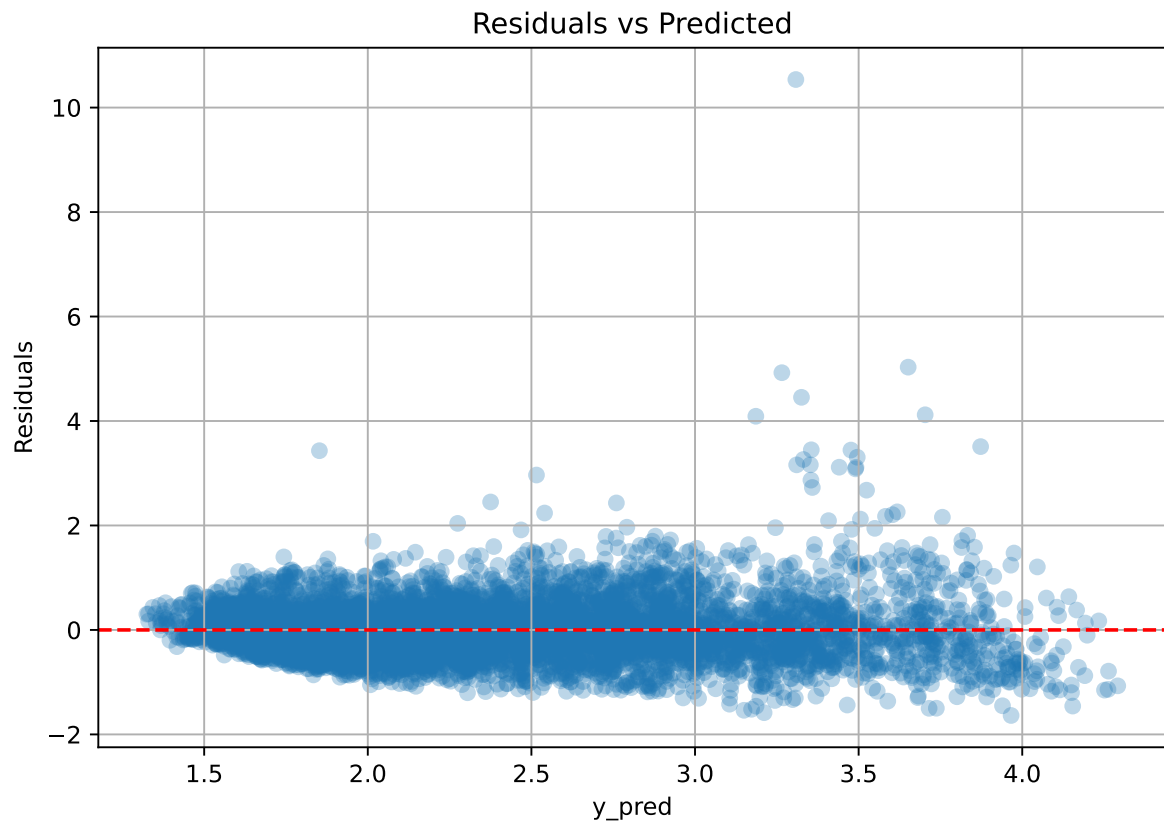
Coefficients:

coef_r_CO2 = -0.0487, coef_r_NH3 = 0.2967, coef_r_NH3_st_12h_diff_last = 0.0014, coef_r_NH3_st_12h_rms = 0.0046, coef_r_NH3_st_12h_slope = 0.0522, coef_r_NH3_st_12h_std = 0.0397, coef_r_NH3_st_2h_diff_last = 0.0014, coef_r_NH3_st_2h_median = -0.3799, coef_r_NH3_st_2h_slope = -0.0059, coef_r_NH3_st_2h_std = 0.0410, coef_r_THI = 0.4076, coef_r_temperature = -0.2061, coef_r_temperature_st_12h_diff_last = -0.0004, coef_r_temperature_st_12h_median = -0.1053, coef_r_temperature_st_12h_min = -0.0683, coef_r_temperature_st_12h_std = -0.0416, coef_r_temperature_st_2h_diff_last = -0.0004, coef_r_temperature_st_2h_std = 0.0748, coef_r_umidity = -1.0378, coef_r_umidity_st_12h_diff_last = 0.0059, coef_r_umidity_st_12h_slope = 0.2180, coef_r_umidity_st_12h_std = -0.0982, coef_r_umidity_st_2h_diff_last = 0.0059, coef_r_umidity_st_2h_mean = 0.5791, coef_r_umidity_st_2h_slope = 0.0297, coef_r_umidity_st_2h_std = 0.0390, intercept = 2.3417,

3.4.2 Scatter



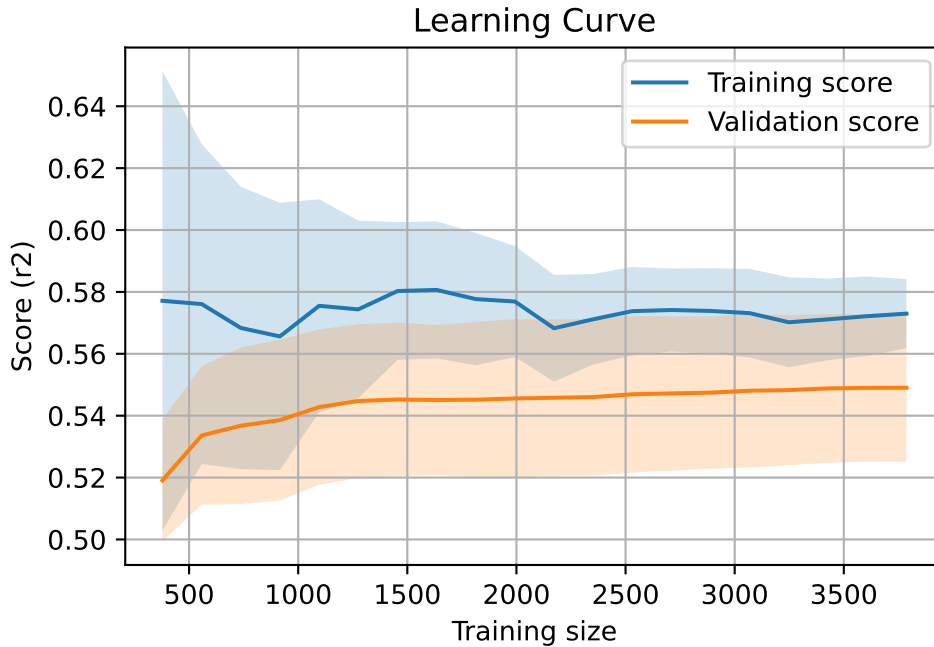
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7730, `Len(training)` : 5411,

`Len(training_real)` : 3788, `Len(test_of_training)` : 1623, `Len(test)` : 2319



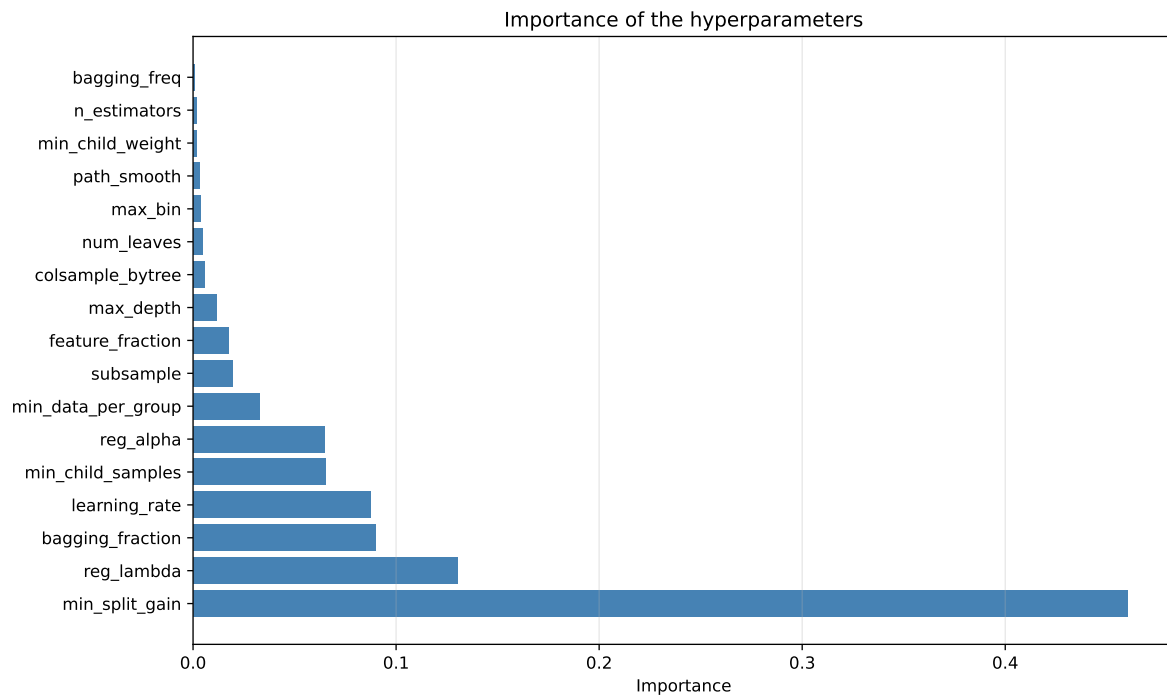
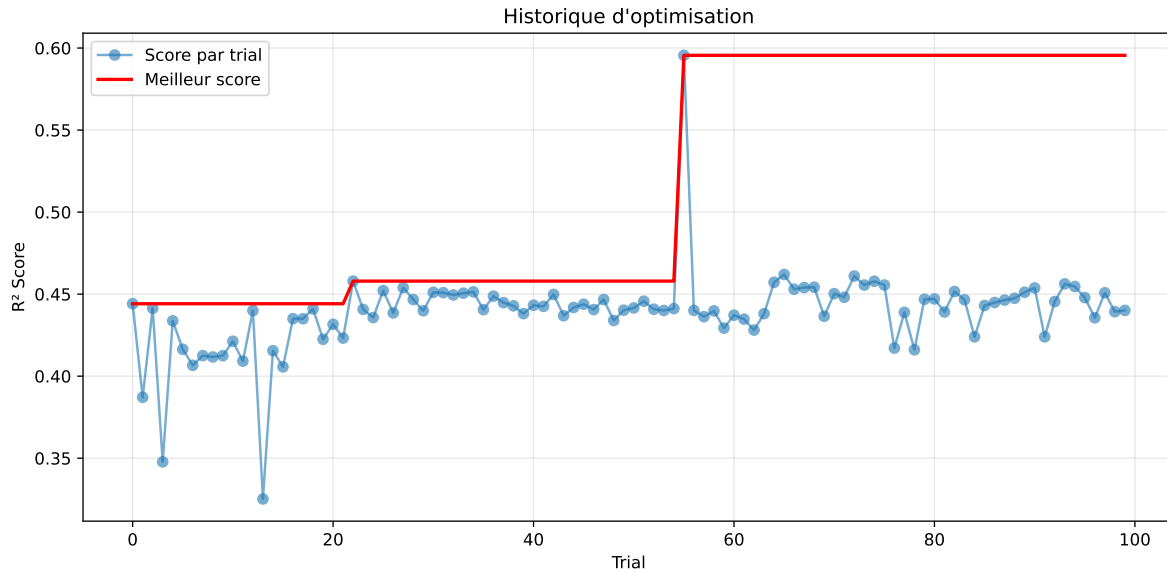
3.5 Model : LGBM

3.5.1 Gridsearch

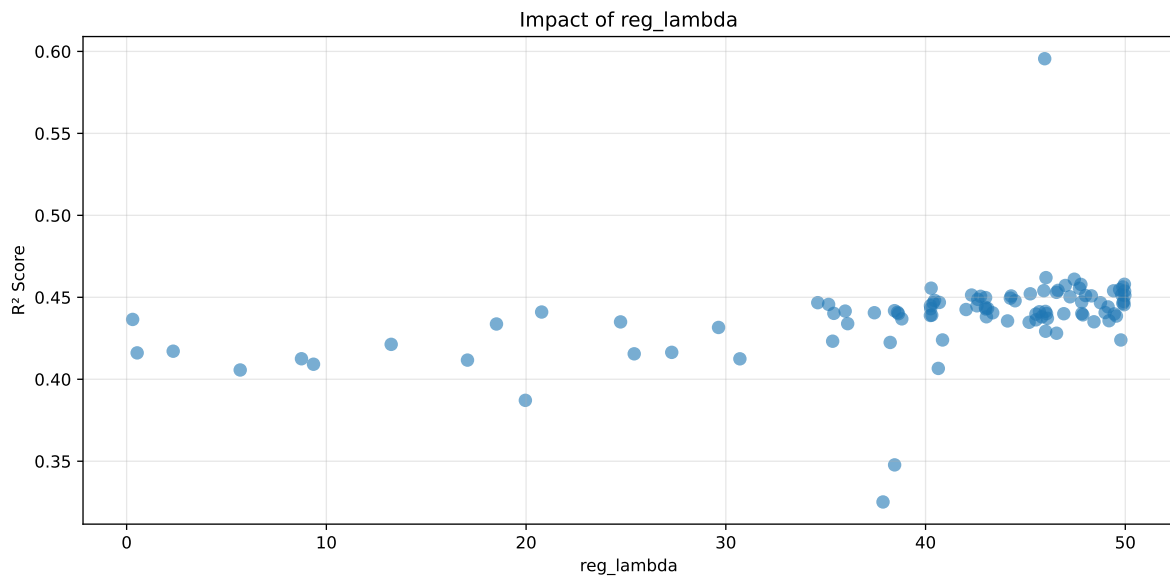
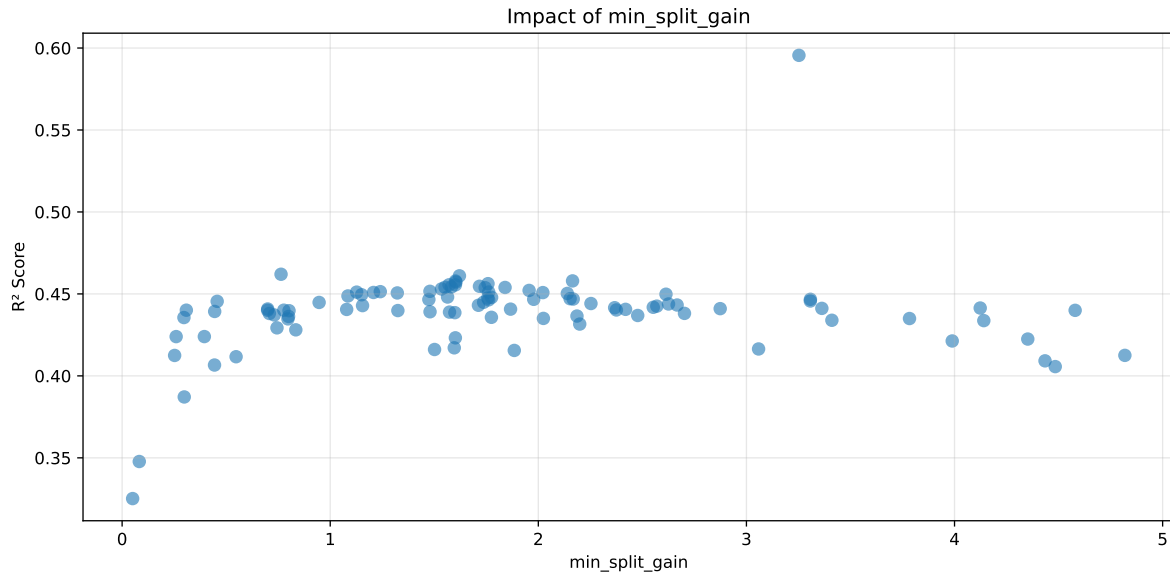
Distribution y_train vs y_test: y_train: mean=2.342, std=0.800, min=0.959, max=13.845
y_test: mean=2.364, std=0.795, min=1.068, max=7.277

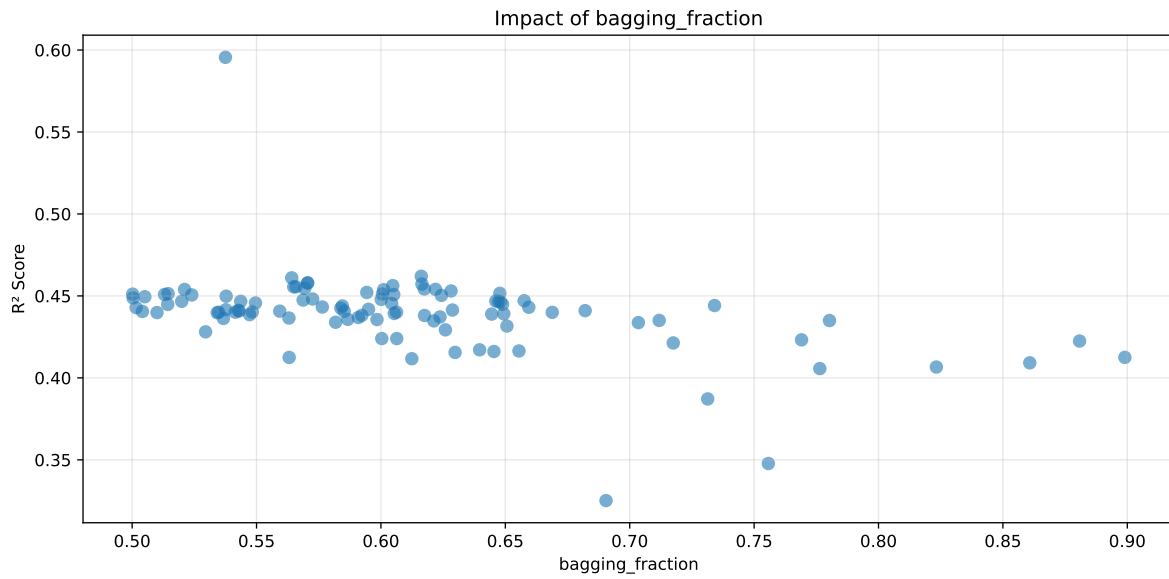
===== TOP Importance FEATURES =====

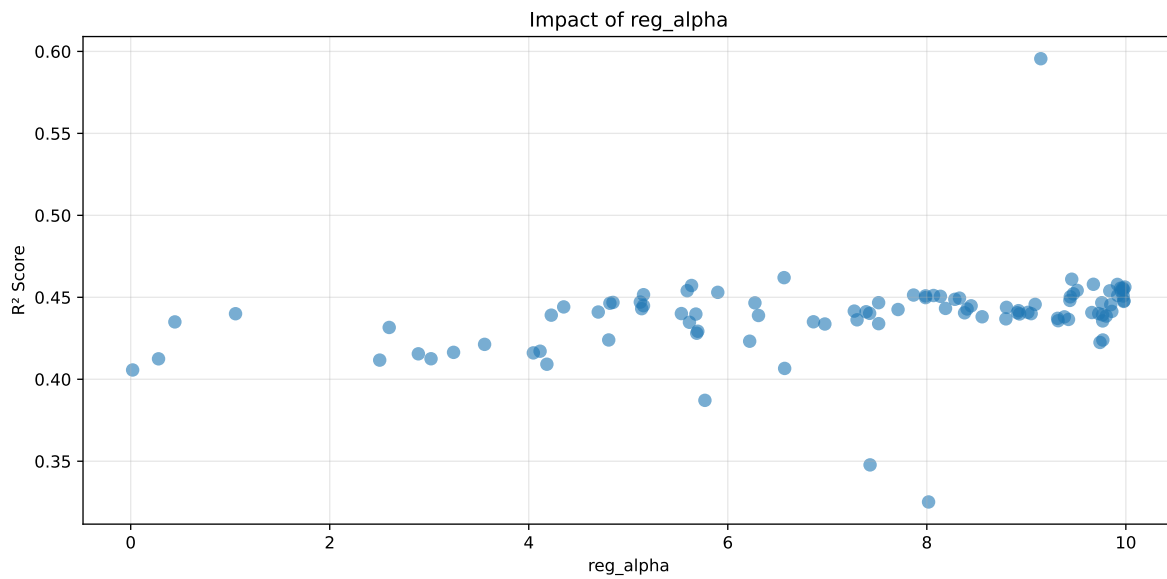
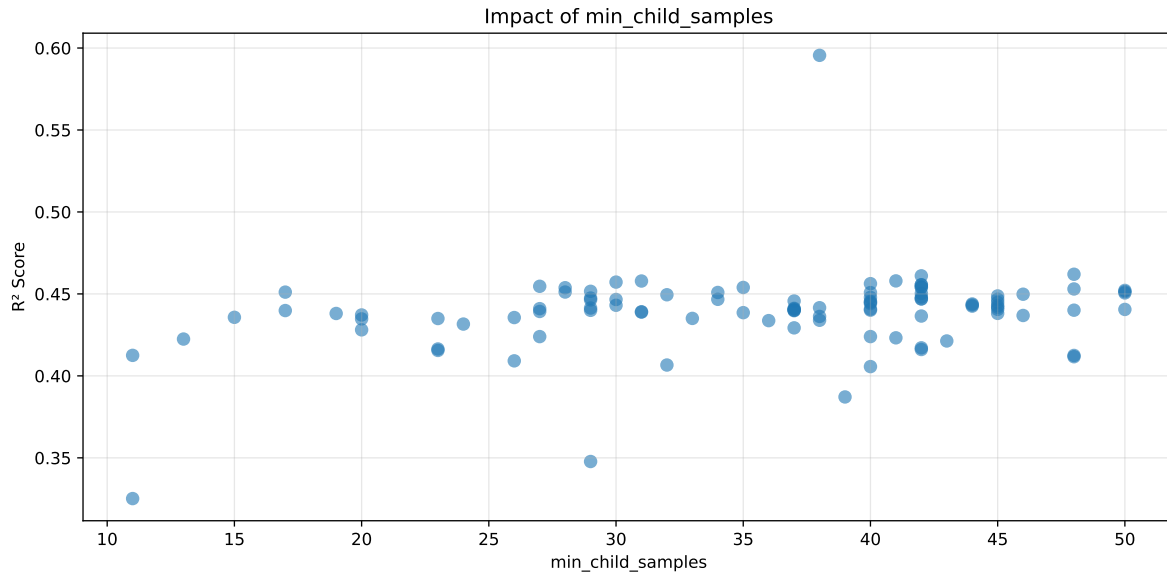
feature	importance
r_CO2	33
r_umidity	27
r_temperature_st_12h_median	26
r_NH3_st_2h_std	20
r_NH3_st_12h_rms	18
r_temperature_st_2h_std	18
r_umidity_st_2h_slope	13
r_temperature_st_2h_diff_last	13
r_temperature	12
r_umidity_st_2h_mean	10
r_umidity_st_12h_std	9
r_umidity_st_2h_diff_last	7
r_NH3_st_12h_slope	5
r_umidity_st_12h_slope	5
r_THI	5
r_NH3_st_2h_diff_last	4
r_temperature_st_12h_diff_last	4
r_temperature_st_12h_std	4
r_NH3	4
r_umidity_st_2h_std	3
r_NH3_st_2h_slope	3
r_umidity_st_12h_diff_last	3
r_NH3_st_12h_diff_last	1
r_NH3_st_12h_std	1
r_NH3_st_2h_median	1
r_temperature_st_12h_min	0



Paramètres avec >5% d'importance : ['min_split_gain', 'reg_lambda', 'bagging_fraction', 'learning_rate', 'min_child_samples', 'reg_alpha']

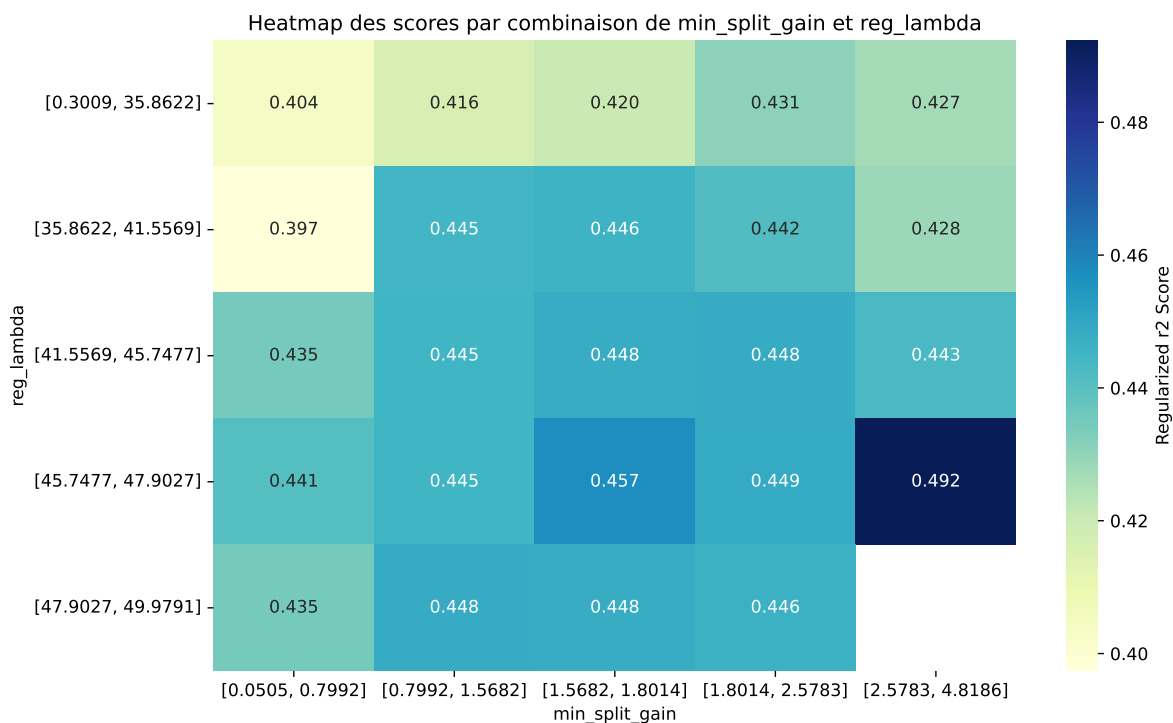






C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6252	0.5955	0.639	0.0262	0.5955	0.6538	0.0582	1.0978	1.0498
2	0.7065	0.6657	0.7088	0.0299	0.462	0.7346	0.0688	1.1034	1.0612
3	0.6662	0.632	0.6745	0.0259	0.461	0.6986	0.0666	1.1054	1.0541
4	0.6487	0.6169	0.6605	0.0284	0.4579	0.6888	0.0719	1.1165	1.0515
5	0.6632	0.629	0.6718	0.0248	0.4579	0.6992	0.0702	1.1116	1.0544
6	0.679	0.642	0.6861	0.0275	0.4572	0.7082	0.0662	1.103	1.0576
7	0.665	0.6302	0.6734	0.0245	0.4563	0.6977	0.0675	1.1072	1.0552
8	0.6616	0.6272	0.6711	0.0256	0.4556	0.699	0.0717	1.1143	1.0547
9	0.6575	0.6238	0.6683	0.0233	0.4555	0.6973	0.0735	1.1178	1.054
10	0.6611	0.6267	0.6701	0.025	0.4547	0.6915	0.0648	1.1035	1.0549
---	---	---	---	---	---	---	---	---	---
100	0.8007	0.7214	0.7668	0.0318	0.3251	0.7926	0.0712	1.0987	1.1099

Hyperparameters:

Rank 1: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.08582026533014522, 'reg__num_leaves': 59, 'reg__subsample': 0.6074937200609373, 'reg__colsample_bytree': 0.7551663106868876, 'reg__min_child_samples': 38, 'reg__reg_alpha': 9.14536837357834, 'reg__reg_lambda': 45.96483570625269, 'reg__min_split_gain': 3.252276729135974, 'reg__path_smooth': 2.6544647926495206, 'reg__min_child_weight':

0.010880603050495562, 'reg__feature_fraction': 0.6337515828254616, 'reg__bagging_fraction': 0.5375035343460679, 'reg__bagging_freq': 5, 'reg__max_bin': 101, 'reg__min_data_per_group': 189}

Rank 2: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.04752586048069002, 'reg__num_leaves': 61, 'reg__subsample': 0.691145324647886, 'reg__colsample_bytree': 0.8233492483201451, 'reg__min_child_samples': 48, 'reg__reg_alpha': 6.565775312943124, 'reg__reg_lambda': 46.02904390990243, 'reg__min_split_gain': 0.763674270791669, 'reg__path_smooth': 2.1863722108128973, 'reg__min_child_weight': 0.12353599502455563, 'reg__feature_fraction': 0.5664508551798428, 'reg__bagging_fraction': 0.6162219848301626, 'reg__bagging_freq': 7, 'reg__max_bin': 76, 'reg__min_data_per_group': 173}

Rank 3: {'reg__n_estimators': 600, 'reg__max_depth': 8, 'reg__learning_rate': 0.0454051646891175, 'reg__num_leaves': 61, 'reg__subsample': 0.5245540261274833, 'reg__colsample_bytree': 0.8186809306290933, 'reg__min_child_samples': 42, 'reg__reg_alpha': 9.45622646083344, 'reg__reg_lambda': 47.44720808893917, 'reg__min_split_gain': 1.620911270925359, 'reg__path_smooth': 2.0306631764965384, 'reg__min_child_weight': 0.004867314128836855, 'reg__feature_fraction': 0.5768647529644046, 'reg__bagging_fraction': 0.5641490756078444, 'reg__bagging_freq': 4, 'reg__max_bin': 97, 'reg__min_data_per_group': 173}

Rank 4: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.05256948401939769, 'reg__num_leaves': 63, 'reg__subsample': 0.5113436232633012, 'reg__colsample_bytree': 0.7351772521756833, 'reg__min_child_samples': 41, 'reg__reg_alpha': 9.673831718362353, 'reg__reg_lambda': 49.95495348031852, 'reg__min_split_gain': 2.164476651793552, 'reg__path_smooth': 1.8839611479482792, 'reg__min_child_weight': 0.01223519106878441, 'reg__feature_fraction': 0.6045466611139206, 'reg__bagging_fraction': 0.5705181740190078, 'reg__bagging_freq': 6, 'reg__max_bin': 67, 'reg__min_data_per_group': 198}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.04994113036460646, 'reg__num_leaves': 46, 'reg__subsample': 0.5255884620830551, 'reg__colsample_bytree': 0.8162516518765772, 'reg__min_child_samples': 31, 'reg__reg_alpha': 9.915309510044825, 'reg__reg_lambda': 47.77584563104878, 'reg__min_split_gain': 1.601968427210473, 'reg__path_smooth': 2.172613525991189, 'reg__min_child_weight': 0.0031810499944928228, 'reg__feature_fraction': 0.5369255819053224, 'reg__bagging_fraction': 0.570533472008863, 'reg__bagging_freq': 4, 'reg__max_bin': 95, 'reg__min_data_per_group': 173}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.08787655477195587, 'reg__num_leaves': 61, 'reg__subsample': 0.6965575720080321, 'reg__colsample_bytree': 0.8229564641428506, 'reg__min_child_samples': 30, 'reg__reg_alpha': 5.63584879108397, 'reg__reg_lambda': 46.99796239678854, 'reg__min_split_gain': 1.6041601834583359, 'reg__path_smooth': 3.3827258784739107, 'reg__min_child_weight':

0.09024557159342879, 'reg__feature_fraction': 0.5787186818680029, 'reg__bagging_fraction': 0.6164711991712278, 'reg__bagging_freq': 7, 'reg__max_bin': 79, 'reg__min_data_per_group': 189}

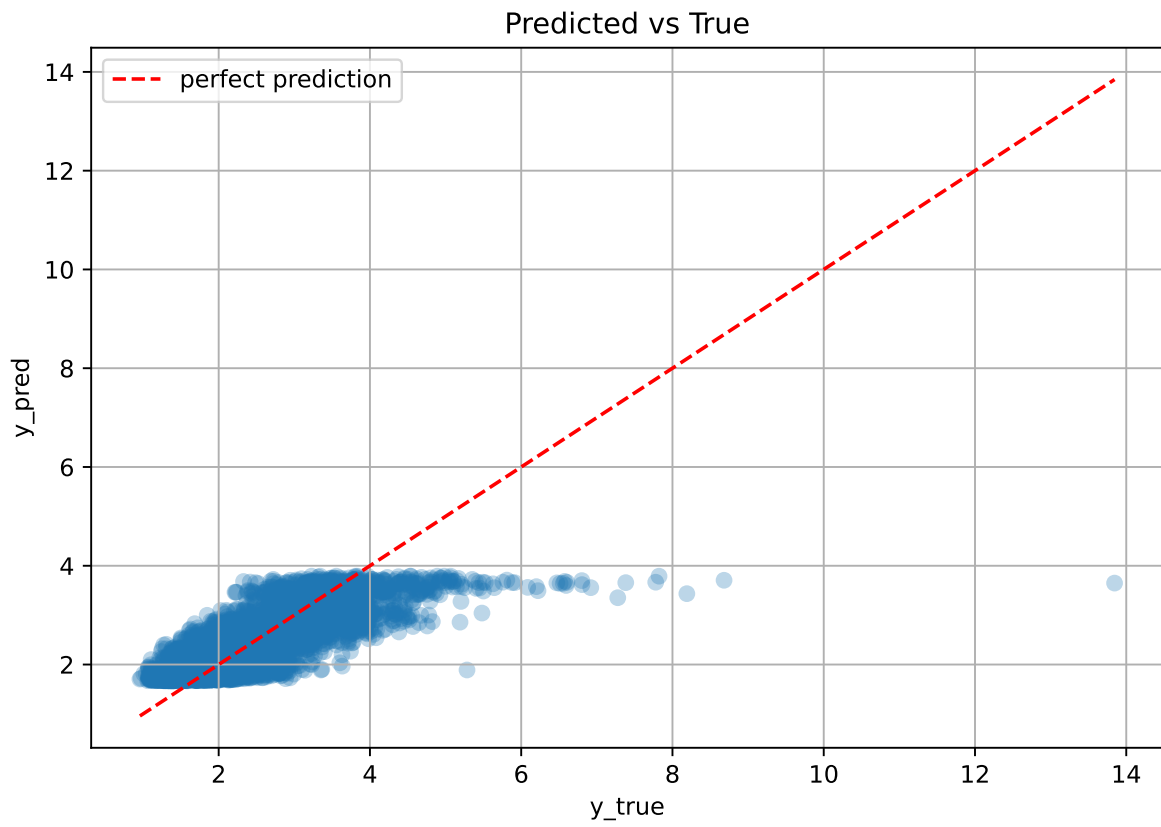
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.042619571043643494, 'reg__num_leaves': 38, 'reg__subsample': 0.7546154310529497, 'reg__colsample_bytree': 0.8608217304536228, 'reg__min_child_samples': 40, 'reg__reg_alpha': 9.99021683269514, 'reg__reg_lambda': 49.8798230989867, 'reg__min_split_gain': 1.7582079499083823, 'reg__path_smooth': 1.9025024204521435, 'reg__min_child_weight': 0.0023083300335546095, 'reg__feature_fraction': 0.5436525219515459, 'reg__bagging_fraction': 0.6047642629631834, 'reg__bagging_freq': 3, 'reg__max_bin': 111, 'reg__min_data_per_group': 148}

Rank 8: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.035198472852081254, 'reg__num_leaves': 61, 'reg__subsample': 0.5244960742572456, 'reg__colsample_bytree': 0.8507935931979386, 'reg__min_child_samples': 42, 'reg__reg_alpha': 9.968717252984218, 'reg__reg_lambda': 40.27952105250667, 'reg__min_split_gain': 1.5710185646475674, 'reg__path_smooth': 2.0566145289985274, 'reg__min_child_weight': 0.005342352245509374, 'reg__feature_fraction': 0.5328311259200106, 'reg__bagging_fraction': 0.5658664448125367, 'reg__bagging_freq': 4, 'reg__max_bin': 98, 'reg__min_data_per_group': 175}

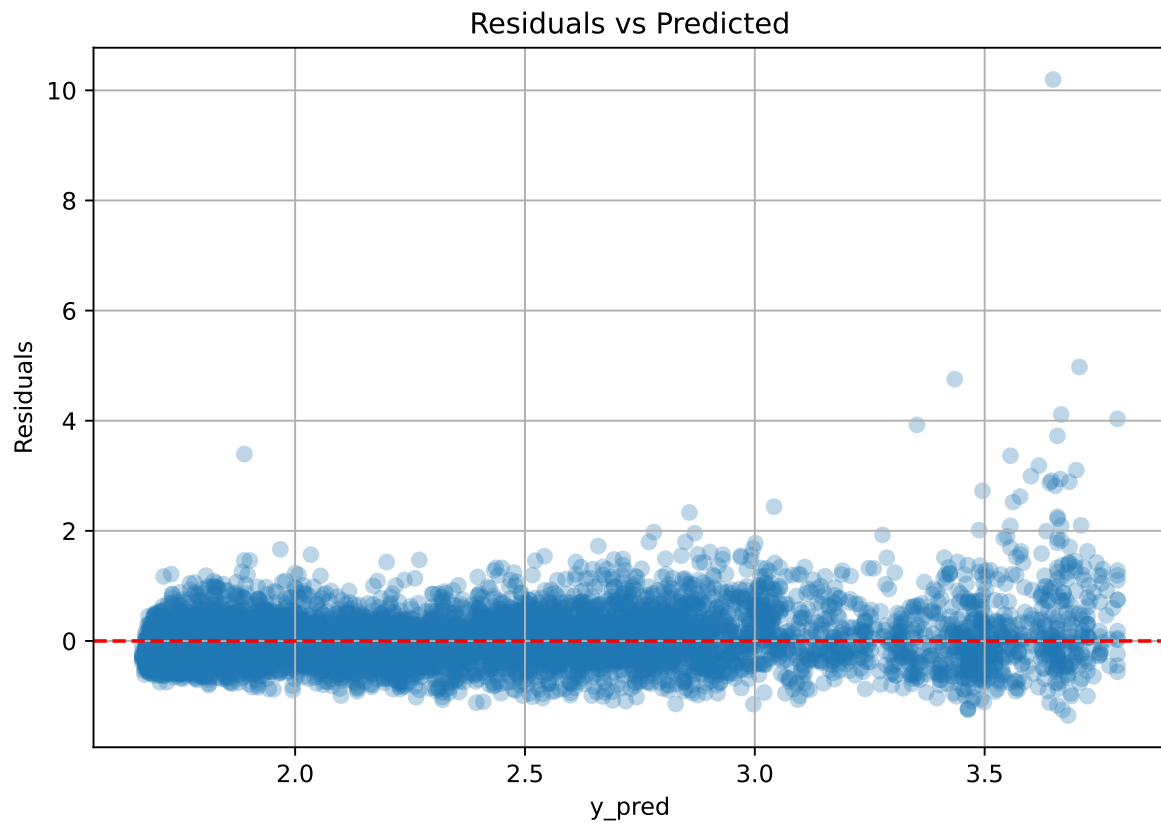
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.0494776679513013, 'reg__num_leaves': 61, 'reg__subsample': 0.5316986176898404, 'reg__colsample_bytree': 0.8148823196722436, 'reg__min_child_samples': 42, 'reg__reg_alpha': 9.946547011697733, 'reg__reg_lambda': 47.70712615399739, 'reg__min_split_gain': 1.6016150265813518, 'reg__path_smooth': 2.2067727991448254, 'reg__min_child_weight': 0.005350909081322934, 'reg__feature_fraction': 0.5767488580862501, 'reg__bagging_fraction': 0.5649370443782976, 'reg__bagging_freq': 6, 'reg__max_bin': 101, 'reg__min_data_per_group': 145}

Rank 10: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.0421525964395641, 'reg__num_leaves': 47, 'reg__subsample': 0.6197318093484372, 'reg__colsample_bytree': 0.8069022723793152, 'reg__min_child_samples': 27, 'reg__reg_alpha': 9.977327588996642, 'reg__reg_lambda': 49.71576729946056, 'reg__min_split_gain': 1.7171760129479614, 'reg__path_smooth': 2.4712244127021656, 'reg__min_child_weight': 0.0011363917818616104, 'reg__feature_fraction': 0.5441782000731513, 'reg__bagging_fraction': 0.5693788844190395, 'reg__bagging_freq': 4, 'reg__max_bin': 93, 'reg__min_data_per_group': 151}

3.5.2 Scatter



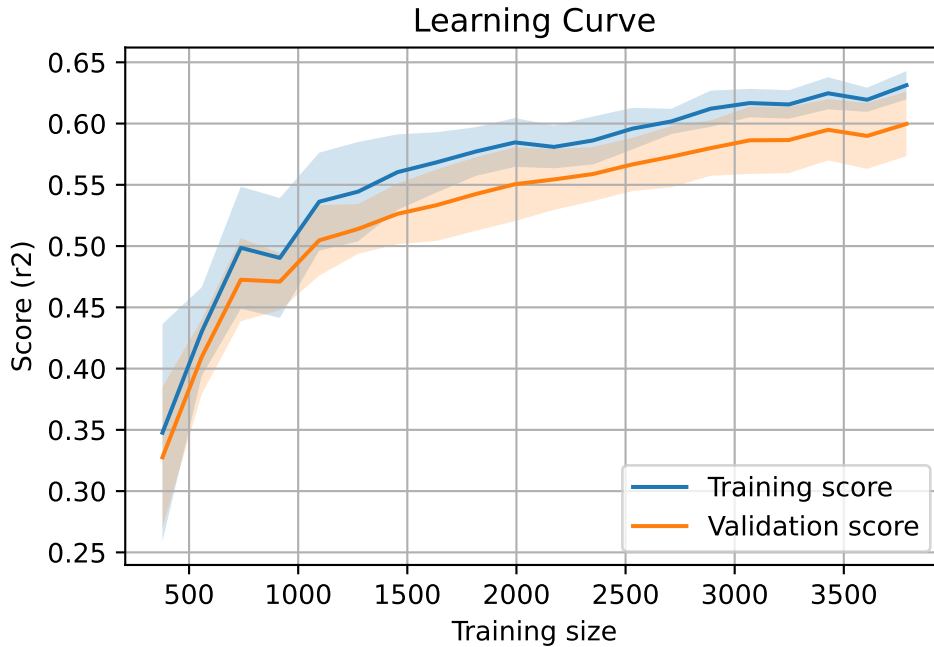
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7730, `Len(training)` : 5411,

`Len(training_real)` : 3788, `Len(test_of_training)` : 1623, `Len(test)` : 2319



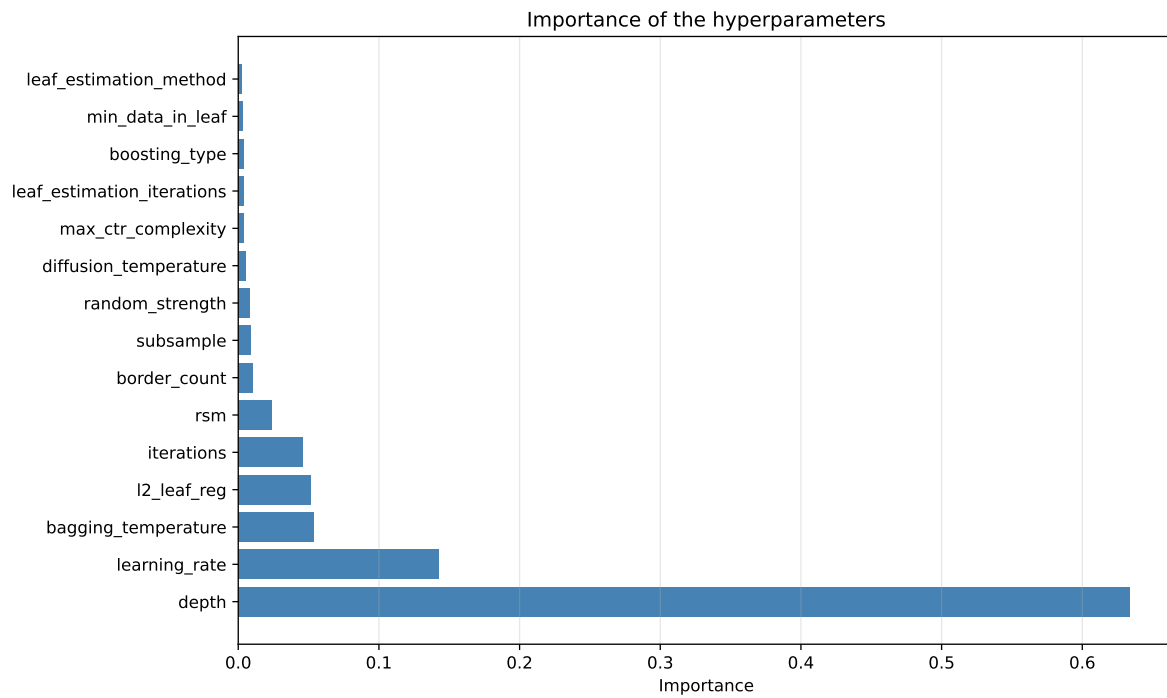
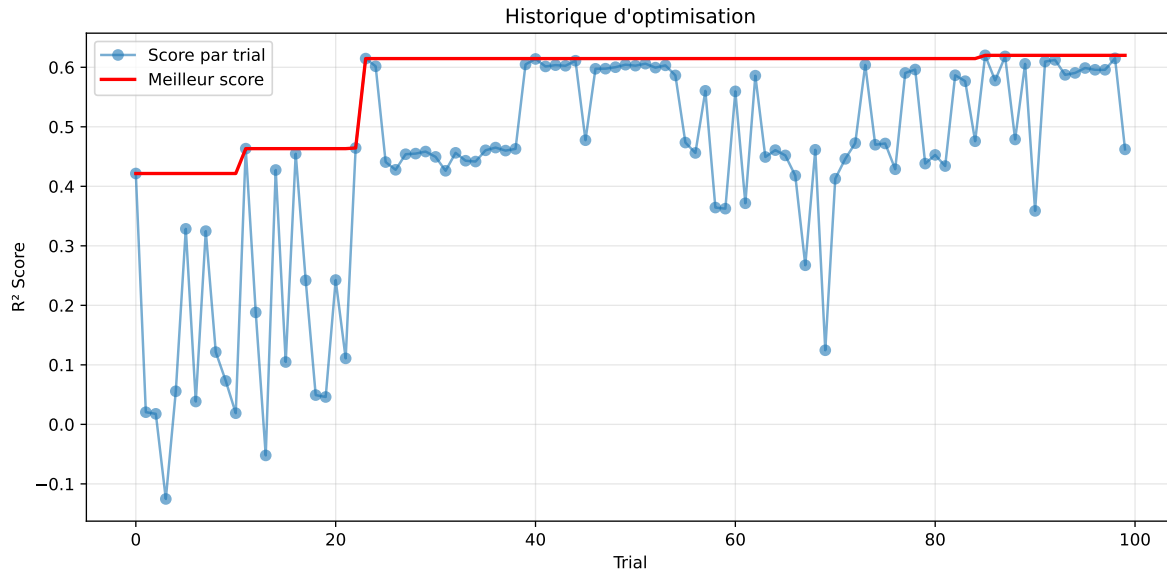
3.6 Model : CatBoost

3.6.1 Gridsearch

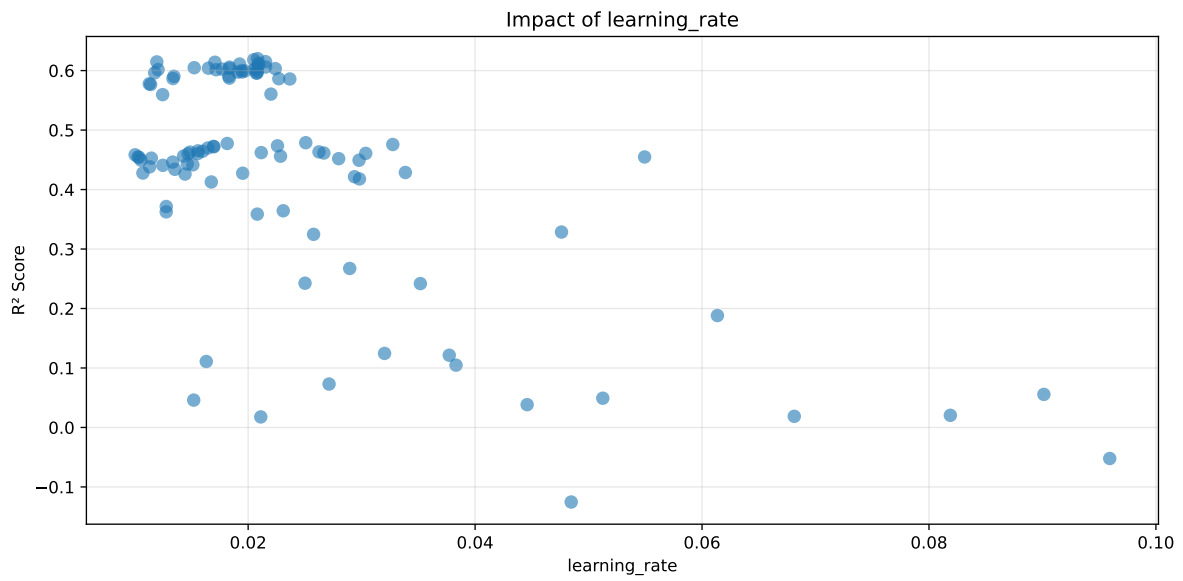
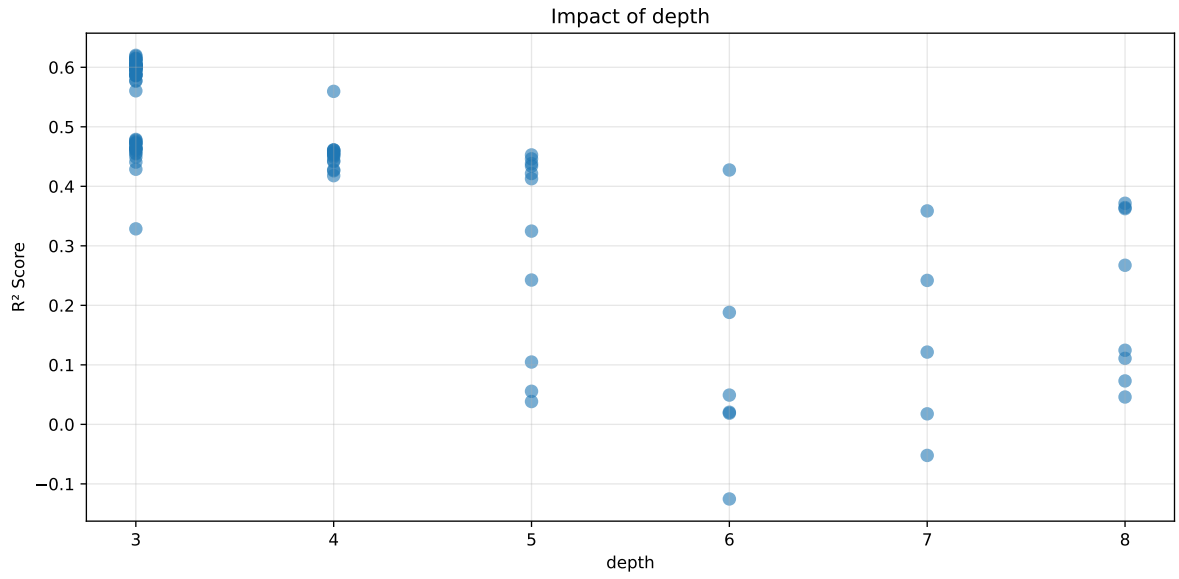
Distribution y_train vs y_test: y_train: mean=2.342, std=0.800, min=0.959, max=13.845
y_test: mean=2.364, std=0.795, min=1.068, max=7.277

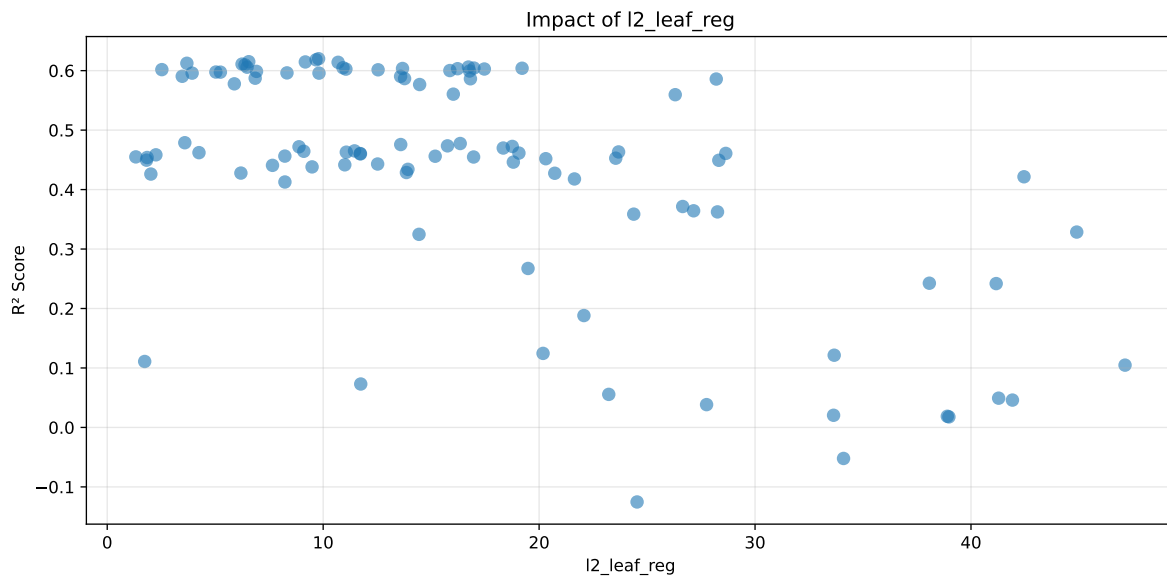
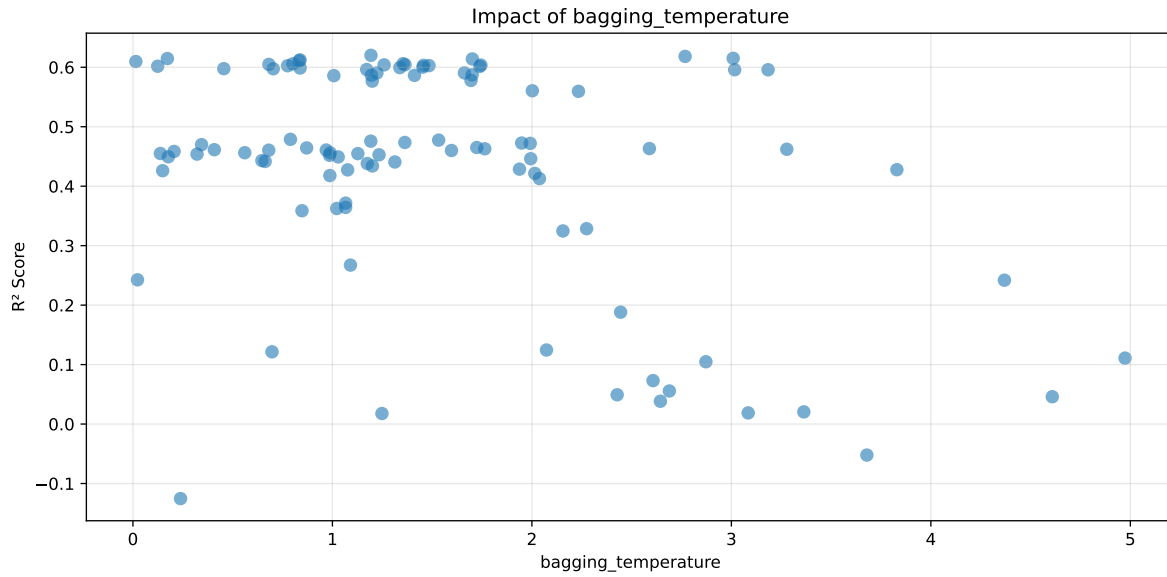
===== TOP Importance FEATURES =====

feature	importance	r_umidity	r_umidity_st_2h_slope	r_temperature_st_2h_std	r_temperature_st_12h_diff_last	r_umidity_st_2h_mean	r_temperature	r_temperature_st_12h_median	r_CO2	r_NH3_st_2h_std	r_temperature_st_2h_diff_last	r_umidity_st_12h_slope	r_NH3_st_12h_rms	r_THI	r_umidity_st_2h_std	r_umidity_st_2h_diff_last	r_NH3	r_NH3_st_2h_median	r_umidity_st_12h_std	r_umidity_st_12h_diff_last	r_temperature_st_12h_std	r_NH3_st_12h_slope	r_NH3_st_2h_slope	r_NH3_st_12h_std	r_temperature_st_12h_min	r_NH3_st_2h_diff_last	r_NH3_st_12h_diff_last																										
feature	importance	r_umidity	20.814896	r_umidity_st_2h_slope	11.671790	r_temperature_st_2h_std	10.838420	r_temperature_st_12h_diff_last	6.837207	r_umidity_st_2h_mean	6.717498	r_temperature	5.599349	r_temperature_st_12h_median	4.625657	r_CO2	4.231566	r_NH3_st_2h_std	4.094919	r_temperature_st_2h_diff_last	3.341445	r_umidity_st_12h_slope	2.648576	r_NH3_st_12h_rms	2.550821	r_THI	2.287199	r_umidity_st_2h_std	2.044969	r_umidity_st_2h_diff_last	1.793818	r_NH3	1.605362	r_NH3_st_2h_median	1.321844	r_umidity_st_12h_std	1.155776	r_umidity_st_12h_diff_last	1.049607	r_temperature_st_12h_std	0.962769	r_NH3_st_12h_slope	0.930135	r_NH3_st_2h_slope	0.780769	r_NH3_st_12h_std	0.664564	r_temperature_st_12h_min	0.638979	r_NH3_st_2h_diff_last	0.465273	r_NH3_st_12h_diff_last	0.326790



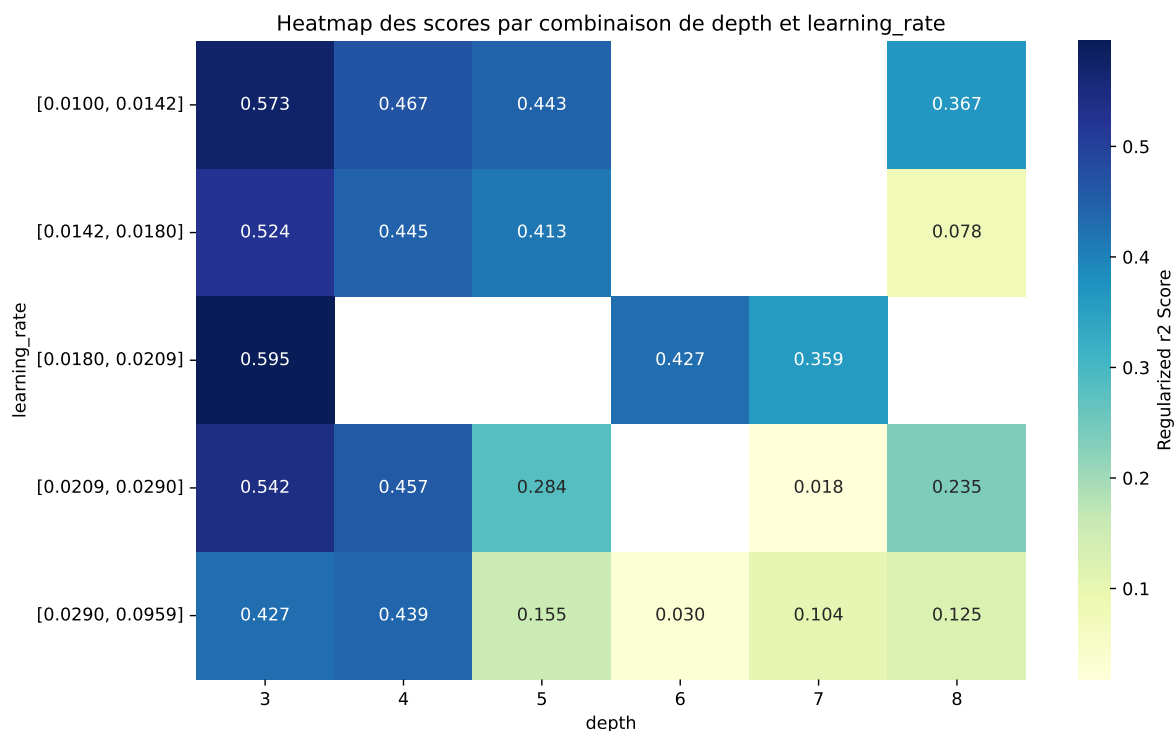
Paramètres avec >5% d'importance : ['depth', 'learning_rate', 'bagging_temperature', 'l2_leaf_reg']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6496	0.6201	0.6651	0.0272	0.6201	0.6704	0.0503	1.0812	1.0477
2	0.648	0.6183	0.6654	0.0257	0.6183	0.6684	0.0501	1.0811	1.0481
3	0.6448	0.615	0.6617	0.0253	0.615	0.6677	0.0527	1.0857	1.0484
4	0.645	0.6146	0.6623	0.0258	0.6146	0.6704	0.0558	1.0907	1.0495
5	0.6423	0.614	0.6603	0.026	0.614	0.6658	0.0518	1.0843	1.046
6	0.6414	0.6124	0.659	0.0246	0.6124	0.6649	0.0526	1.0858	1.0473
7	0.6407	0.611	0.6581	0.0245	0.611	0.6619	0.0508	1.0832	1.0486
8	0.6387	0.6096	0.6564	0.0259	0.6096	0.6653	0.0557	1.0913	1.0476
9	0.6339	0.6061	0.6518	0.0258	0.6061	0.6609	0.0548	1.0905	1.0459
10	0.6352	0.6058	0.6536	0.0245	0.6058	0.6591	0.0532	1.0879	1.0486
---	---	---	---	---	---	---	---	---	---
100	0.9677	0.7855	0.8276	0.034	-0.1253	0.8434	0.0579	1.0737	1.2319

Hyperparameters:

Rank 1: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.02083425965777953, 'reg__l2_leaf_reg': 9.788140895754347, 'reg__bagging_temperature': 1.1933066138122426, 'reg__random_strength': 4.683439161985013, 'reg__subsample': 0.6970841696019842, 'reg__min_data_in_leaf': 33, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.58119085572373, 'reg__border_count': 114, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 3597.0446594675677}

Rank 2: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.020494025974980984, 'reg__l2_leaf_reg': 9.671830531357422, 'reg__bagging_temperature': 2.7680833702456447, 'reg__random_strength': 4.294144071563847, 'reg__subsample': 0.694340515750745, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.5854739839696177, 'reg__border_count': 119, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 2147.3535052090165}

Rank 3: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.021533155932177278, 'reg__l2_leaf_reg': 6.55380554886607, 'reg__bagging_temperature': 3.009074546143161, 'reg__random_strength': 2.433470209933736, 'reg__subsample': 0.7200582194166812, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.6207819953742169, 'reg__border_count': 119, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 986.5847632929303}

Rank 4: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.011951996895721528, 'reg__l2_leaf_reg': 9.173836967749228, 'reg__bagging_temperature': 0.17279394164097583, 'reg__random_strength': 1.9815794496153845, 'reg__subsample': 0.7596309339312365, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.6178654324715376, 'reg__border_count': 126, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3515.4524637468185}

Rank 5: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.01706999563983117, 'reg__l2_leaf_reg': 10.692928782596335, 'reg__bagging_temperature': 1.7016019804641622, 'reg__random_strength': 3.26493992957415, 'reg__subsample': 0.8036908893905002, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.6011560085276558, 'reg__border_count': 127, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 2708.4318189916594}

Rank 6: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.02092170336759102, 'reg__l2_leaf_reg': 3.6886384982435914, 'reg__bagging_temperature': 0.8378625706234173, 'reg__random_strength': 2.581227064017572, 'reg__subsample': 0.7134224110017963, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.616289522485633, 'reg__border_count': 114, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4308.377473140525}

Rank 7: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.01924546331965651, 'reg__l2_leaf_reg': 6.243036760597769, 'reg__bagging_temperature': 0.833346050153143, 'reg__random_strength': 2.0888852452690814, 'reg__subsample': 0.7998623613633185,

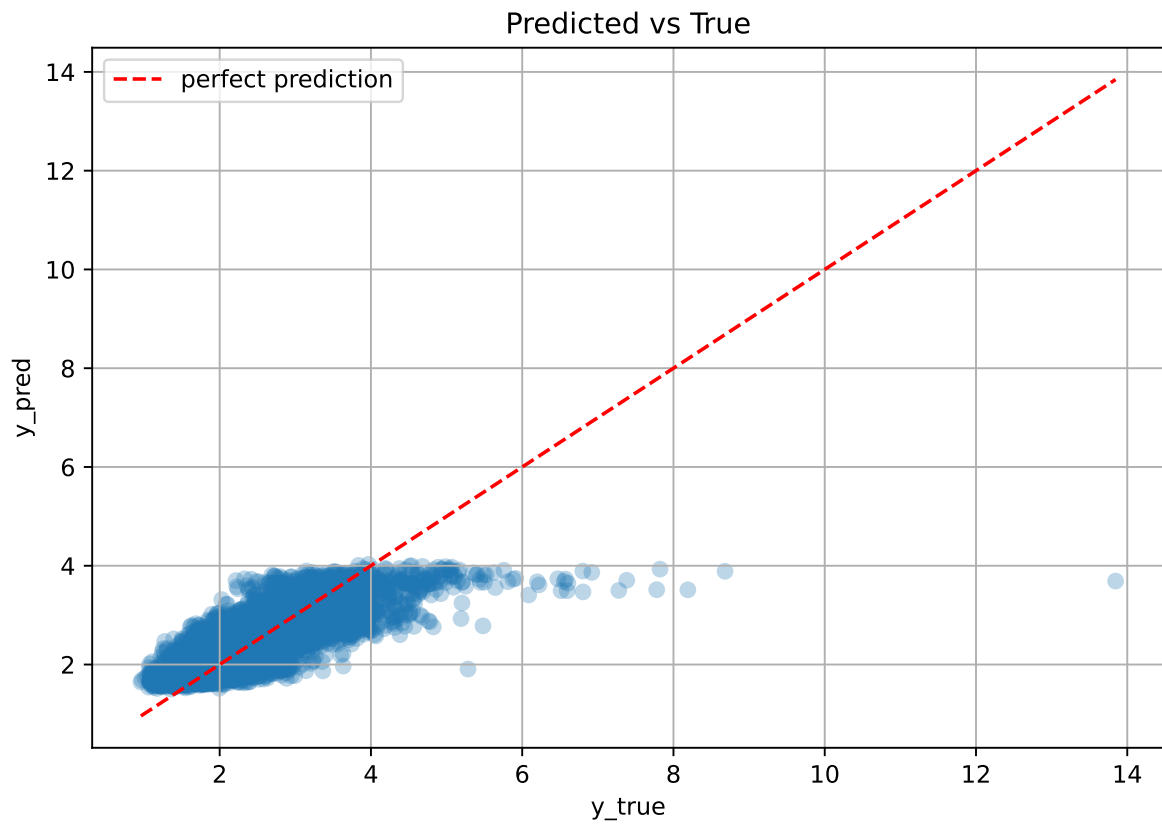
'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.5324942096425564, 'reg__border_count': 118, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 2920.0180895450994}

Rank 8: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.020918058106509946, 'reg__l2_leaf_reg': 6.380143474175099, 'reg__bagging_temperature': 0.014579921725477174, 'reg__random_strength': 2.652721896711021, 'reg__subsample': 0.6552521677111327, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.6201908062798912, 'reg__border_count': 114, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 865.5172565601815}

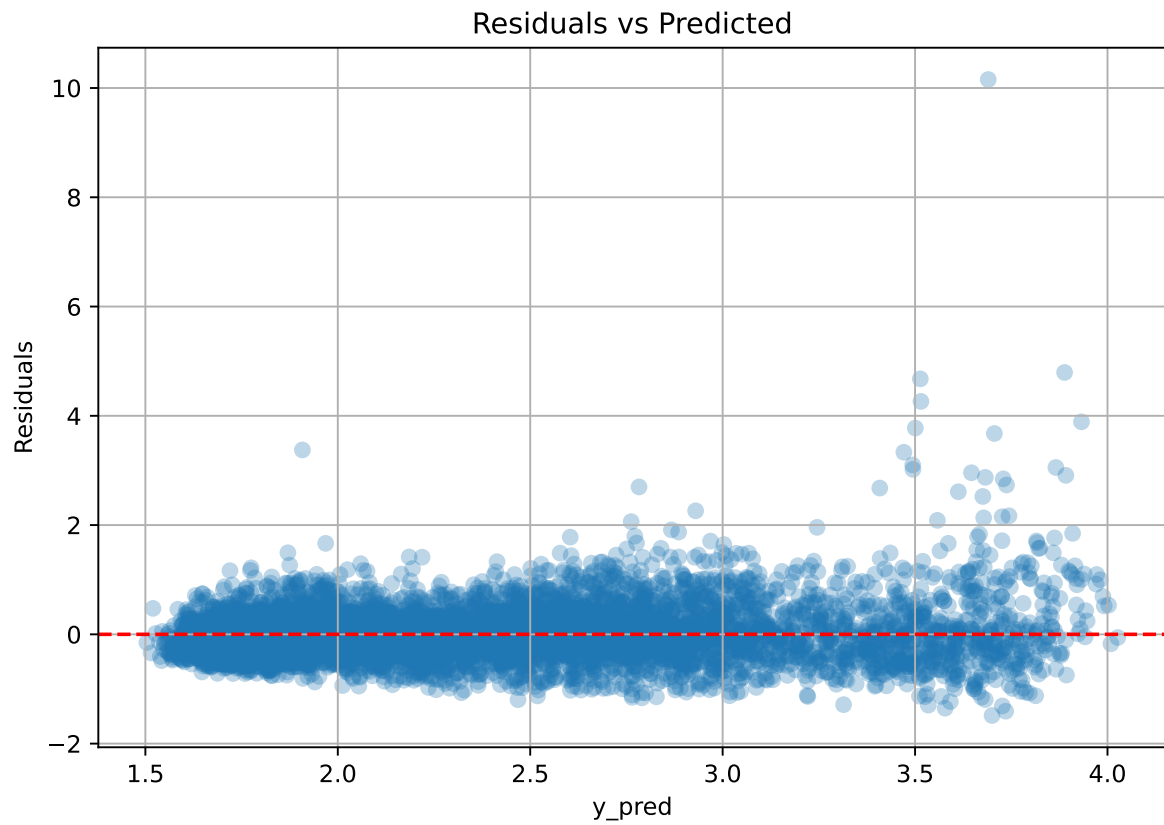
Rank 9: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.021540726182561826, 'reg__l2_leaf_reg': 16.727425555528793, 'reg__bagging_temperature': 1.354190744214579, 'reg__random_strength': 3.198592150318758, 'reg__subsample': 0.6813200065157542, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.530942577799667, 'reg__border_count': 108, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1574.027933483384}

Rank 10: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.018361062689947766, 'reg__l2_leaf_reg': 6.4765302676836765, 'reg__bagging_temperature': 0.8016924534996807, 'reg__random_strength': 2.569179288056805, 'reg__subsample': 0.7560538427764457, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.6487200948964897, 'reg__border_count': 119, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 2532.8471976251135}

3.6.2 Scatter



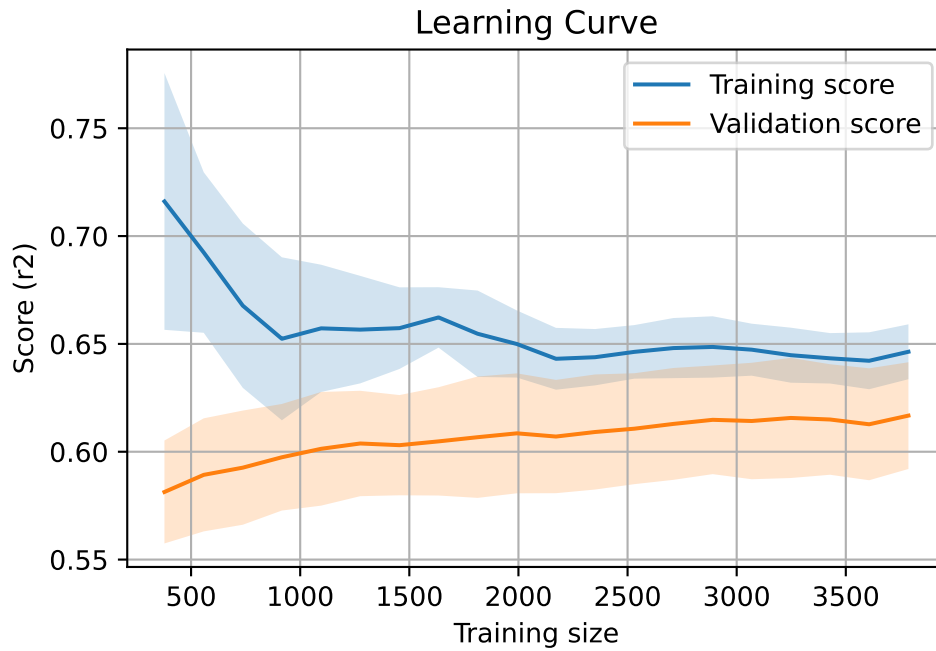
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7730, `Len(training)` : 5411,

`Len(training_real)` : 3788, `Len(test_of_training)` : 1623, `Len(test)` : 2319



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.573	0.549	0.024	0.589	0.59	0.042	1.076	1.044	nan
LGBM	0.625	0.596	0.026	0.639	0.654	0.058	1.098	1.05	0.5955
CatBoost	0.65	0.62	0.027	0.665	0.67	0.05	1.081	1.048	0.6201

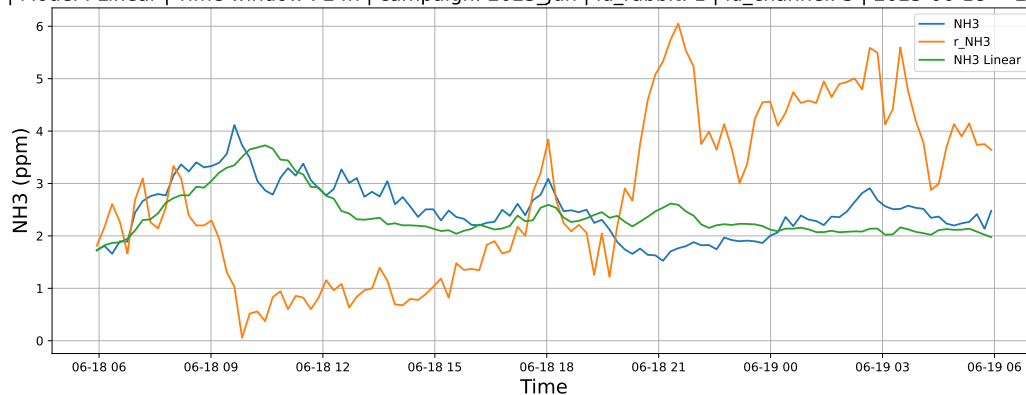
4 Plot

4.1 Standalone plots

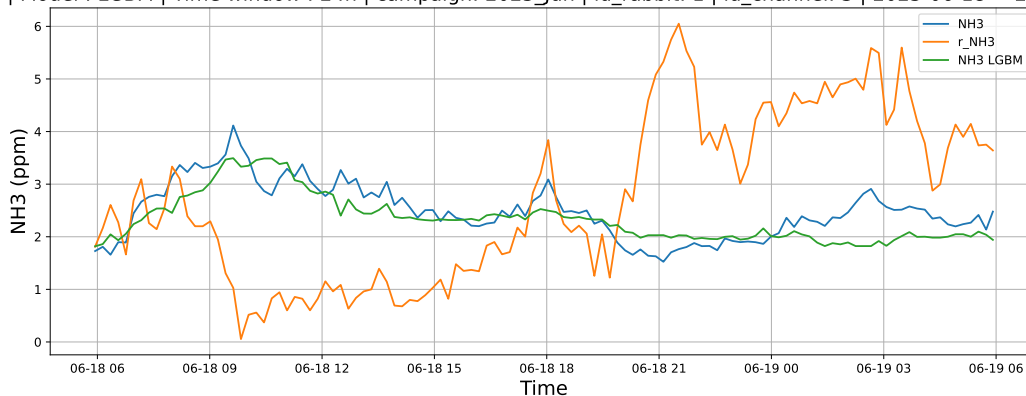
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

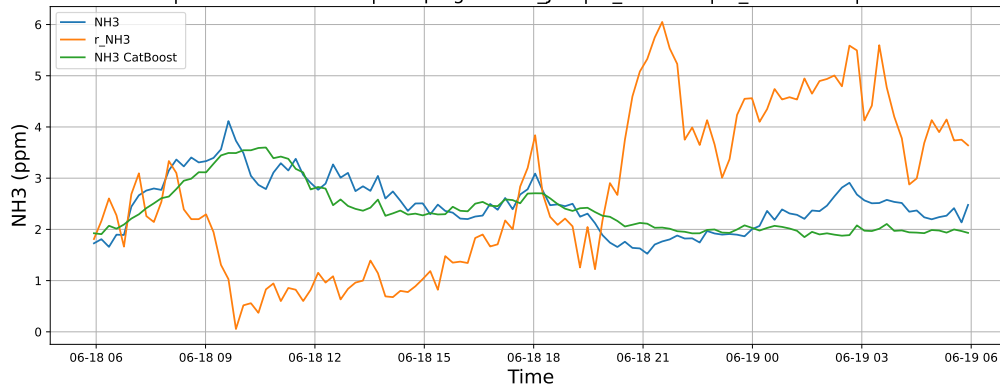
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

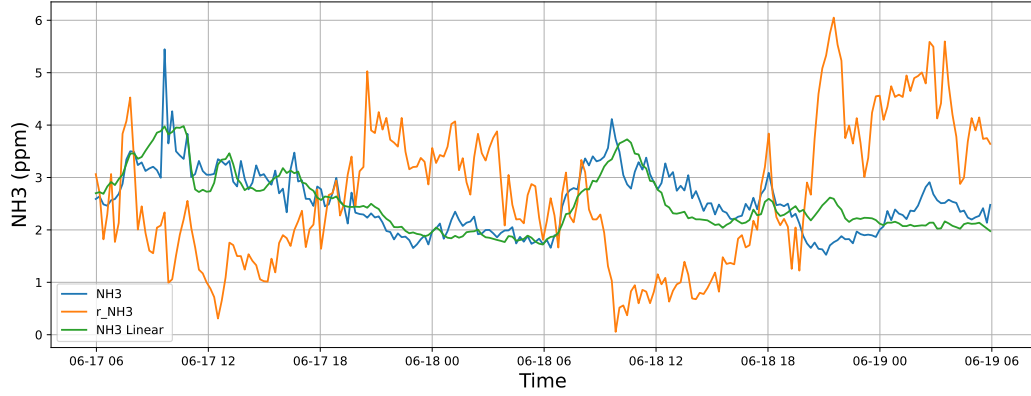


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

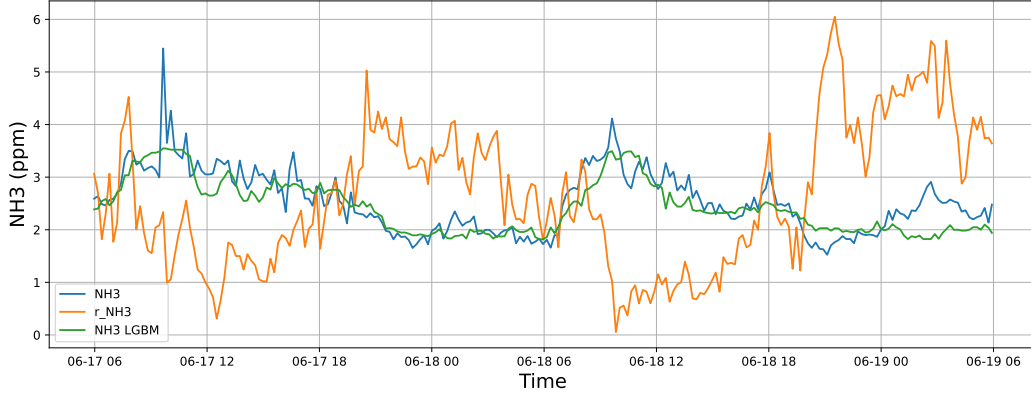


4.1.1.2 Time window : 48

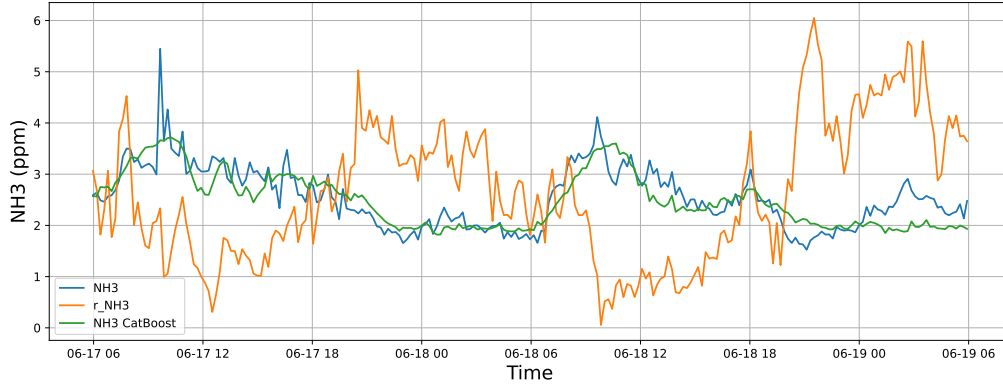
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

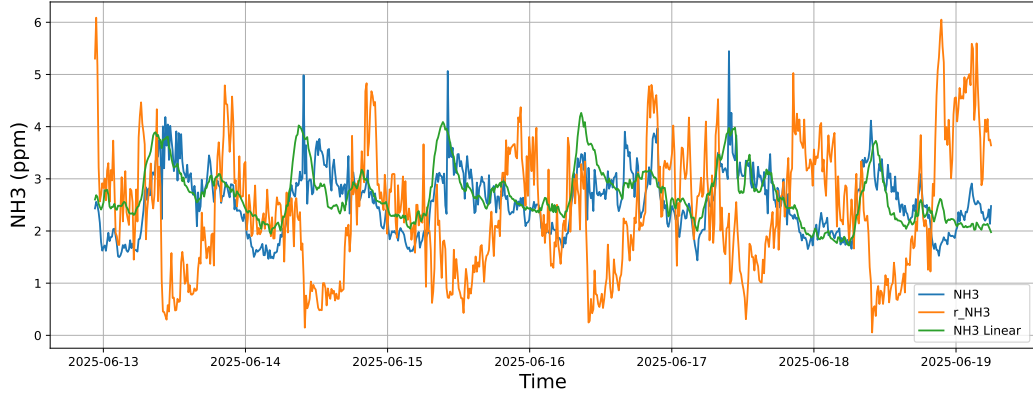


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

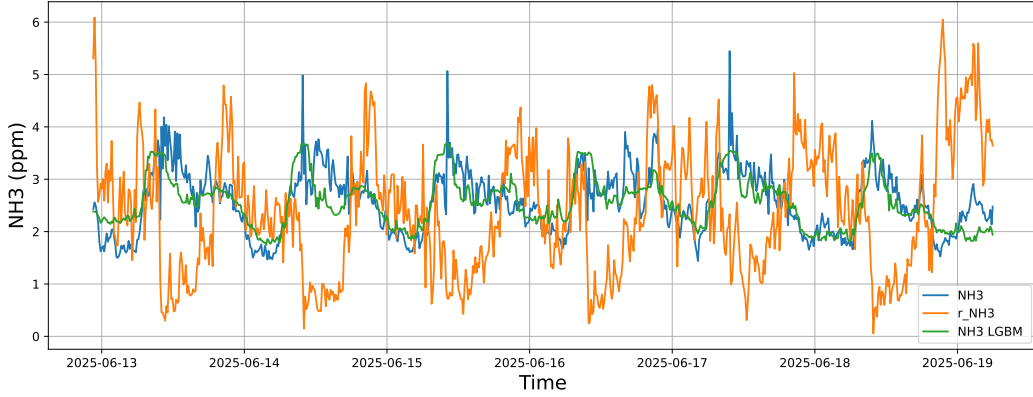


4.1.1.3 Time window : ALL

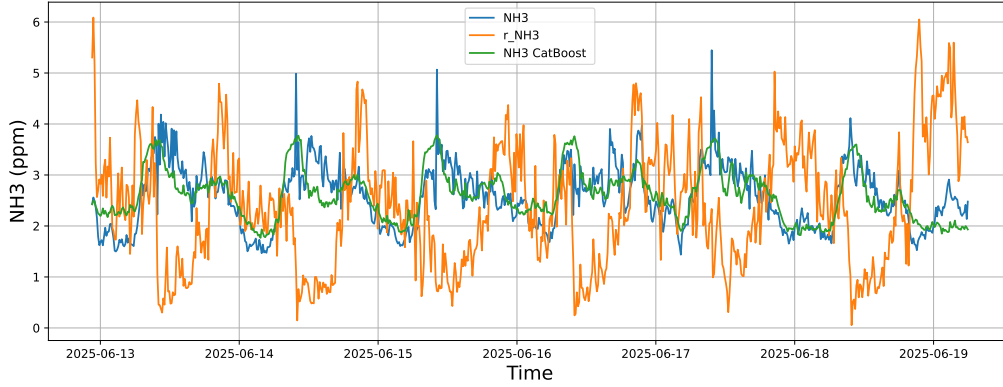
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



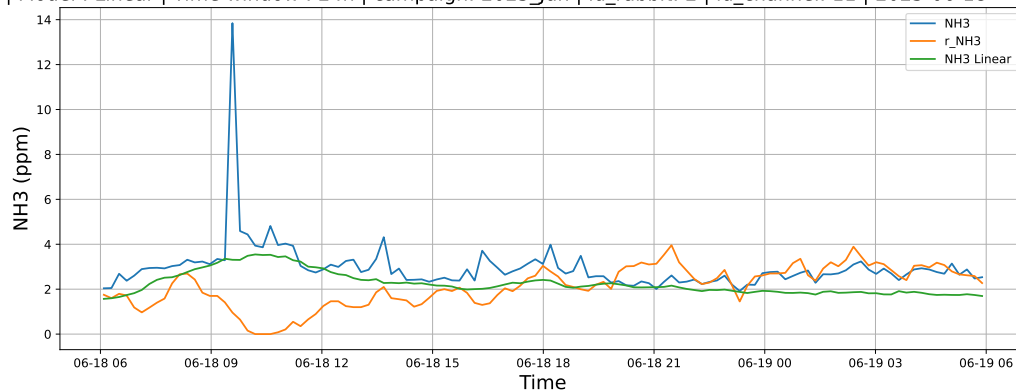
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



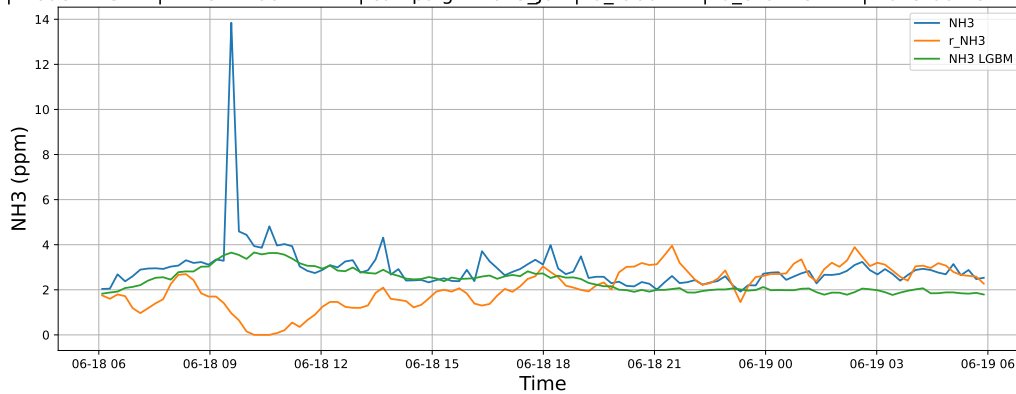
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

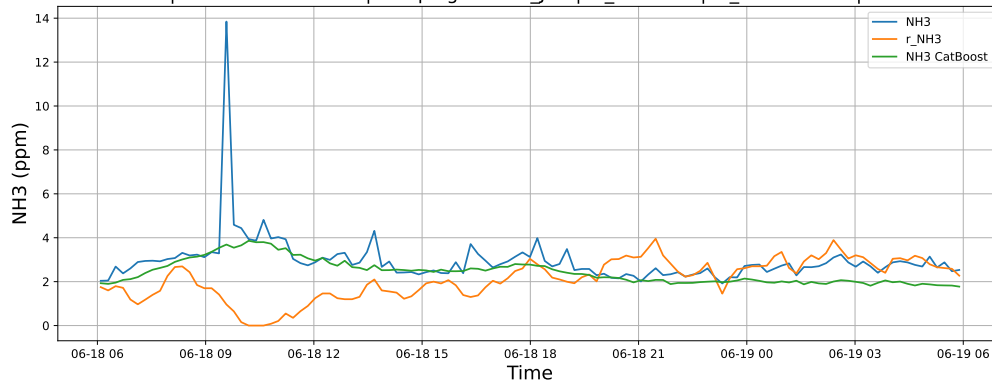
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

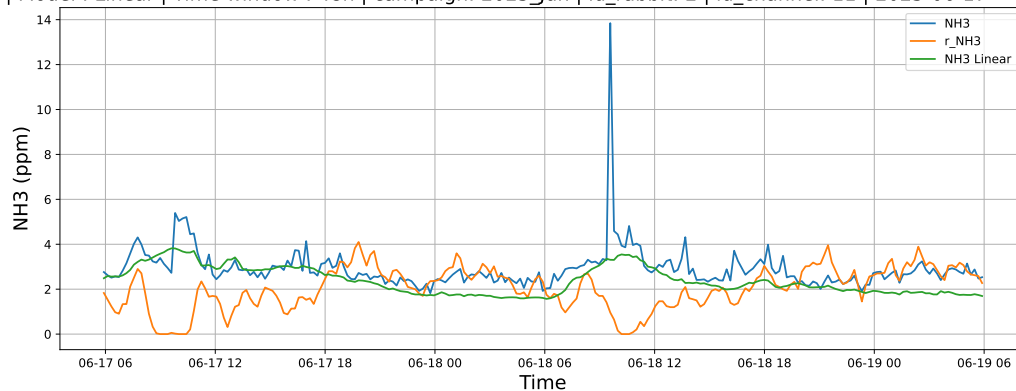


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

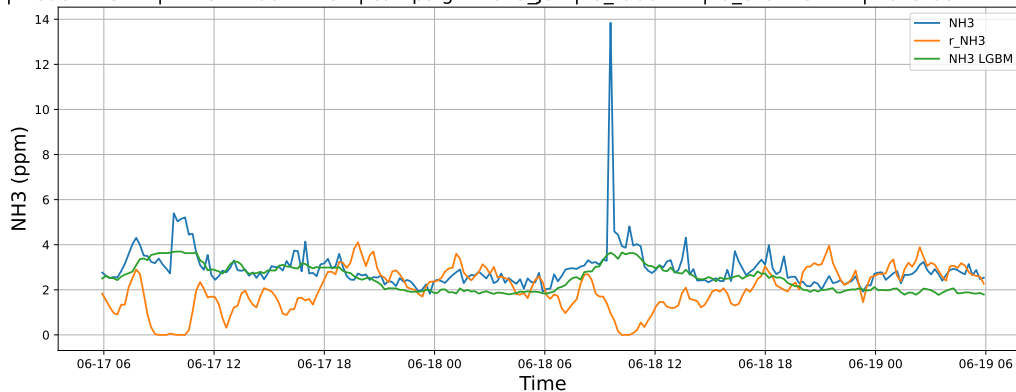


4.1.2.2 Time window : 48

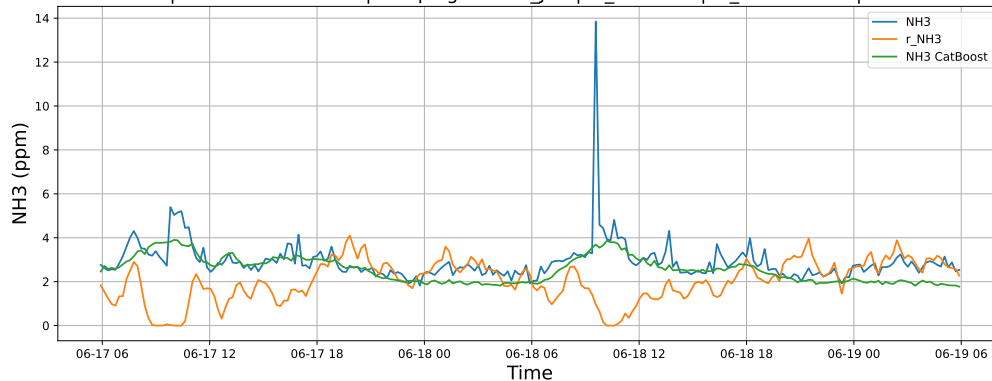
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

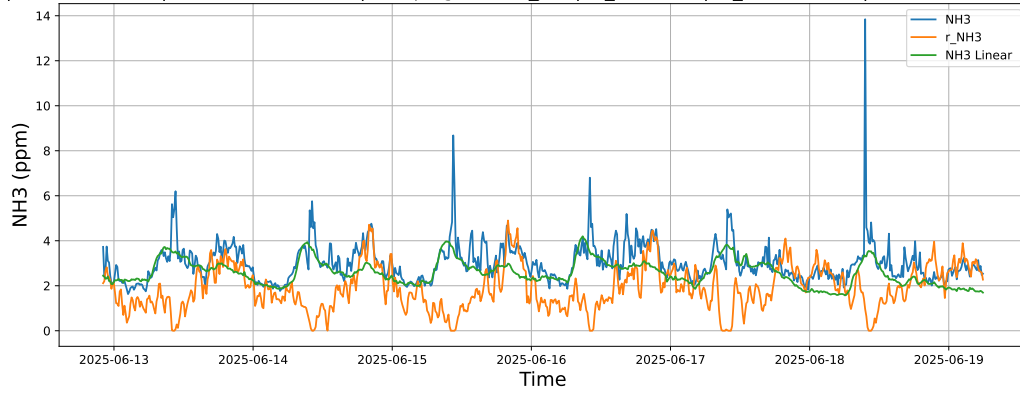


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

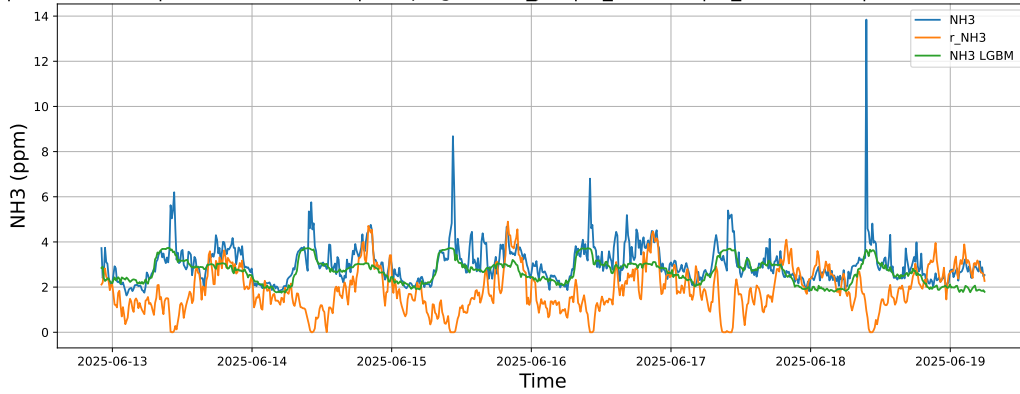


4.1.2.3 Time window : ALL

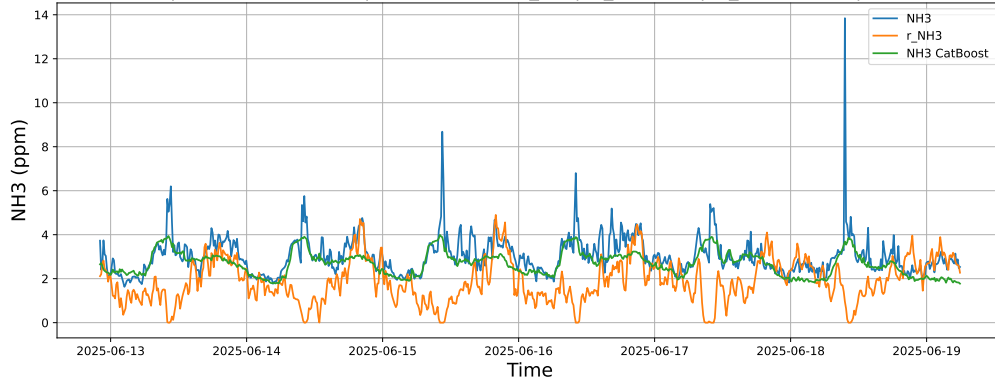
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



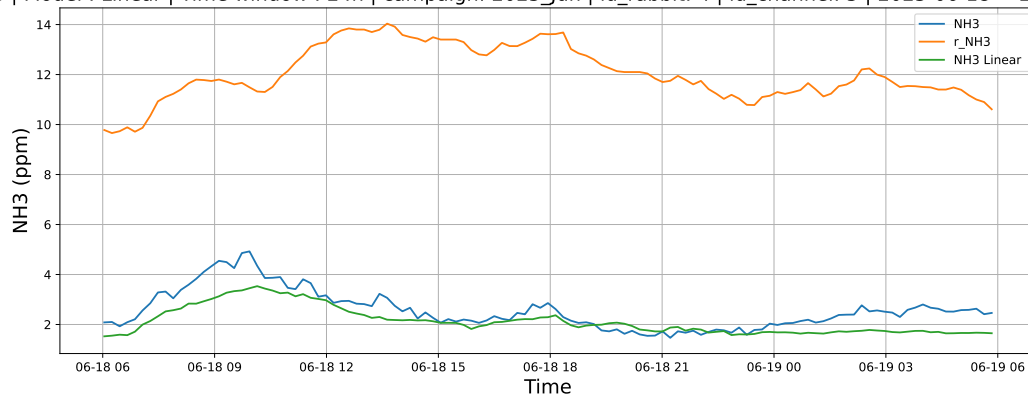
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



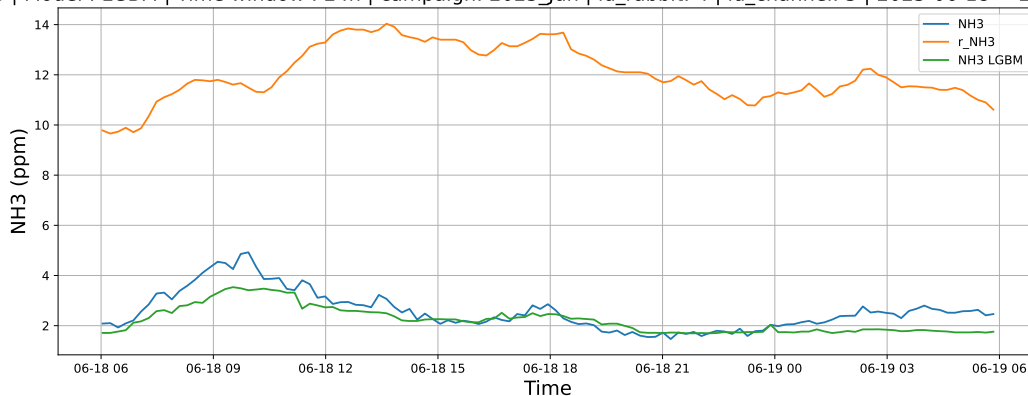
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

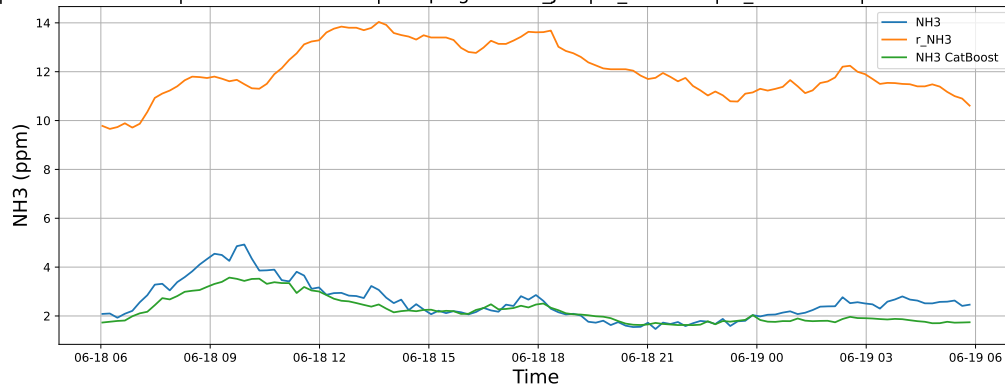
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

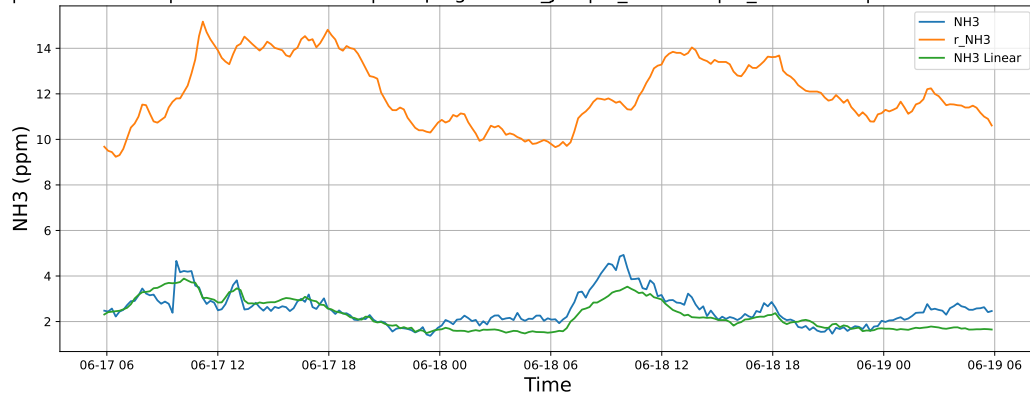


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

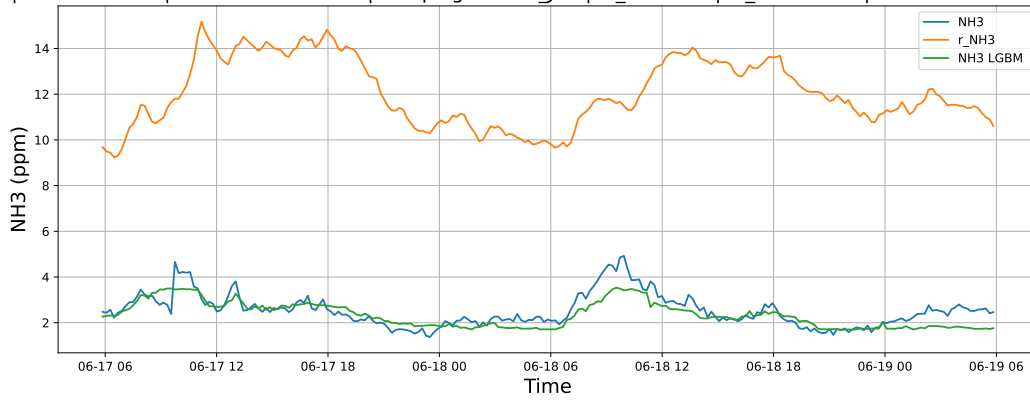


4.1.3.2 Time window : 48

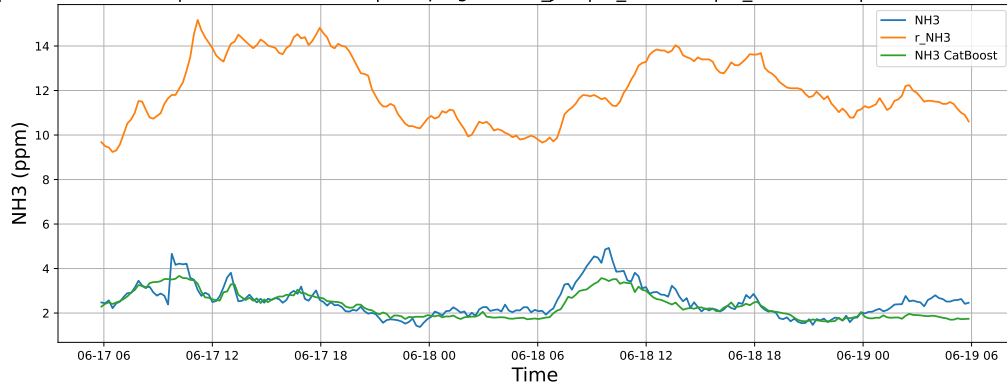
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

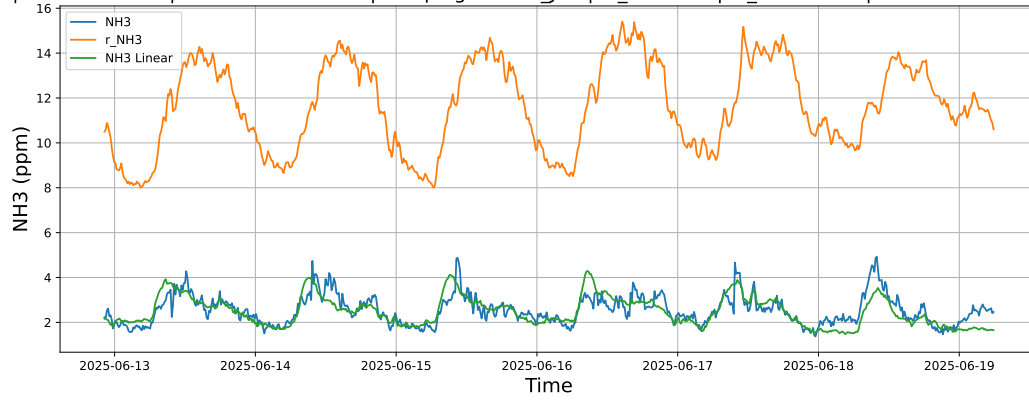


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

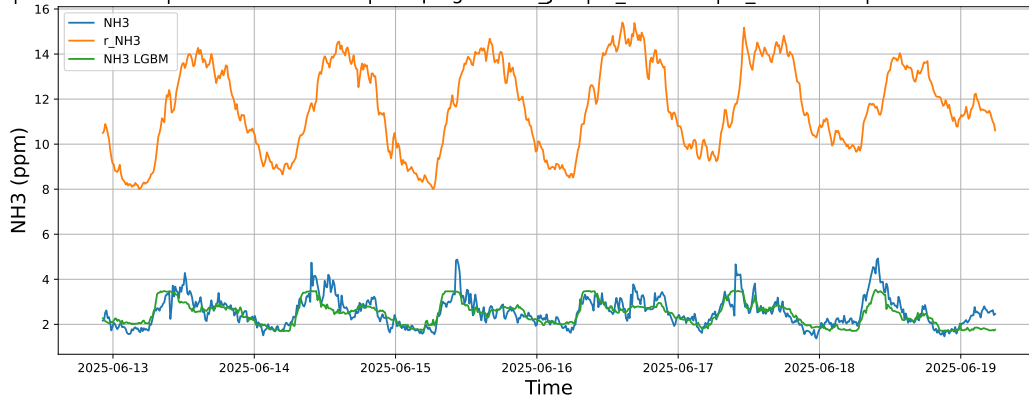


4.1.3.3 Time window : ALL

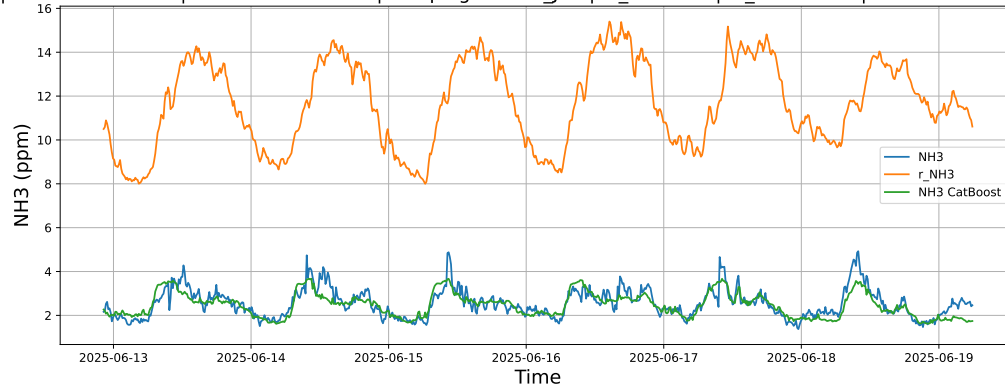
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



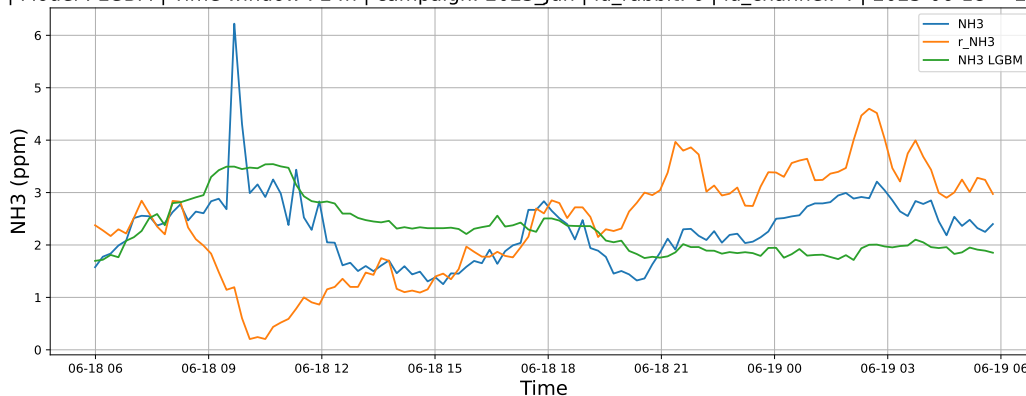
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

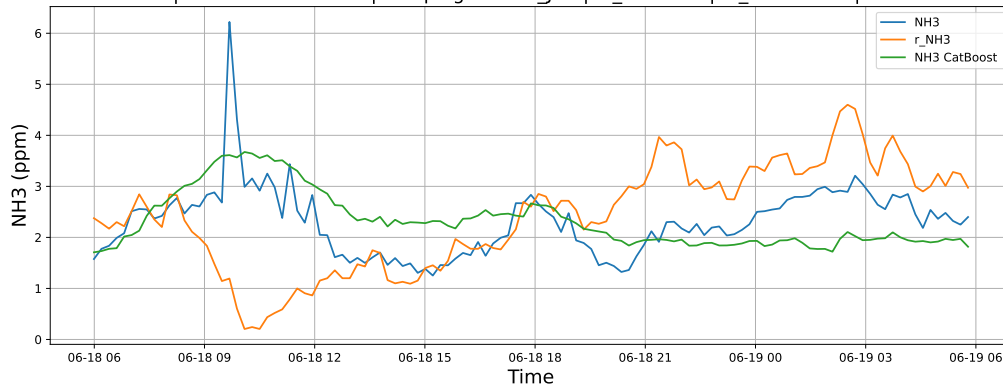
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

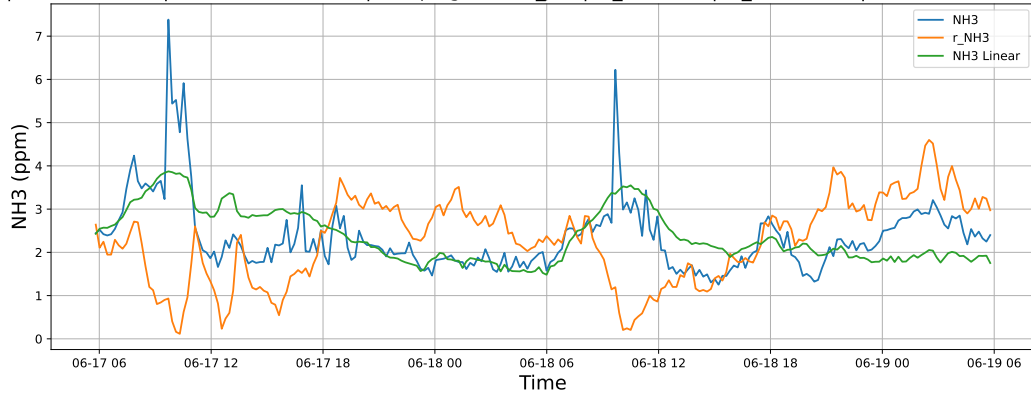


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

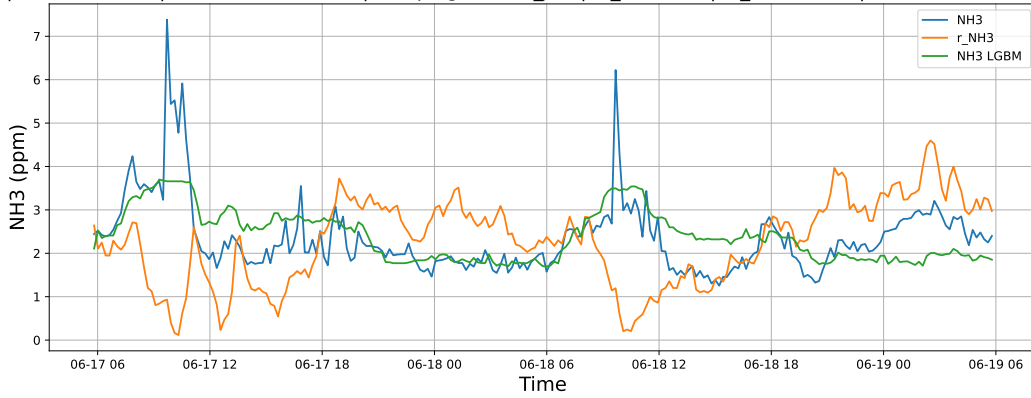


4.1.4.2 Time window : 48

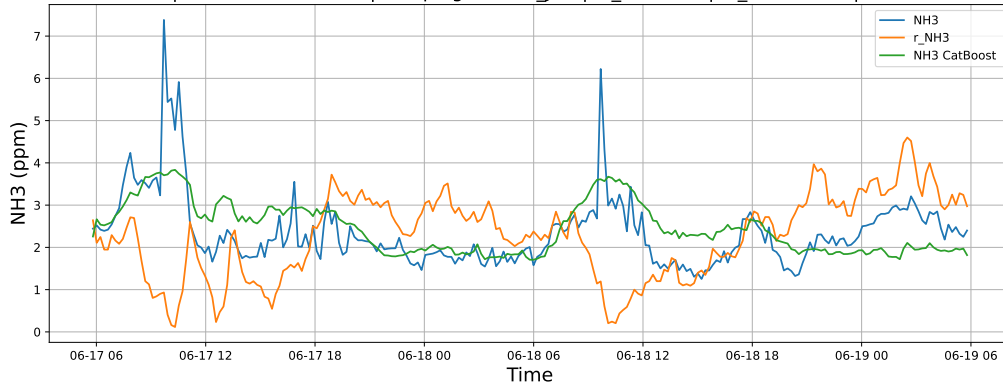
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

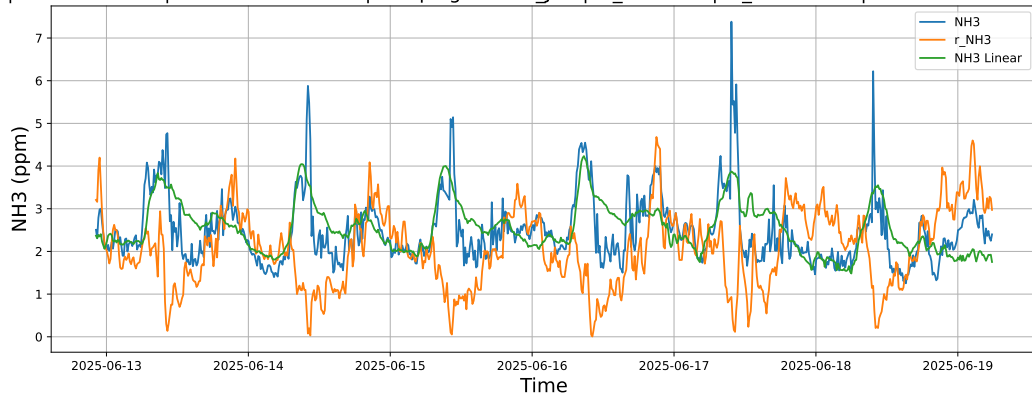


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

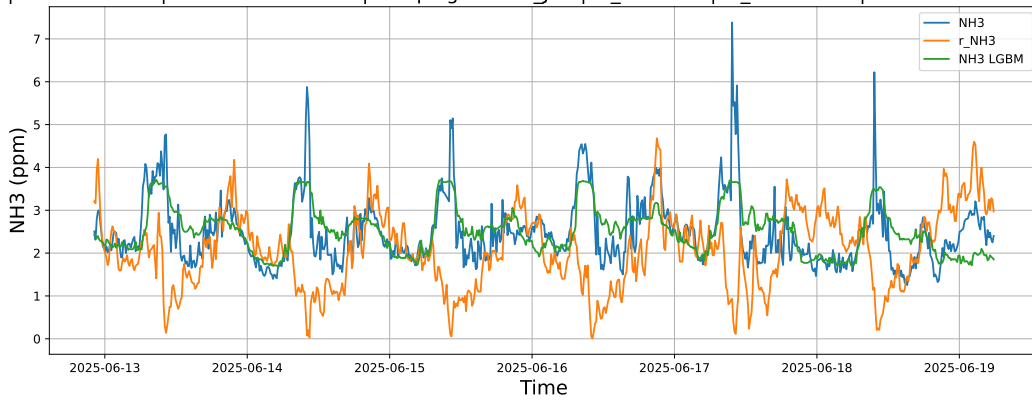


4.1.4.3 Time window : ALL

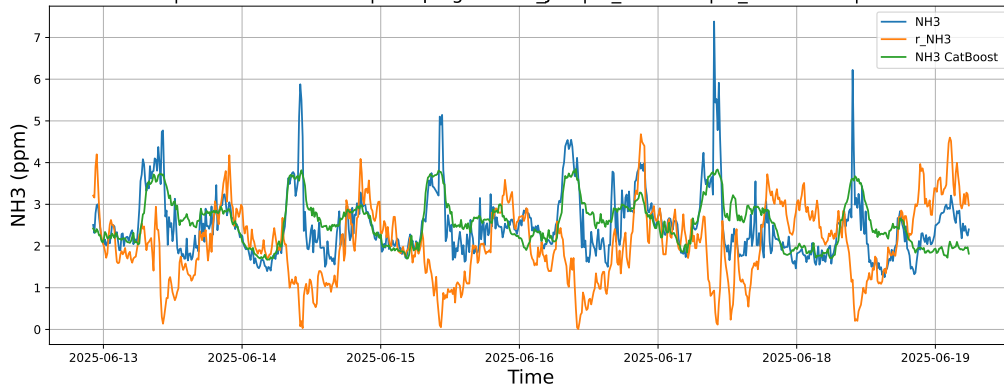
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



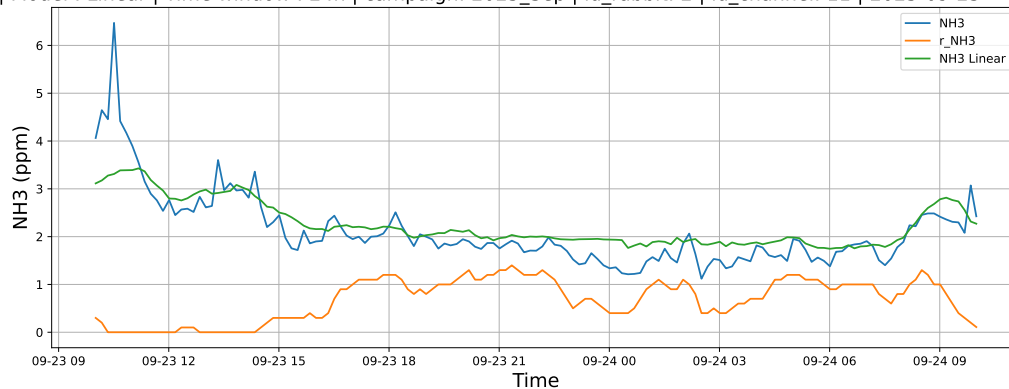
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



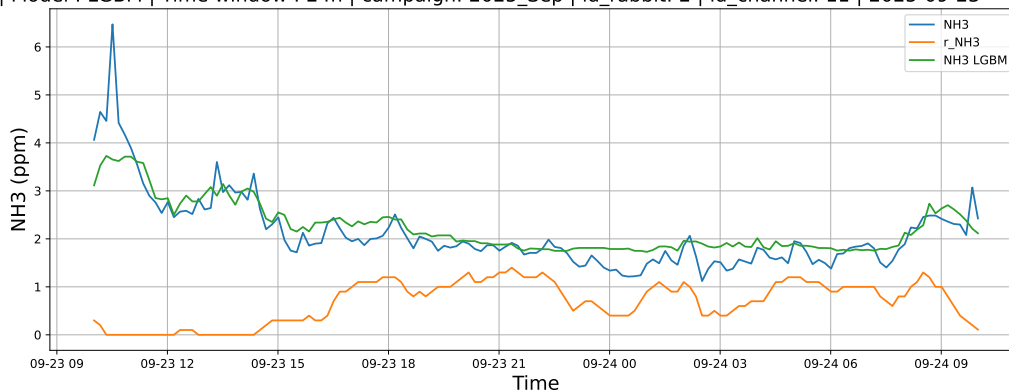
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

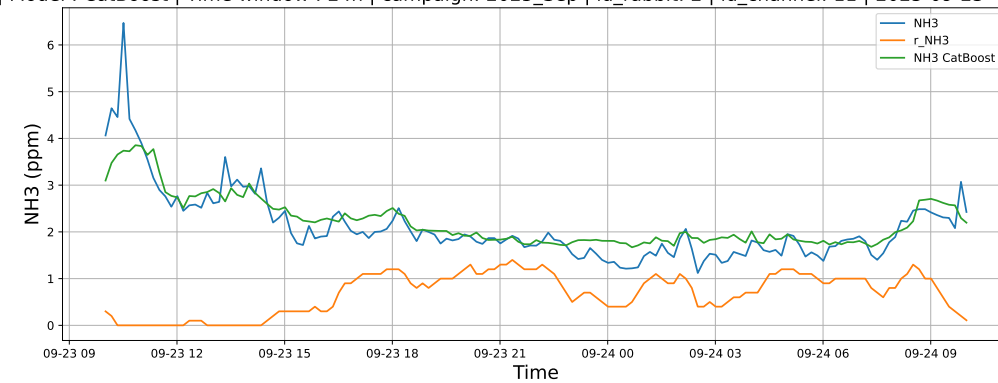
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

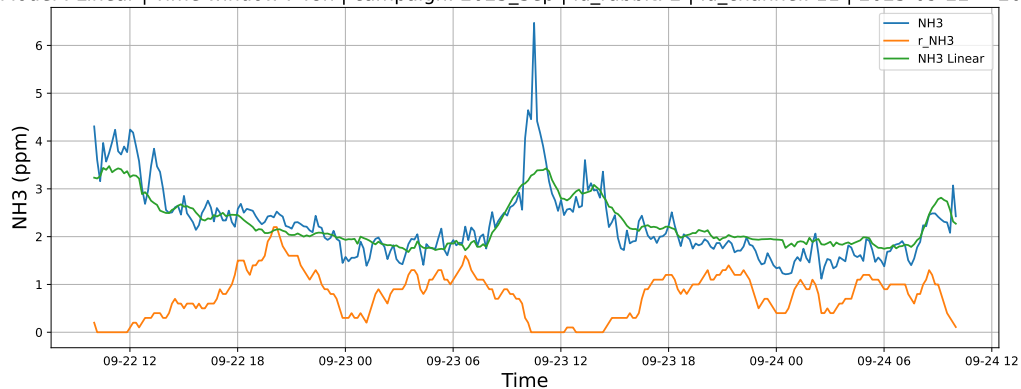


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

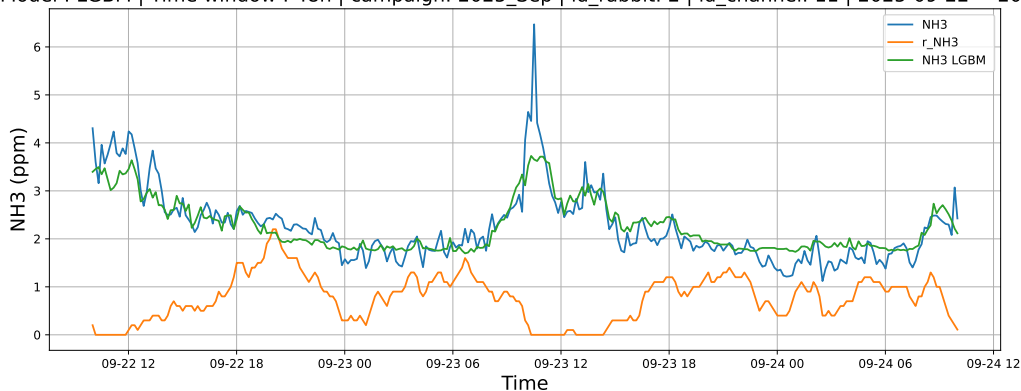


4.1.5.2 Time window : 48

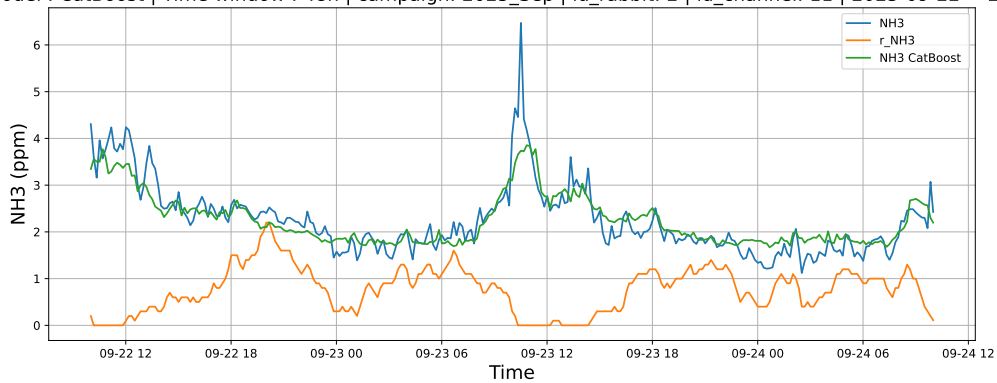
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

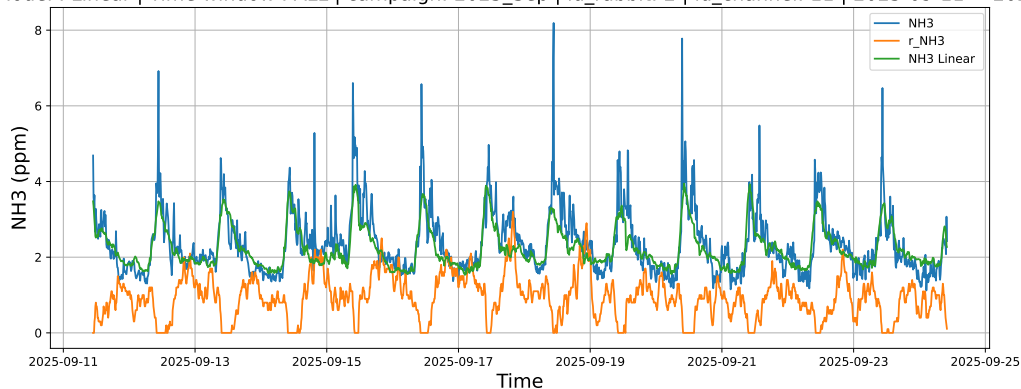


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

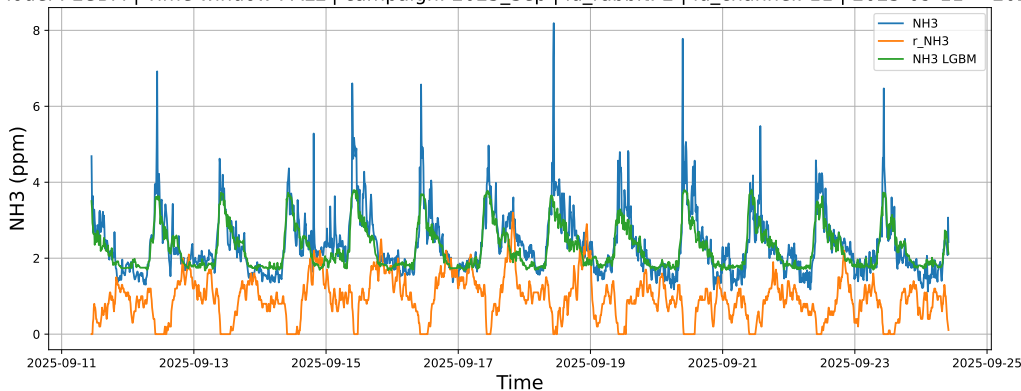


4.1.5.3 Time window : ALL

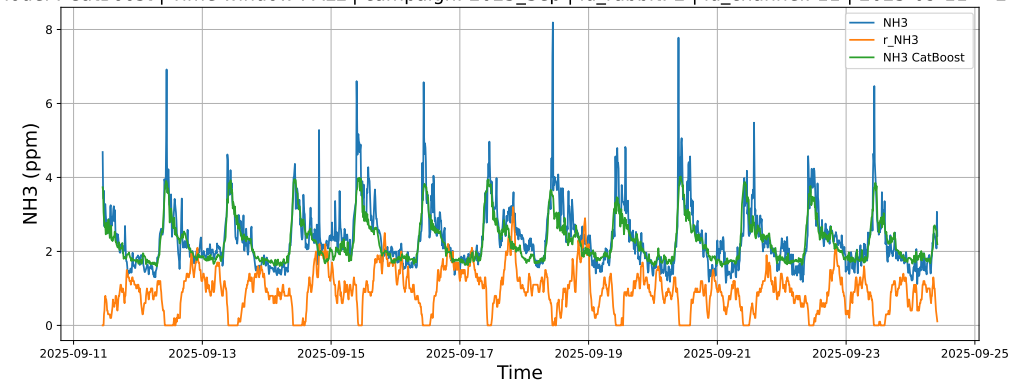
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



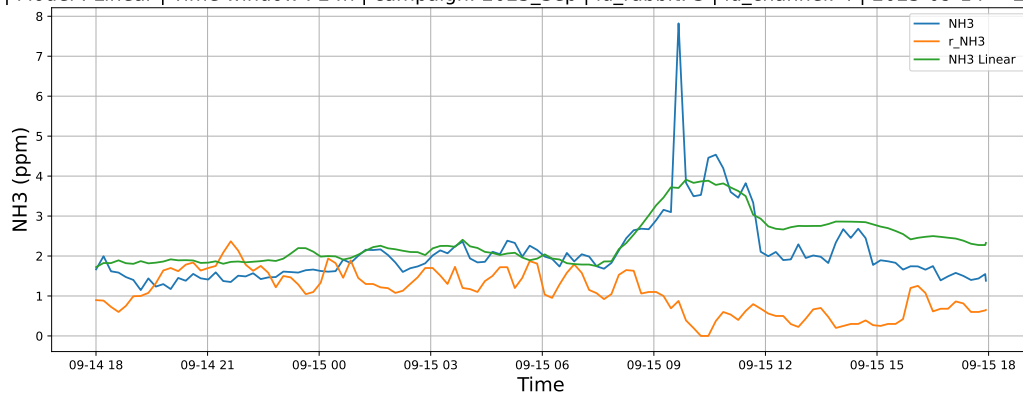
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



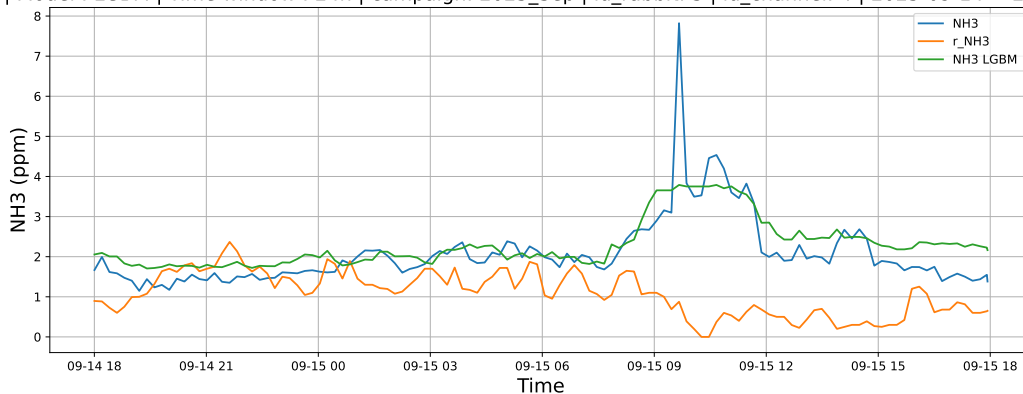
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

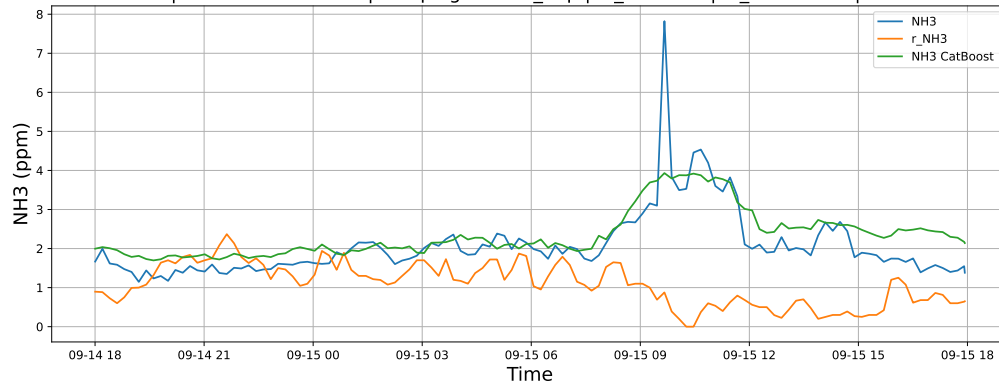
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

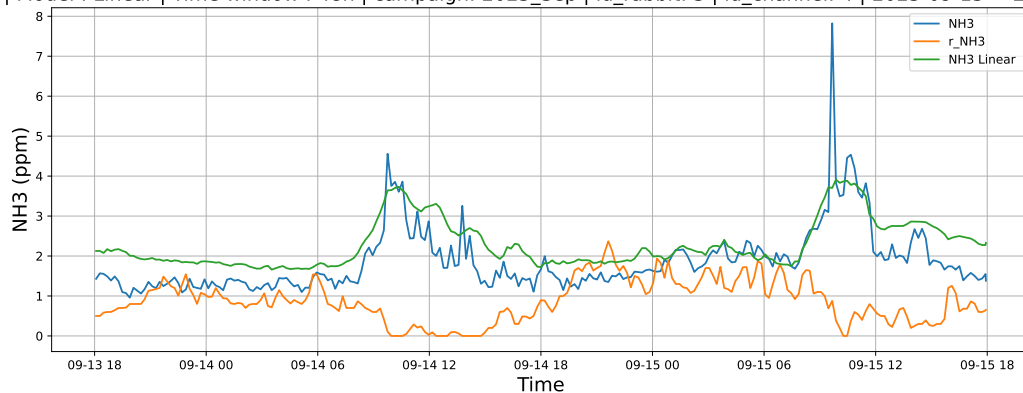


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

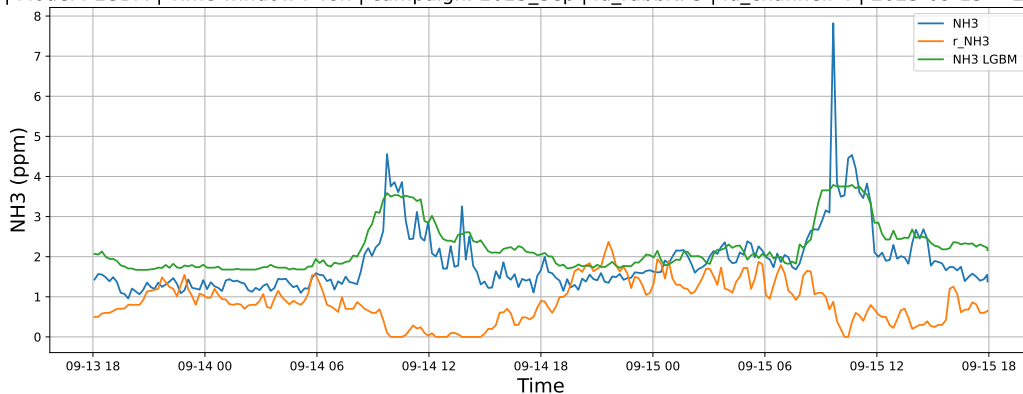


4.1.6.2 Time window : 48

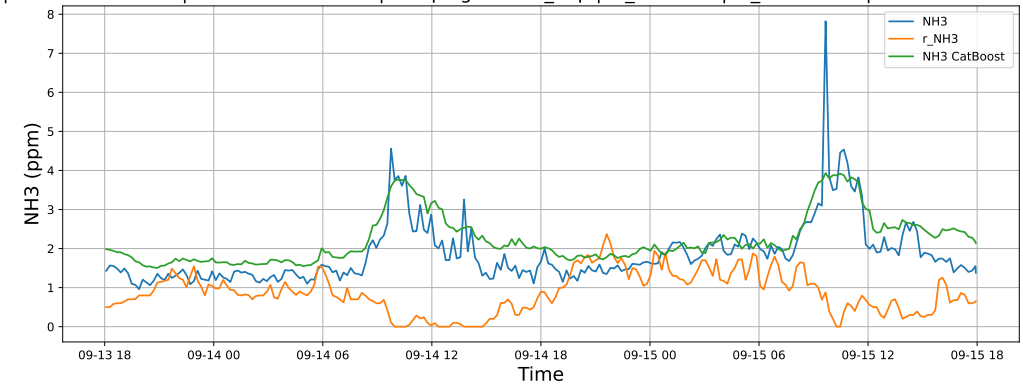
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

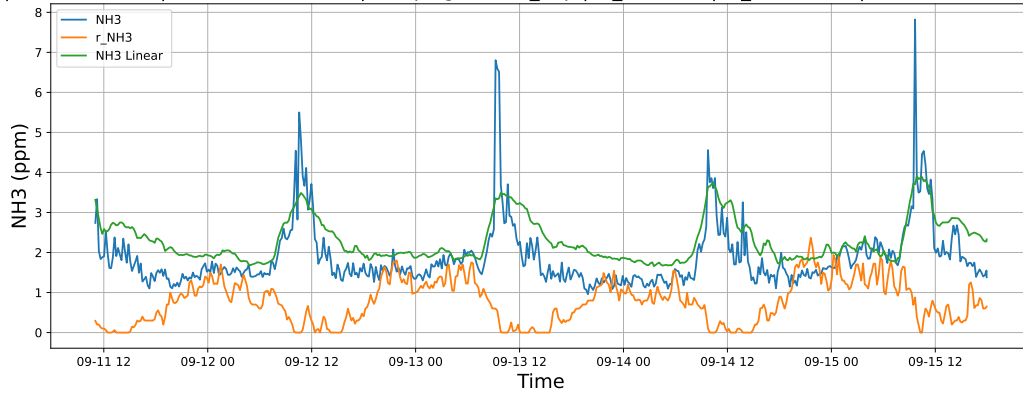


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

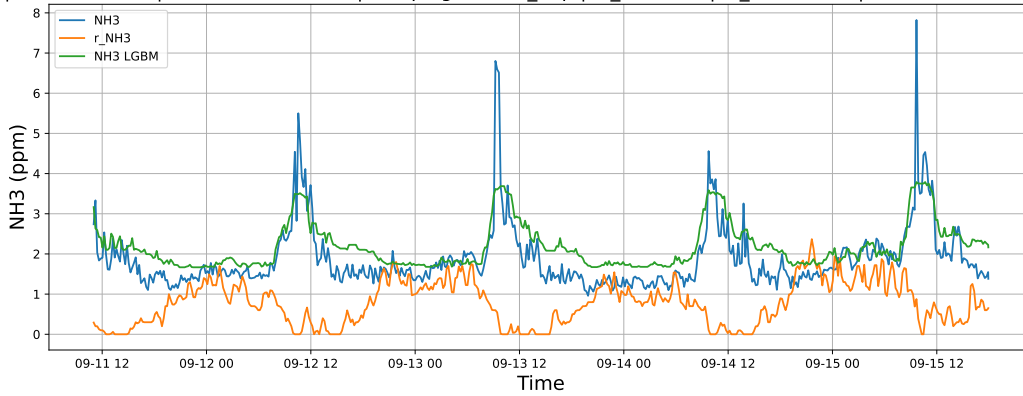


4.1.6.3 Time window : ALL

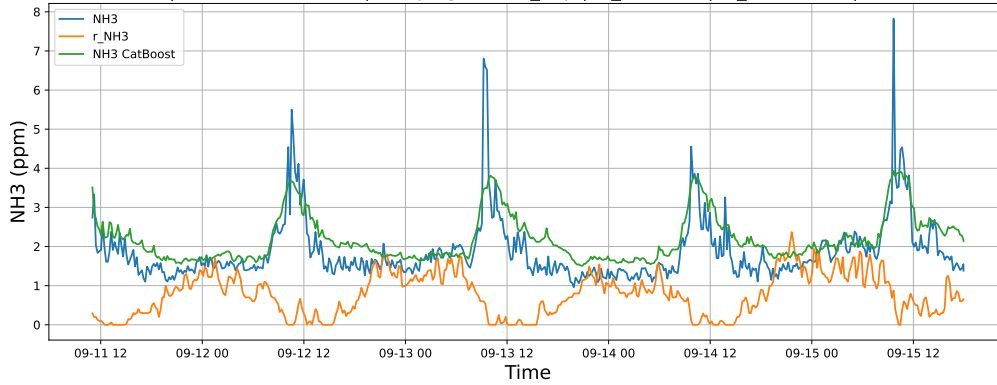
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



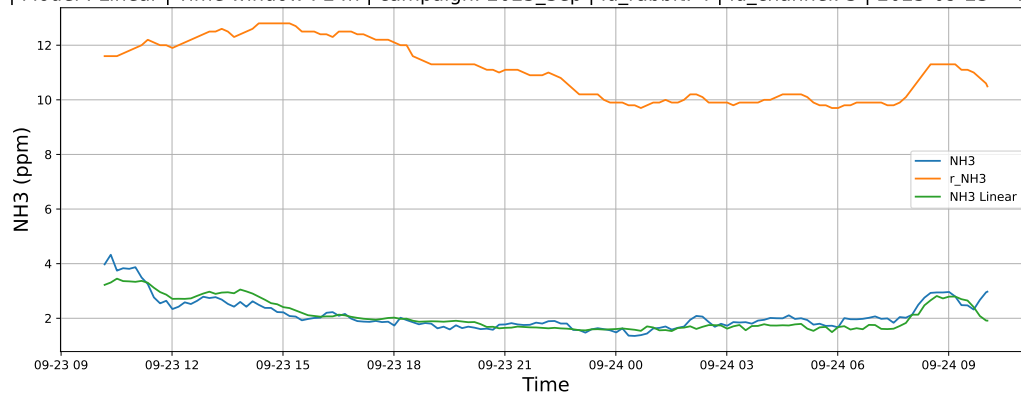
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



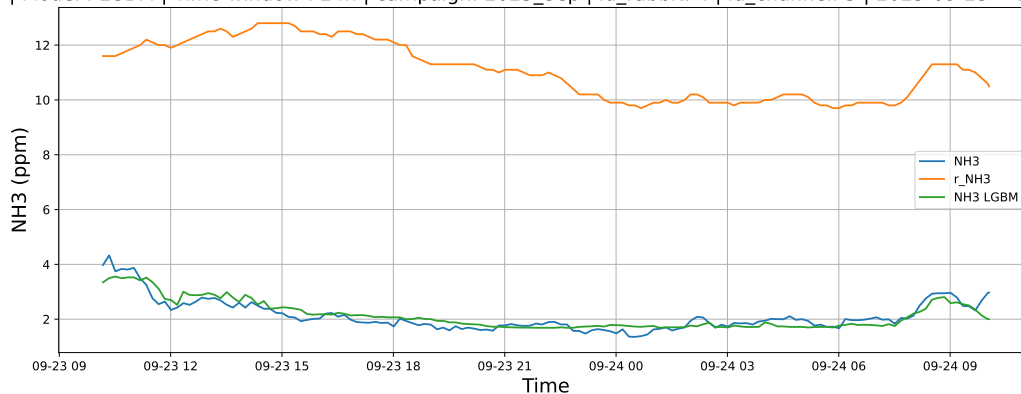
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

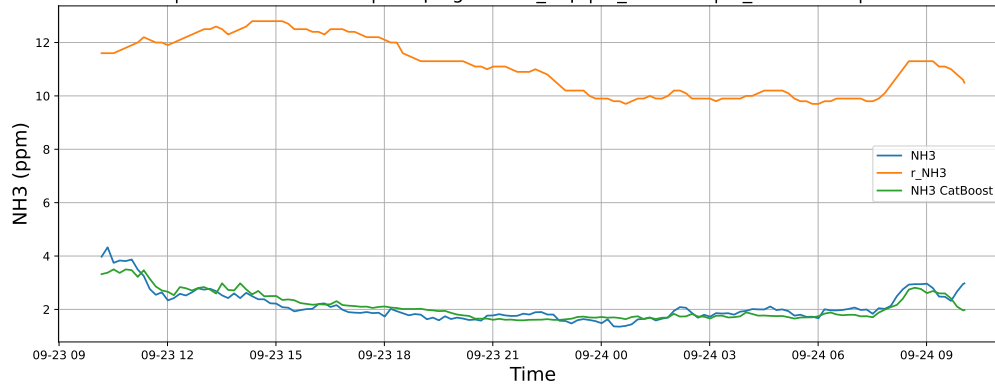
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

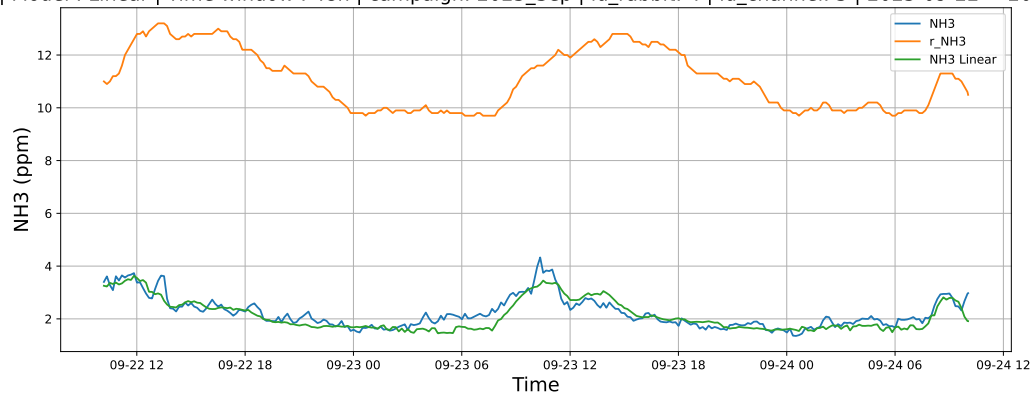


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

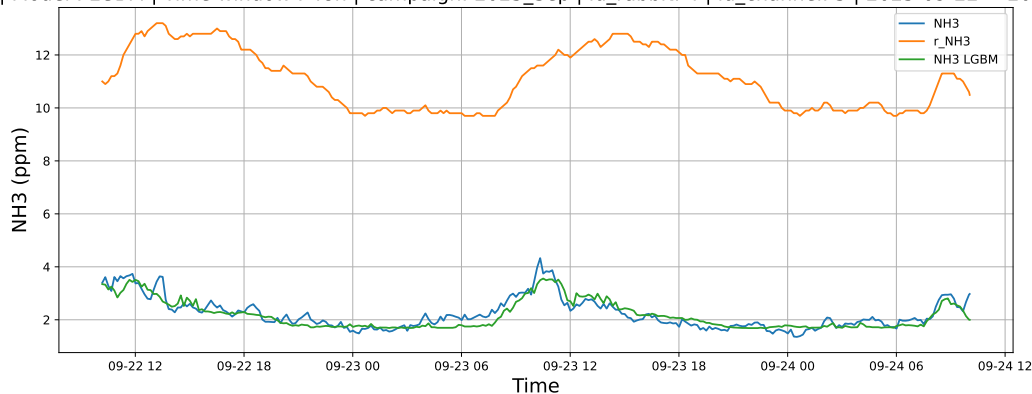


4.1.7.2 Time window : 48

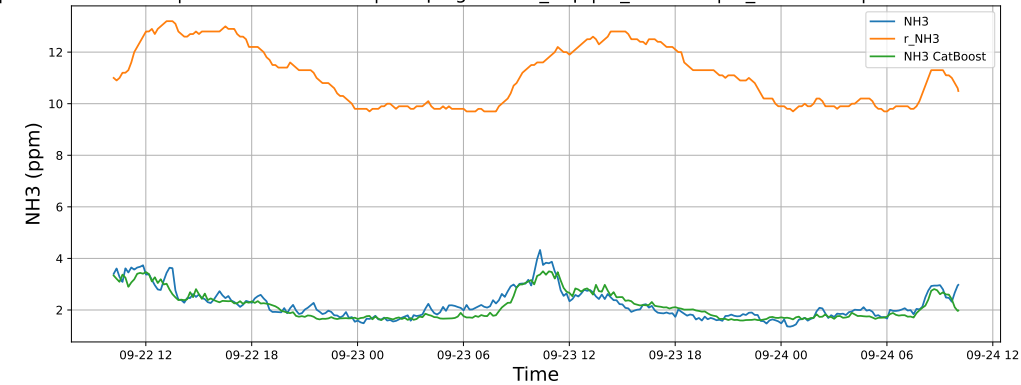
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

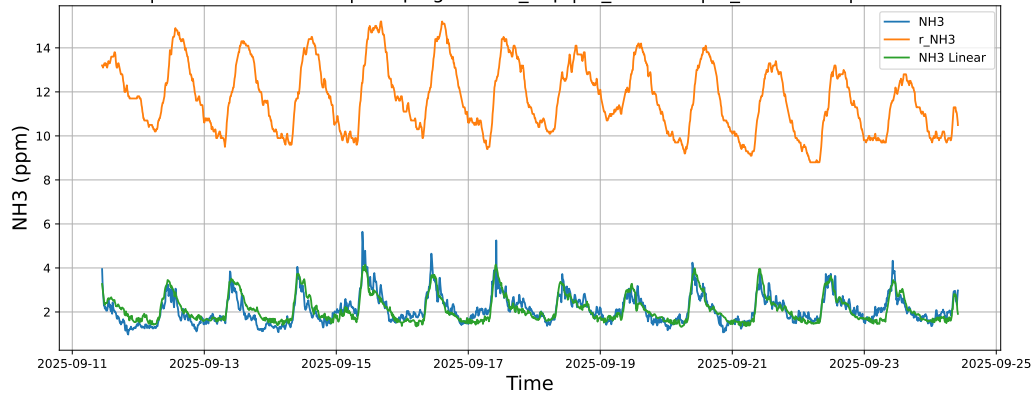


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

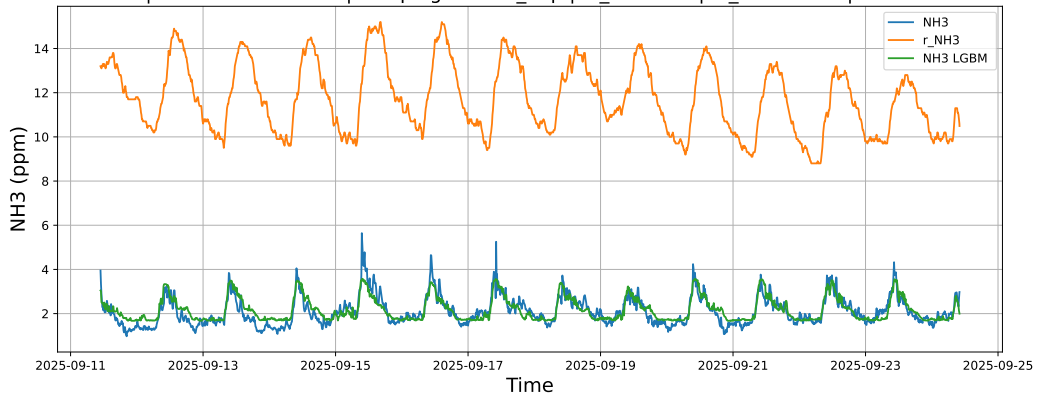


4.1.7.3 Time window : ALL

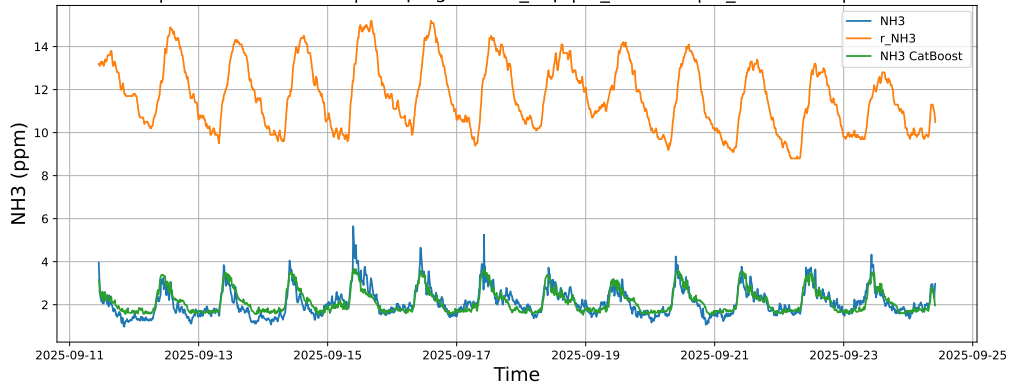
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



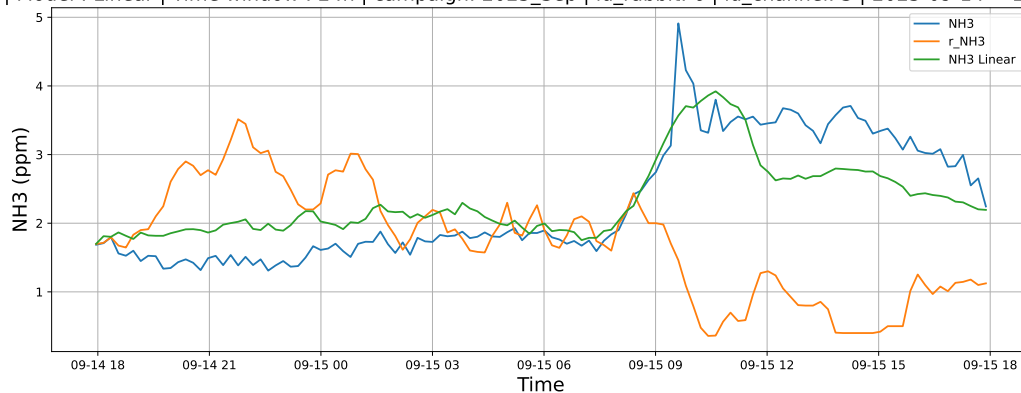
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



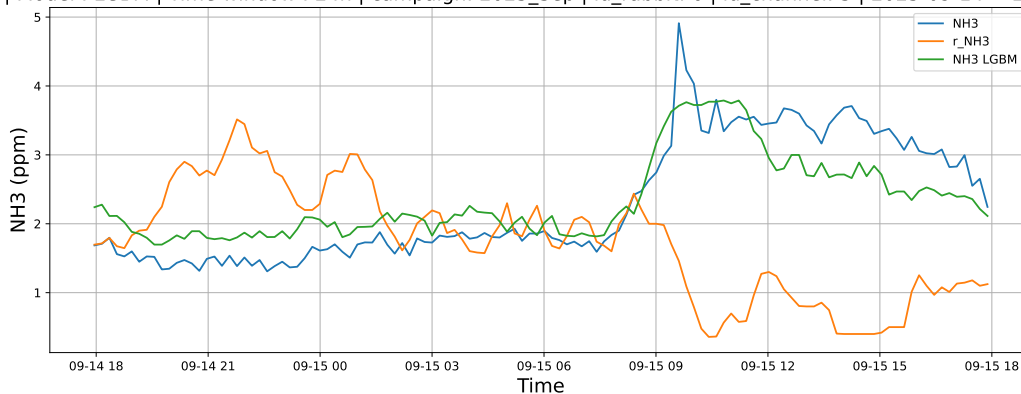
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

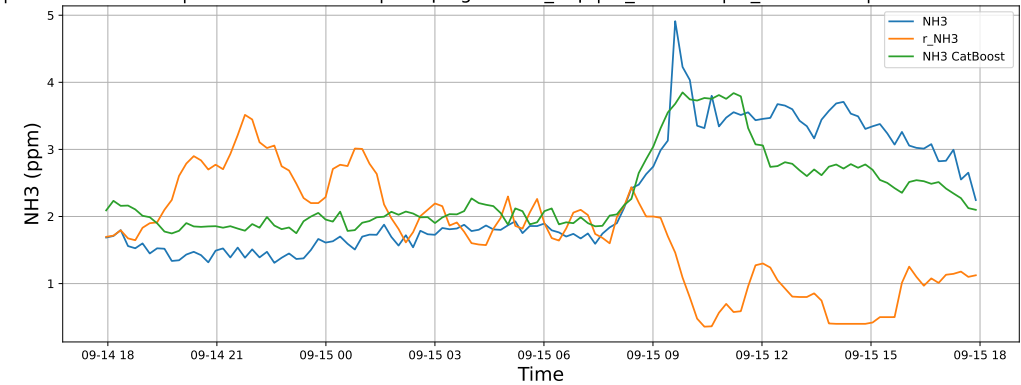
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

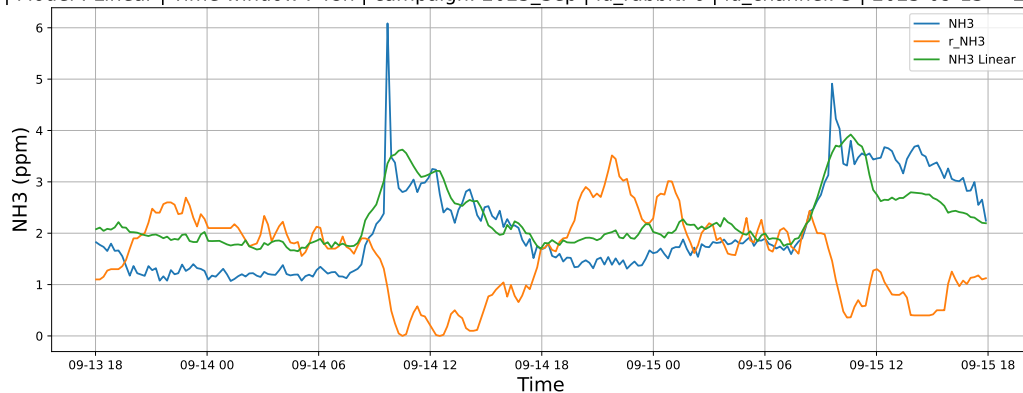


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

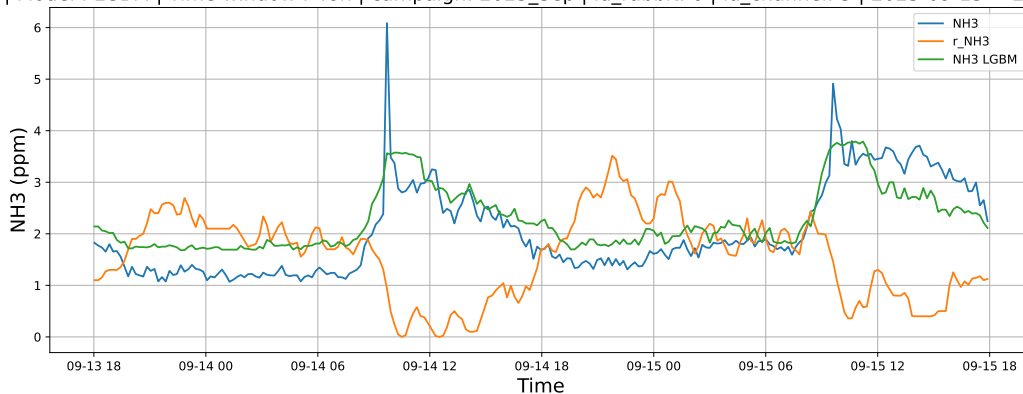


4.1.8.2 Time window : 48

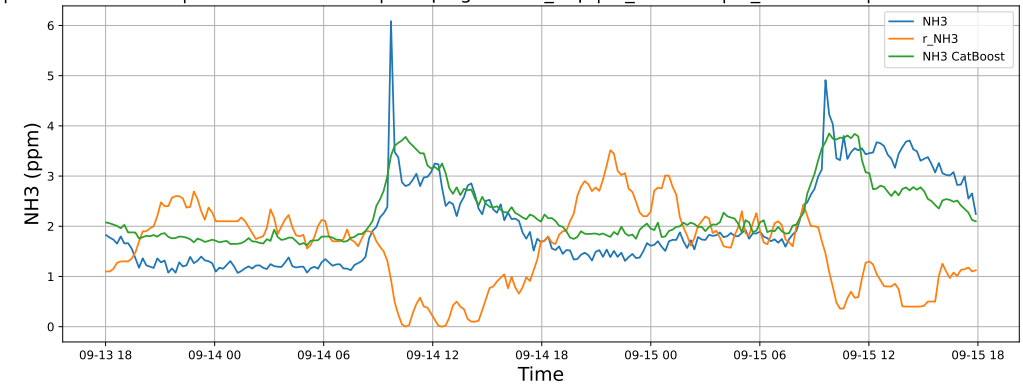
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

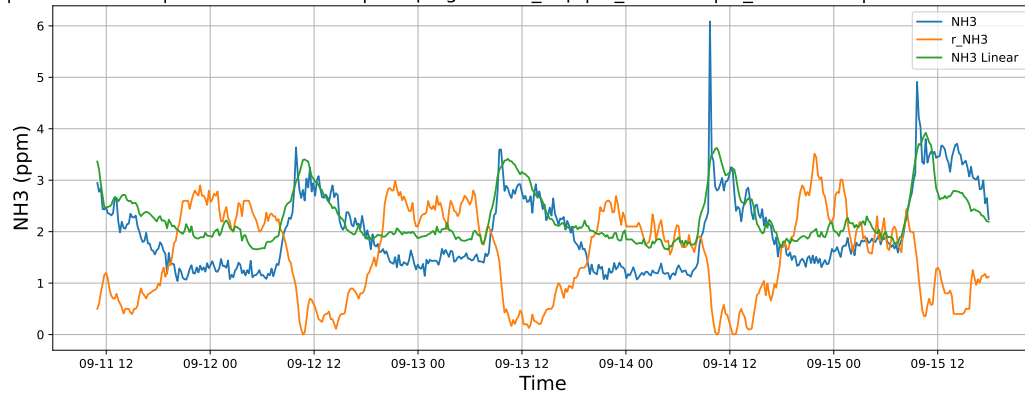


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

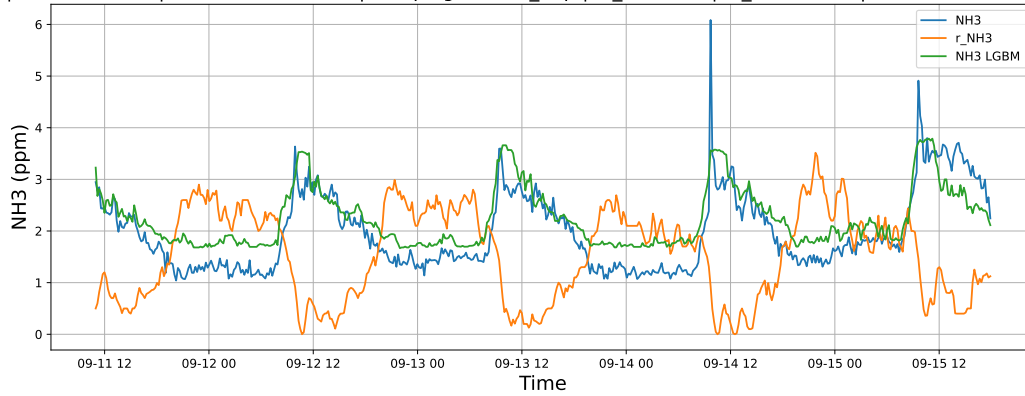


4.1.8.3 Time window : ALL

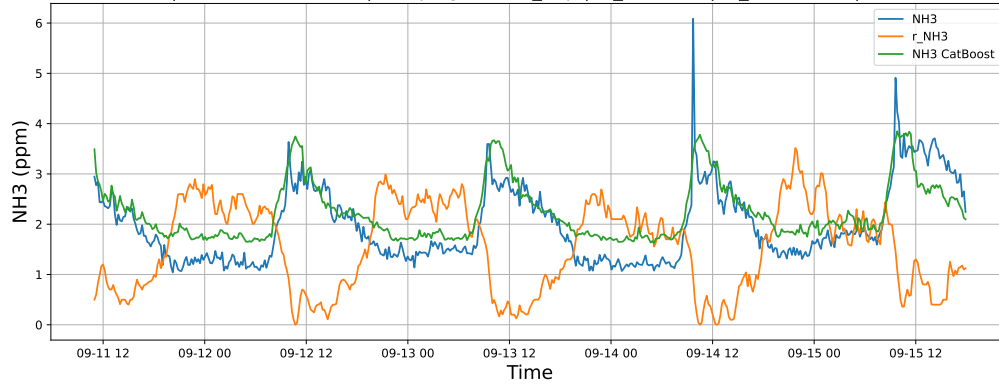
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

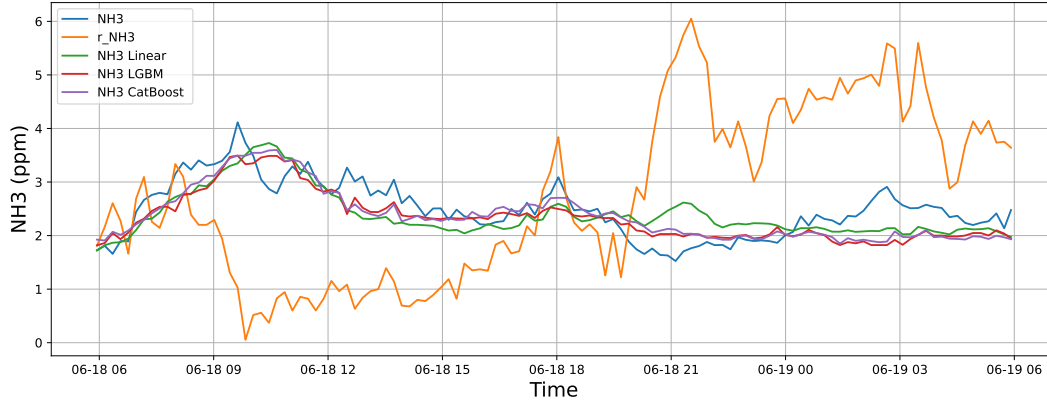


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

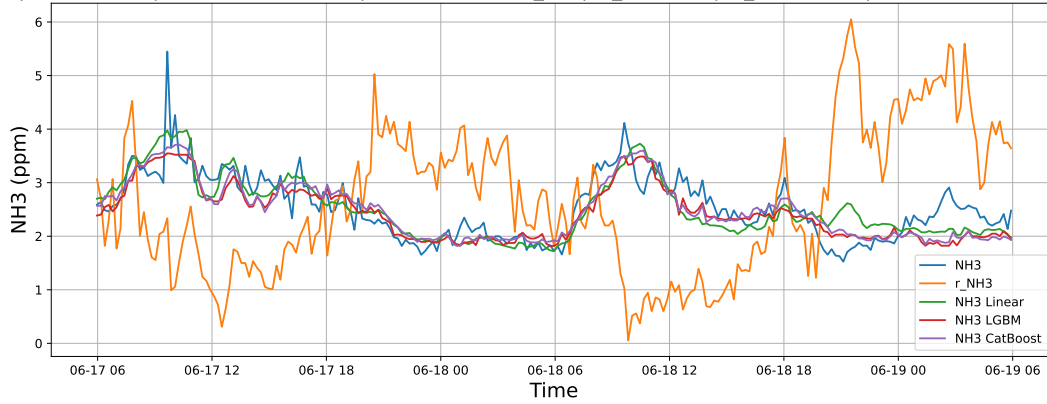
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



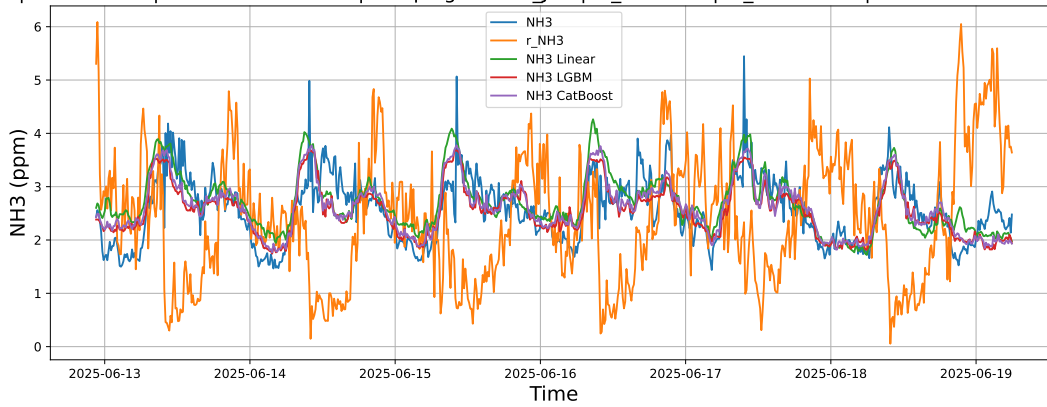
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

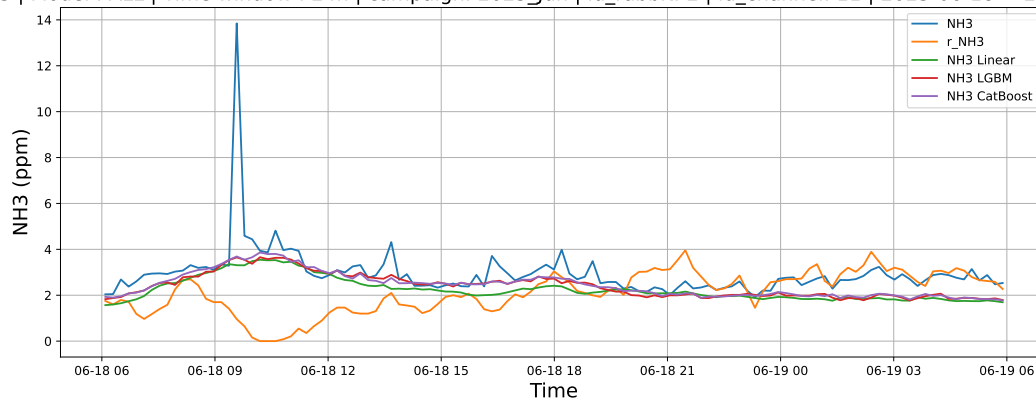
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

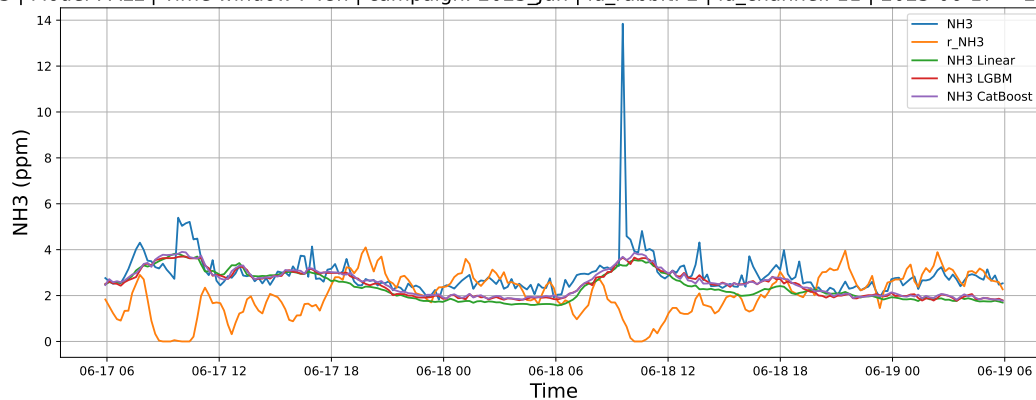
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



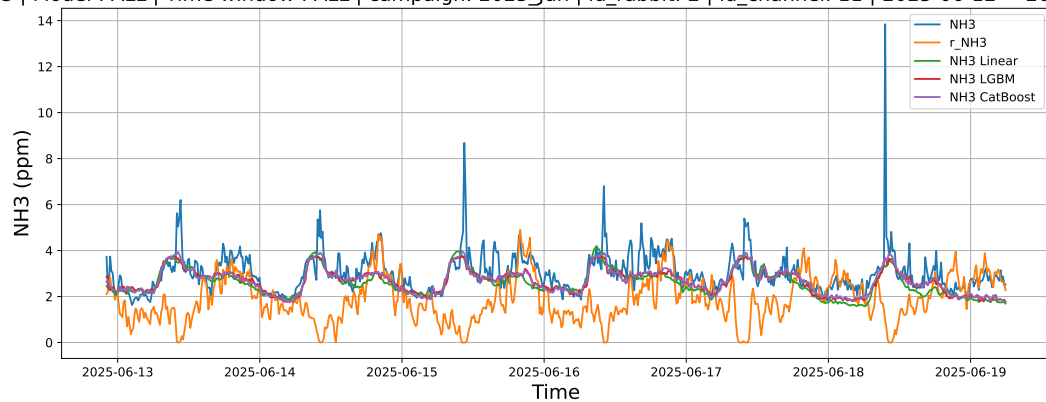
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

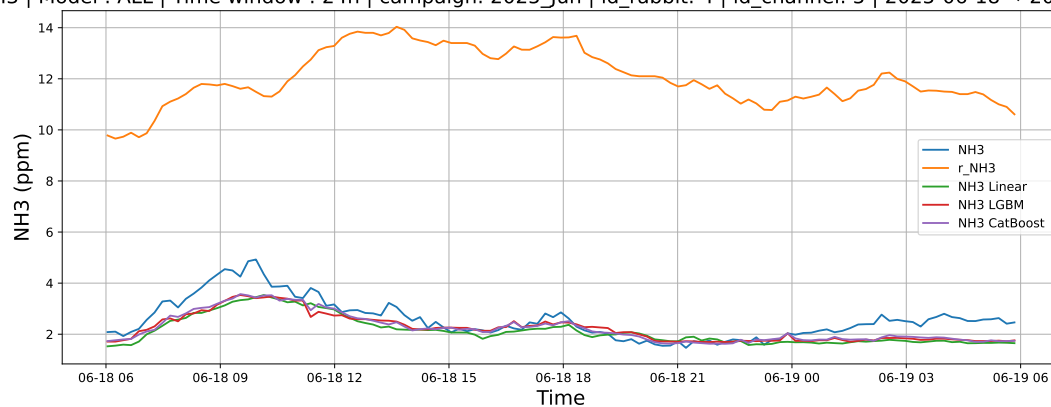
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

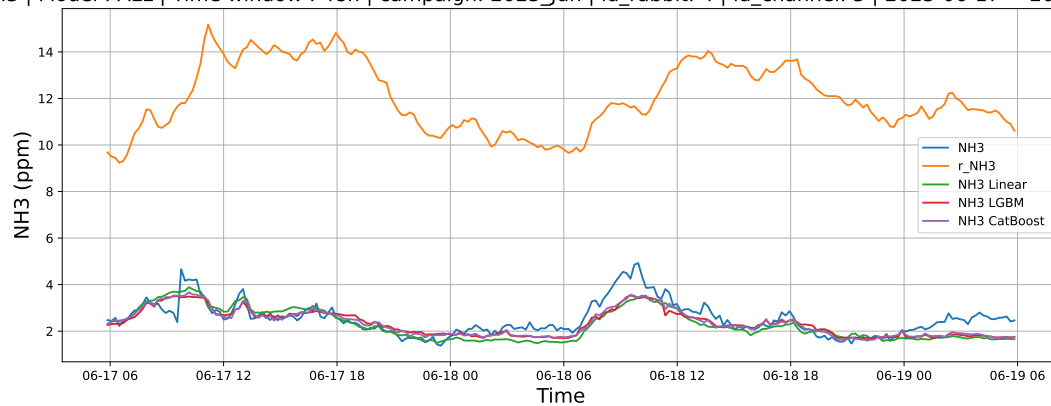
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



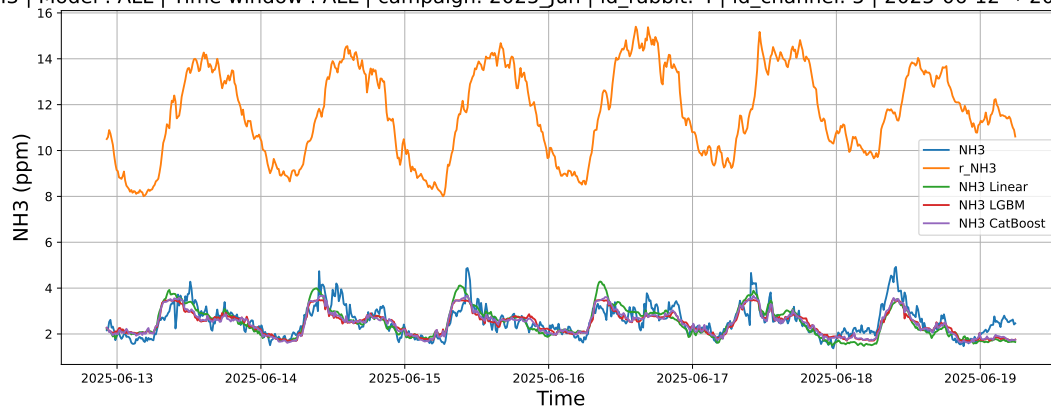
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

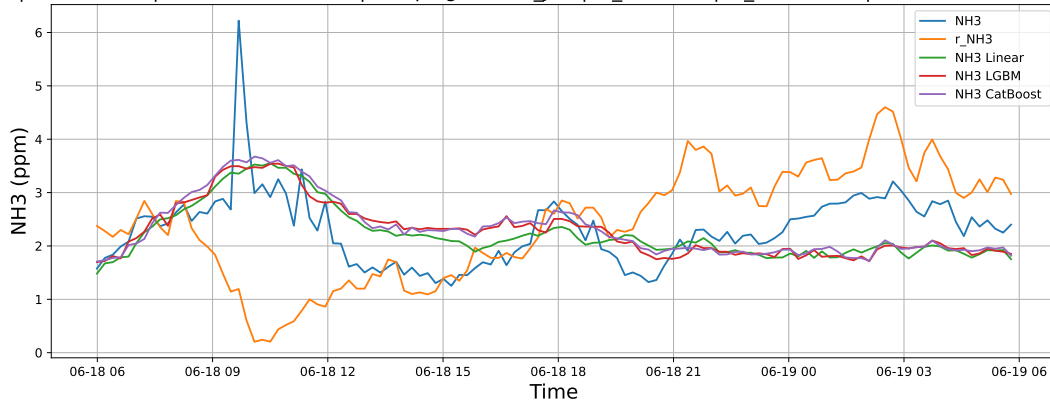
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

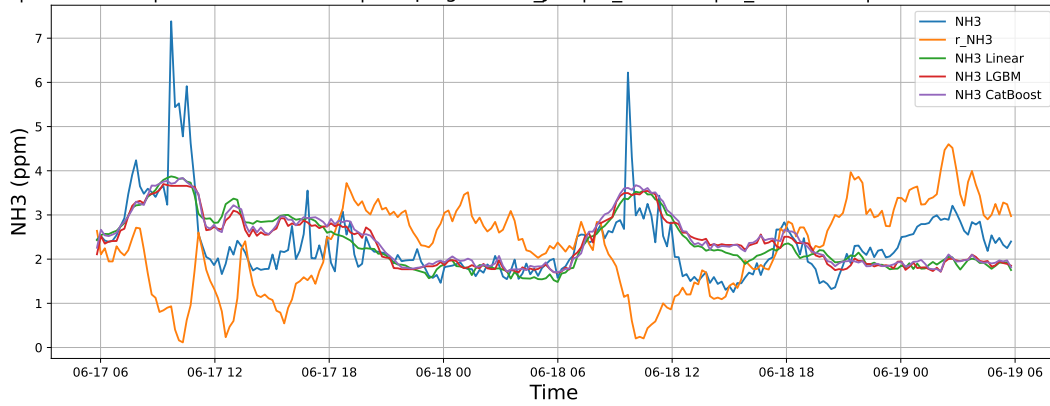
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



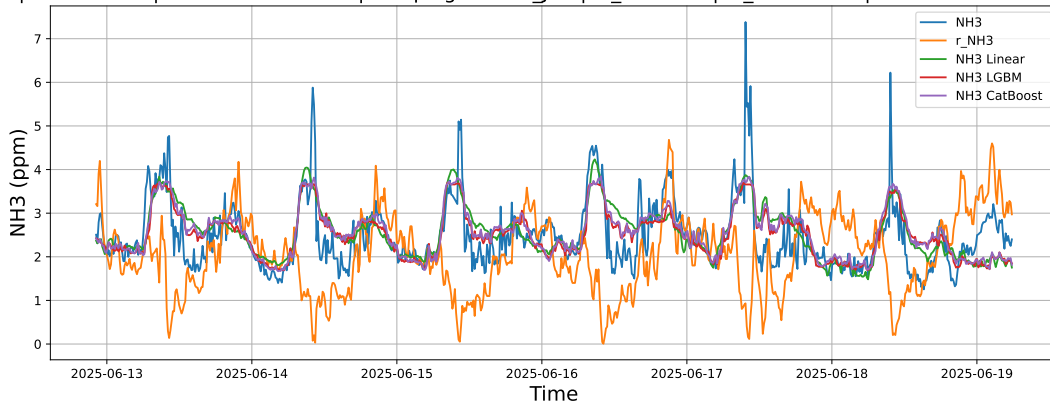
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

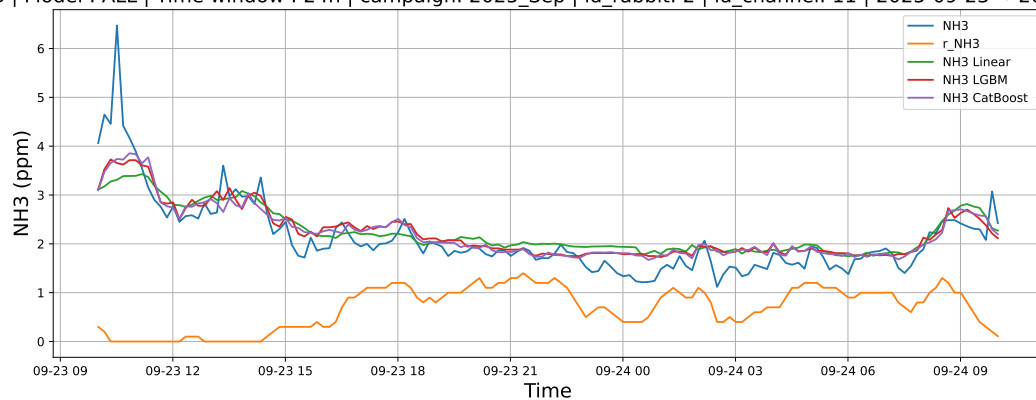
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

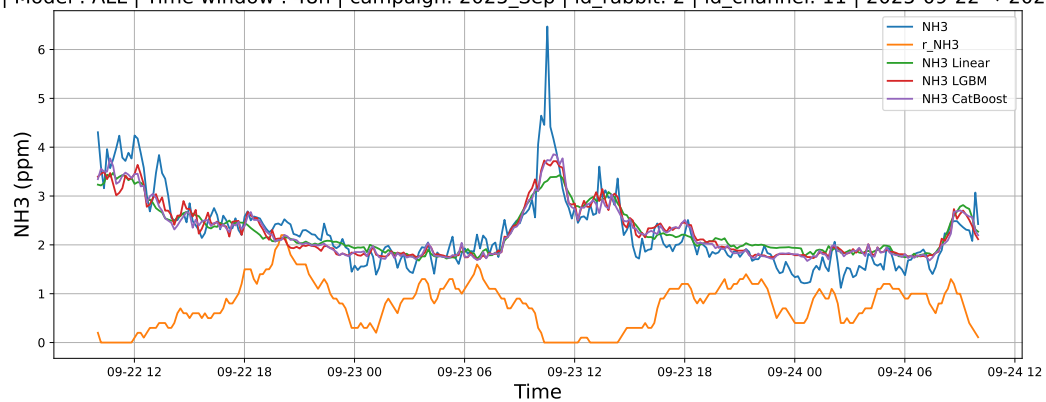
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



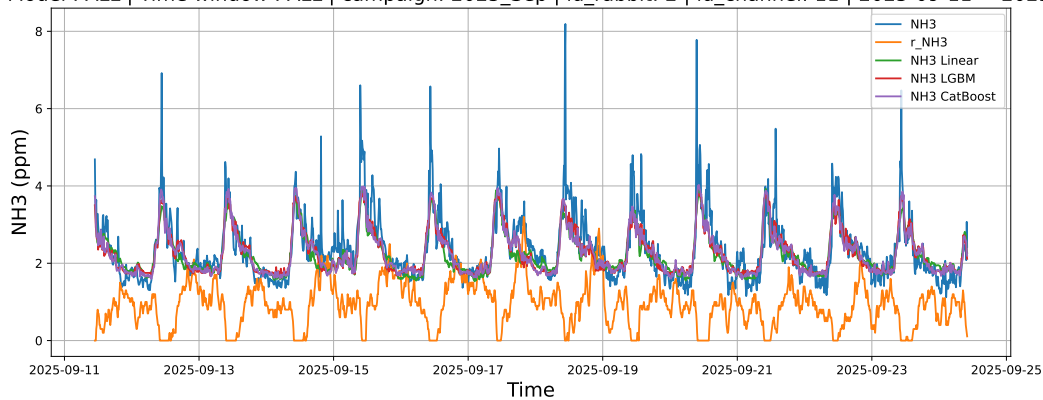
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

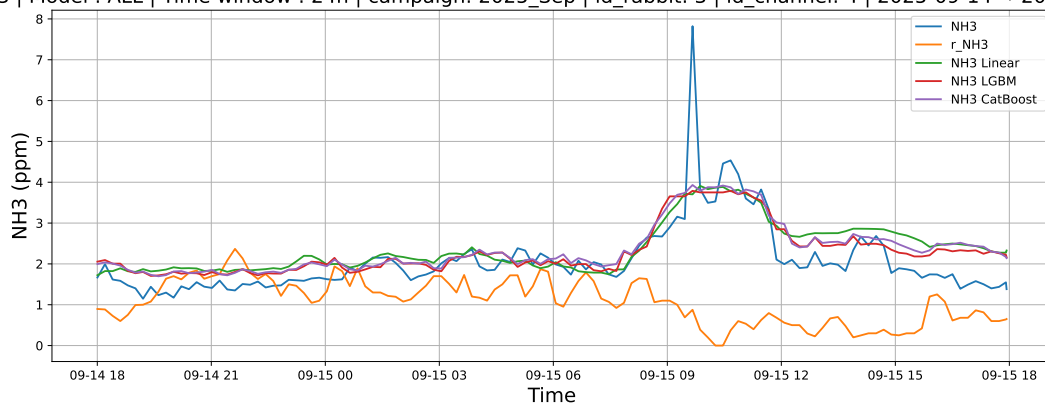
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

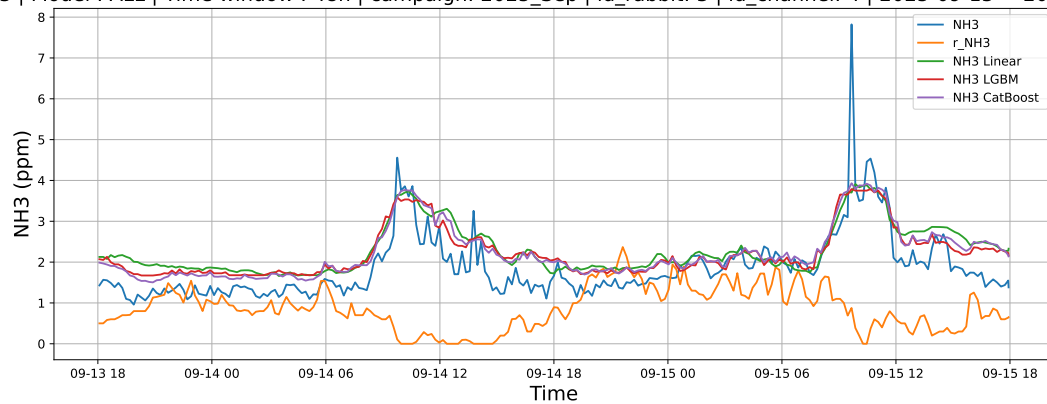
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



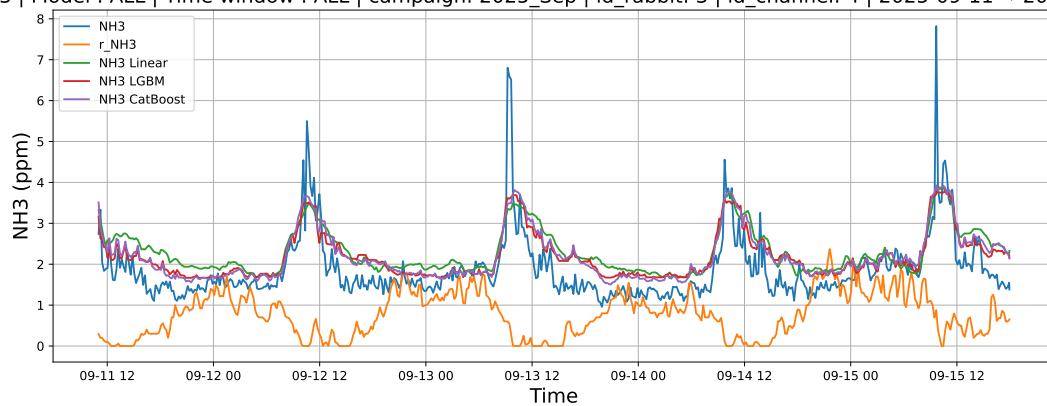
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

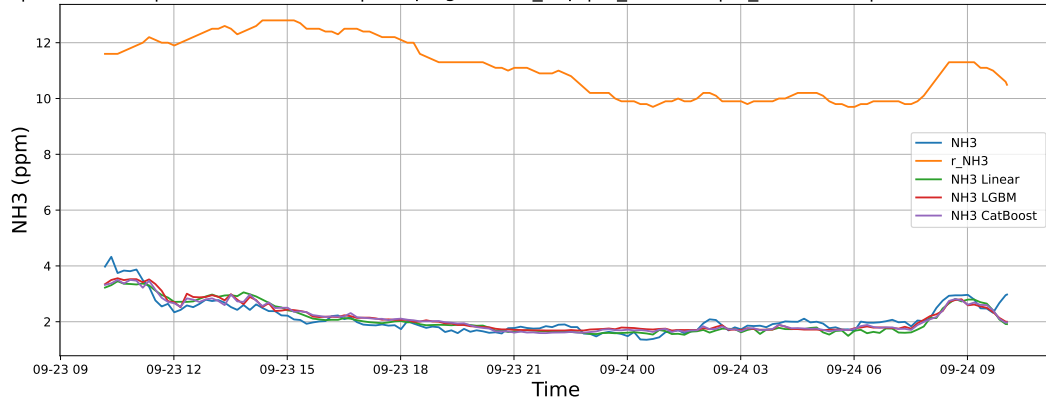
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

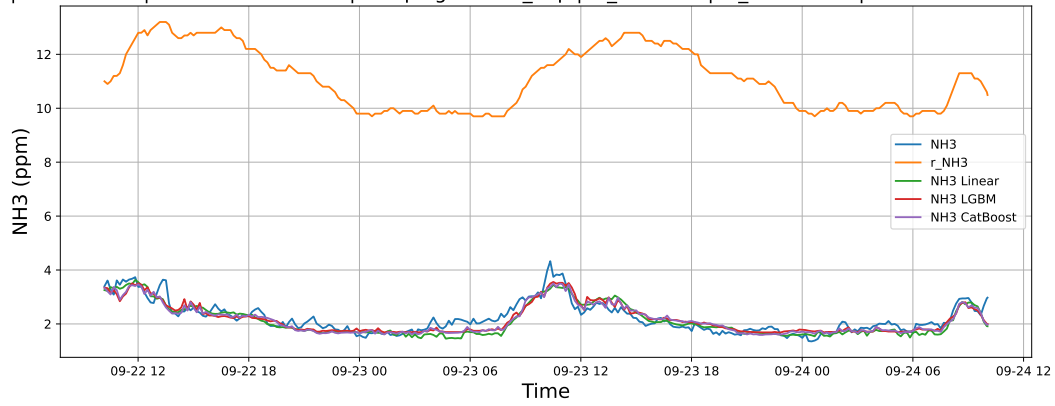
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



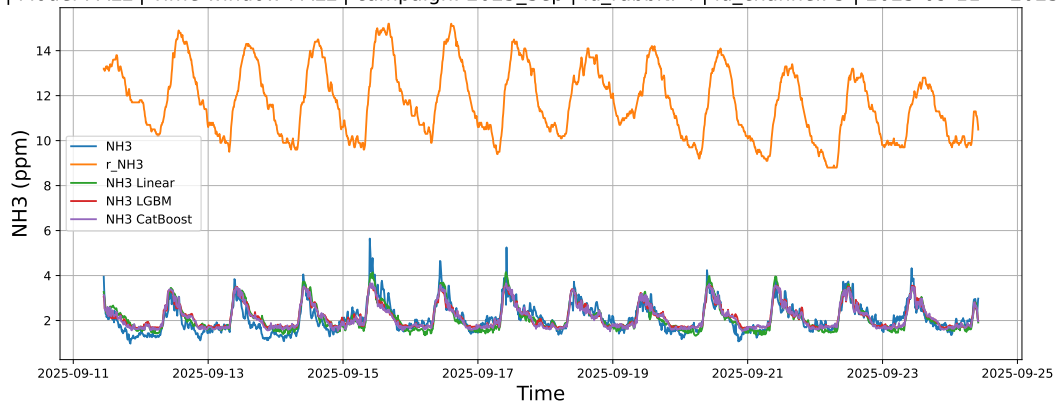
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

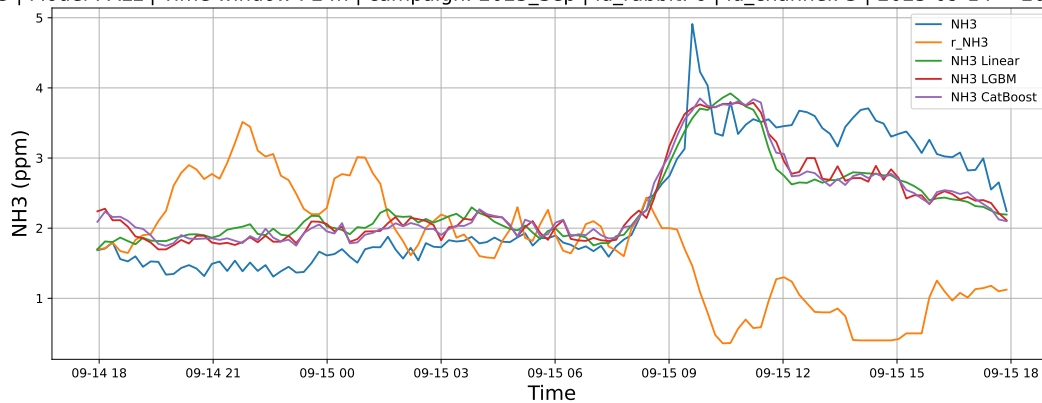
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

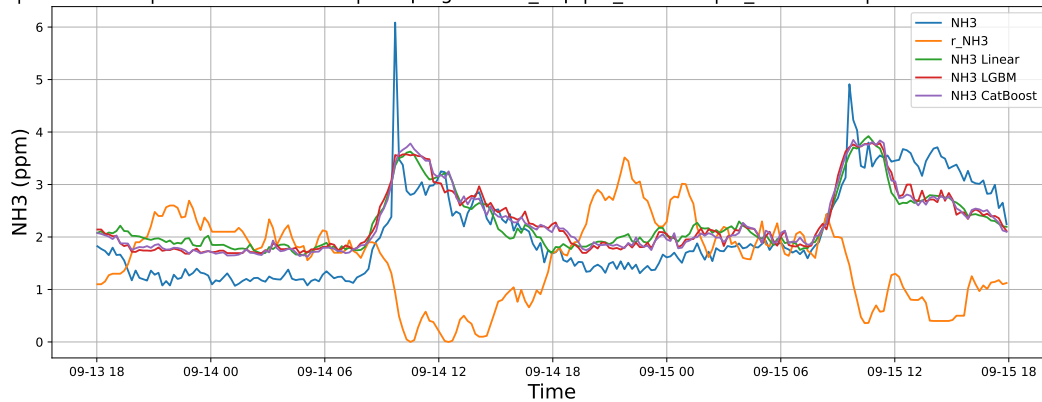
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

